table2_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	98.7	99.0	98.5	98.8	98.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.1	94.1	94.6	94.0	94.8	81.5	98.8	98.5	98.5	97.2	98.6
20%	95.2	95.4	95.7	95.1	93.8	83.9	99.3	98.8	99.0	98.7	98.3
30%	95.4	95.9	93.9	95.8	94.4	83.4	98.5	99.2	97.8	99.0	97.9
40%	95.6	96.0	96.6	97.0	97.0	86.6	99.1	98.9	99.2	99.6	99.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.6	92.4	92.2	94.6	78.3	98.1	97.8	97.6	97.2	98.2
20%	91.8	91.3	92.2	92.6	91.7	76.1	97.2	96.5	97.3	97.0	96.8
30%	90.3	90.0	87.4	90.3	88.6	69.6	96.2	96.7	94.4	96.0	94.9
40%	88.2	89.5	88.6	89.5	88.4	66.3	94.3	94.9	94.9	95.9	95.5
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.5	92.0	92.6	92.6	93.6	77.5	97.8	97.8	97.8	97.1	97.9
20%	91.4	90.8	91.6	91.8	91.8	74.8	97.3	97.2	97.3	97.3	96.6
30%	90.7	89.5	87.3	90.0	88.0	68.1	95.9	96.4	94.3	96.2	94.4
40%	87.3	90.8	87.8	89.8	87.5	64.7	94.9	96.0	95.2	96.3	94.1
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.6	98.7	97.9	98.6	98.5
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.7	94.3	95.0	94.2	94.9	81.9	98.1	98.5	98.4	98.0	98.6
20%	94.3	93.9	94.0	94.6	96.2	82.5	98.4	98.4	98.1	98.3	98.8
30%	95.5	95.8	96.6	95.3	96.0	84.8	99.0	99.1	99.4	98.7	99.1
40%	96.0	96.3	96.1	96.4	95.7	84.7	98.3	99.2	99.2	99.5	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	92.7	94.4	92.6	93.6	79.1	97.7	98.1	98.2	97.1	98.3
20%	90.7	90.8	91.0	91.7	93.4	75.2	97.6	96.0	96.3	96.6	97.9
30%	89.9	90.9	91.5	89.9	91.4	71.6	95.5	96.8	96.7	95.9	96.2
40%	88.1	87.8	87.8	86.6	88.2	64.3	94.8	94.3	94.1	93.8	94.8
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.2	93.5	94.1	92.8	93.0	78.4	97.9	97.9	98.1	97.3	97.9
20%	91.5	90.0	90.8	91.7	92.5	74.2	97.1	96.2	96.3	96.7	97.3
30%	89.6	91.1	91.1	89.4	89.7	68.6	95.9	96.1	96.7	95.1	95.2
40%	88.2	87.5	87.2	87.0	85.4	62.5	94.0	94.0	93.6	93.8	93.0

table2_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	94.4
Imputation using linear regression	NA
10%	93.5
20%	95.0
30%	94.8
40%	96.8
Imputation using predictive mean matching method	NA
10%	91.9
20%	89.2
30%	86.3
40%	83.7
Imputation using propensity−score method	NA
10%	91.5
20%	89.6
30%	84.7
40%	82.9
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.9
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.1
20%	94.2
30%	96.1
40%	95.4
Imputation using predictive mean matching method	NA
10%	92.7
20%	89.3
30%	87.5
40%	81.3
Imputation using propensity−score method	NA
10%	92.4
20%	88.5
30%	85.9
40%	79.7

table2_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	98.7	99.0	98.5	98.8	98.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.9	95.5	93.9	93.5	95.0	81.6	98.7	98.8	98.6	97.4	98.6
20%	95.4	95.0	96.5	94.6	94.3	83.6	98.6	98.6	99.0	99.1	98.9
30%	95.6	96.5	94.8	96.0	94.9	85.3	98.8	99.2	98.7	98.9	98.7
40%	96.1	95.7	96.3	96.1	96.5	85.7	99.0	99.2	99.0	99.3	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.8	93.0	90.5	93.7	77.6	97.8	97.9	97.8	96.9	97.4
20%	89.6	89.6	90.0	88.9	89.0	70.3	96.0	95.8	95.3	95.4	95.2
30%	84.9	85.5	82.9	86.3	85.3	62.2	92.9	94.1	91.7	94.7	93.0
40%	78.3	79.9	80.1	80.6	80.1	51.8	88.7	89.1	90.3	90.3	89.4
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.6	92.5	91.5	93.5	76.7	97.8	98.0	97.3	96.5	97.7
20%	90.4	88.6	89.9	88.8	88.9	68.7	95.7	95.5	96.4	95.6	95.6
30%	85.7	87.0	84.6	86.4	85.8	62.8	93.4	93.7	92.5	94.3	93.1
40%	80.4	83.0	83.3	82.5	81.9	53.8	90.0	91.3	91.6	91.2	89.3
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.6	98.7	97.9	98.6	98.5
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.8	94.7	96.1	94.6	95.1	83.5	97.9	98.8	98.6	98.1	98.8
20%	94.4	93.0	94.2	94.8	95.5	82.4	98.0	97.2	97.7	98.4	98.7
30%	95.0	95.4	96.1	96.0	96.5	86.2	98.1	98.5	99.6	98.7	99.3
40%	96.0	95.8	96.3	96.4	95.7	86.7	98.5	98.7	98.6	98.7	98.9
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.2	93.3	92.5	93.4	78.6	97.0	98.2	98.2	97.7	97.7
20%	89.1	88.1	88.5	88.3	90.2	70.2	95.1	94.1	94.2	94.9	96.3
30%	84.9	86.0	87.3	85.0	85.5	63.3	91.6	93.2	93.8	93.1	93.4
40%	79.5	78.0	78.2	78.3	79.1	50.8	88.7	87.9	87.8	88.9	88.8
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.6	93.6	92.9	92.2	92.4	77.9	97.4	98.6	98.0	97.7	97.6
20%	89.5	86.7	88.5	88.6	89.3	68.9	94.8	93.7	94.2	95.2	96.4
30%	85.5	86.3	86.3	85.5	86.0	62.2	92.3	92.5	93.8	93.2	93.4
40%	79.0	80.0	79.5	79.9	79.4	53.1	89.1	87.6	88.3	88.5	88.0

table2_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	94.4
Imputation using linear regression	NA
10%	94.1
20%	94.9
30%	95.7
40%	96.5
Imputation using predictive mean matching method	NA
10%	91.0
20%	85.3
30%	80.4
40%	70.7
Imputation using propensity−score method	NA
10%	90.6
20%	85.8
30%	79.9
40%	72.0
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.9
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
10%	94.0
20%	93.0
30%	95.7
40%	95.5
Imputation using predictive mean matching method	NA
10%	92.0
20%	85.7
30%	79.4
40%	68.9
Imputation using propensity−score method	NA
10%	91.6
20%	84.7
30%	79.1
40%	69.4

table2_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	98.4	98.6	98.4	98.7	97.9
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.2	94.3	94.6	93.6	94.7	82.7	98.2	98.4	98.3	96.9	98.6
20%	94.3	94.8	95.6	94.8	93.8	84.8	98.7	98.6	98.9	98.6	98.2
30%	95.3	95.2	93.8	95.7	94.4	84.2	98.5	98.9	97.8	99.1	97.9
40%	96.1	96.3	96.5	97.3	97.0	88.3	98.7	98.6	99.2	99.5	99.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.7	93.0	92.6	92.0	94.6	80.3	97.4	97.6	97.4	97.3	98.2
20%	91.2	91.2	91.9	92.9	91.8	77.8	97.0	96.3	97.2	97.0	96.7
30%	90.9	90.5	87.9	90.3	88.6	71.9	96.3	96.6	94.8	95.7	94.7
40%	89.1	88.4	88.9	89.8	88.6	69.1	94.9	95.1	95.1	95.8	95.3
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.4	92.5	92.6	93.0	93.8	79.2	97.4	97.5	97.2	97.2	97.8
20%	91.4	91.3	91.6	91.7	91.7	77.1	96.7	96.7	97.1	97.2	96.5
30%	89.7	90.2	87.4	90.7	88.2	70.8	96.6	96.2	94.0	96.1	94.2
40%	89.2	91.1	88.4	90.0	87.8	68.5	95.6	96.1	95.4	96.4	94.1
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	98.5	98.5	97.6	98.5	98.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.0	94.1	95.0	93.6	94.9	83.0	98.0	98.5	98.5	97.9	98.5
20%	94.5	93.6	94.0	94.6	96.1	84.1	97.8	98.4	98.1	98.2	98.5
30%	95.5	95.8	96.9	95.3	95.9	86.1	98.6	98.8	99.4	98.6	99.0
40%	95.5	96.5	96.2	96.5	95.7	85.6	98.6	99.0	99.3	99.5	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.9	92.8	94.0	92.4	93.6	80.4	97.4	97.6	97.8	97.1	98.0
20%	91.2	91.1	91.7	91.9	93.3	77.8	96.5	96.4	96.1	96.6	97.5
30%	90.5	91.0	91.7	90.1	91.2	74.1	95.8	96.5	97.0	96.1	96.0
40%	88.1	87.6	87.7	86.8	88.2	66.7	94.6	94.0	94.3	93.8	94.8
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.2	94.0	92.7	92.9	79.0	97.6	97.6	97.6	97.0	97.7
20%	91.0	90.1	91.1	92.5	92.4	76.4	96.3	96.2	95.8	96.3	97.3
30%	90.6	91.3	91.0	90.1	90.0	72.3	95.9	96.1	96.9	95.2	95.3
40%	88.5	88.3	88.0	87.3	86.6	66.5	94.0	94.2	93.9	93.8	93.1

table2_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	94.0
Imputation using linear regression	NA
10%	93.5
20%	94.6
30%	94.8
40%	96.5
Imputation using predictive mean matching method	NA
10%	92.1
20%	89.2
30%	86.8
40%	85.4
Imputation using propensity−score method	NA
10%	91.5
20%	89.5
30%	85.3
40%	84.5
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.8
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.4
20%	93.8
30%	95.7
40%	95.9
Imputation using predictive mean matching method	NA
10%	92.5
20%	89.8
30%	88.5
40%	82.8
Imputation using propensity−score method	NA
10%	92.3
20%	88.7
30%	87.3
40%	80.8

table2_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	98.4	98.6	98.4	98.7	97.9
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.7	95.1	94.3	93.7	95.1	82.7	98.1	98.7	98.3	97.4	98.4
20%	95.3	94.6	96.2	94.6	94.2	84.1	98.6	98.8	99.0	98.9	98.9
30%	95.8	96.2	94.9	96.0	95.0	86.5	98.8	98.8	98.6	98.9	98.5
40%	96.3	96.0	96.4	96.0	96.5	87.0	98.8	99.2	99.0	99.2	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.1	93.2	90.5	93.6	78.8	97.5	97.9	97.7	97.0	97.3
20%	89.7	90.0	90.1	89.0	89.0	72.5	95.4	95.5	95.1	94.9	94.9
30%	85.8	86.1	83.2	86.5	85.4	65.1	92.4	93.7	91.3	94.5	92.8
40%	78.9	79.3	80.5	80.5	79.8	55.1	88.4	88.7	90.1	89.8	89.2
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.3	92.6	91.6	93.5	78.1	97.2	97.9	97.1	96.4	97.7
20%	89.6	89.1	90.0	89.1	89.0	70.7	95.9	95.3	96.2	95.8	95.2
30%	85.3	87.2	85.3	86.2	85.9	64.6	92.4	93.1	92.4	93.8	93.0
40%	80.2	82.9	83.4	82.3	82.0	57.0	90.3	90.8	90.8	91.0	89.0
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	98.5	98.5	97.6	98.5	98.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.5	94.3	95.8	94.1	95.2	84.2	97.7	98.5	98.6	98.1	98.7
20%	93.7	92.9	94.3	94.7	95.5	83.4	97.9	97.1	97.6	98.1	98.7
30%	95.6	95.9	96.2	95.9	96.6	87.3	98.4	98.2	99.5	98.4	99.1
40%	95.6	96.0	96.2	96.2	95.7	87.5	98.4	98.8	98.6	98.7	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.0	93.2	92.3	93.4	80.2	96.5	97.4	97.9	97.7	97.1
20%	88.6	87.9	88.8	88.1	90.2	72.4	94.7	94.5	94.0	95.0	96.0
30%	85.6	85.3	86.8	85.0	85.4	65.6	91.7	93.2	93.8	92.9	93.2
40%	79.7	78.7	78.2	78.0	79.2	53.5	87.7	87.4	87.3	88.4	88.3
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.3	92.8	92.8	92.1	92.5	79.1	96.9	98.0	97.7	97.4	97.4
20%	88.5	87.4	88.0	88.4	90.0	71.1	94.5	94.1	93.8	94.5	96.2
30%	85.3	85.8	86.3	85.5	86.6	64.1	92.2	93.1	93.6	93.3	93.5
40%	79.3	79.5	80.3	80.6	80.0	55.8	88.8	86.8	88.3	88.7	88.2

table2_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	94.0
Imputation using linear regression	NA
10%	93.8
20%	95.3
30%	95.7
40%	96.4
Imputation using predictive mean matching method	NA
10%	91.8
20%	85.5
30%	80.8
40%	72.3
Imputation using propensity−score method	NA
10%	91.1
20%	86.1
30%	80.5
40%	73.8
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.8
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
10%	93.9
20%	93.2
30%	95.5
40%	95.4
Imputation using predictive mean matching method	NA
10%	91.3
20%	86.5
30%	80.5
40%	70.9
Imputation using propensity−score method	NA
10%	90.9
20%	85.5
30%	80.2
40%	71.3

table2_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	98.1	98.4	98.4	98.7	97.7
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.4	94.4	94.5	93.4	94.8	83.6	98.0	98.0	98.1	97.2	98.5
20%	93.6	94.4	95.4	94.9	93.8	85.0	98.2	98.5	98.9	98.6	98.2
30%	94.7	95.2	93.7	95.9	94.4	84.8	98.3	98.7	97.8	99.1	97.8
40%	95.7	95.8	96.4	97.0	97.0	88.2	98.8	98.5	99.2	99.4	99.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.6	93.2	92.8	92.0	94.5	81.6	97.2	97.7	97.3	97.1	98.1
20%	91.3	91.4	91.7	92.9	91.9	78.8	96.3	96.3	97.0	97.0	96.7
30%	90.4	90.2	88.0	90.5	88.5	73.1	96.0	96.4	94.8	95.6	94.6
40%	90.3	89.3	89.8	89.8	88.8	71.7	95.0	94.9	94.9	95.5	95.0
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.8	92.6	93.1	93.6	80.5	97.3	97.4	97.3	97.1	97.5
20%	91.2	91.7	91.4	91.7	91.7	78.1	96.2	96.2	97.0	96.9	96.6
30%	90.5	89.8	87.8	90.6	88.1	72.1	96.3	95.7	94.2	96.1	94.2
40%	90.1	91.4	88.6	90.2	87.9	71.3	95.5	96.0	95.6	95.9	94.0
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.3	98.4	97.6	98.4	98.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.3	94.1	94.8	93.4	94.8	83.4	97.7	98.3	98.5	97.8	98.5
20%	94.2	94.0	94.2	94.4	96.1	85.4	97.8	98.2	97.9	98.1	98.5
30%	95.5	95.5	96.9	95.5	95.8	86.9	98.6	98.5	99.2	98.5	98.9
40%	95.1	95.8	96.0	96.6	95.7	85.9	98.3	99.0	99.2	99.5	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	91.8	92.6	94.0	92.5	93.8	81.1	97.2	97.6	97.7	96.6	97.8
20%	90.9	91.1	91.6	91.8	93.4	79.2	96.2	96.5	95.8	96.4	97.5
30%	90.6	91.7	91.8	90.2	91.2	75.8	95.6	96.3	97.0	96.0	96.0
40%	88.2	88.6	88.0	87.1	88.2	68.8	94.1	93.9	94.2	93.9	94.7
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.9	93.9	93.1	92.7	80.3	97.0	97.6	97.5	96.9	97.5
20%	91.2	90.1	91.0	92.6	92.8	77.7	96.1	96.1	95.7	96.2	97.3
30%	90.7	91.3	91.4	90.4	90.5	74.6	96.0	95.8	96.8	95.3	95.3
40%	88.5	88.3	88.5	87.6	86.7	68.6	93.8	94.4	93.8	93.7	93.4

table2_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.9
Imputation using linear regression	NA
10%	93.7
20%	94.4
30%	94.6
40%	96.5
Imputation using predictive mean matching method	NA
10%	92.1
20%	89.7
30%	86.7
40%	85.4
Imputation using propensity−score method	NA
10%	91.8
20%	89.1
30%	85.6
40%	84.8
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.8
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.6
20%	94.1
30%	95.4
40%	95.7
Imputation using predictive mean matching method	NA
10%	92.7
20%	90.4
30%	88.7
40%	83.1
Imputation using propensity−score method	NA
10%	92.2
20%	89.4
30%	87.9
40%	81.8

table2_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	98.1	98.4	98.4	98.7	97.7
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.6	95.4	94.1	94.0	95.1	83.7	97.8	98.3	98.0	97.3	98.3
20%	94.5	94.2	96.0	94.5	94.3	84.2	98.7	98.6	99.0	98.9	98.8
30%	95.3	96.2	94.8	95.9	95.0	87.2	98.3	98.5	98.5	98.9	98.5
40%	96.4	96.2	96.4	96.1	96.5	88.3	98.7	99.1	99.0	99.0	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.7	92.5	90.7	93.5	79.3	97.0	97.8	97.2	96.8	97.2
20%	89.3	89.7	90.1	89.2	89.2	74.0	95.4	95.2	94.9	94.8	94.9
30%	85.5	85.8	83.4	86.8	85.6	66.8	92.4	93.0	91.1	94.1	92.6
40%	79.3	79.3	80.6	80.0	79.8	57.2	88.3	88.5	89.9	89.3	88.9
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.3	92.7	91.8	93.5	79.3	96.9	97.7	96.8	96.2	97.6
20%	89.5	89.1	89.9	89.4	89.0	72.1	95.9	95.4	95.8	95.6	95.1
30%	85.4	86.6	85.0	86.2	85.9	65.9	92.6	93.6	92.2	93.7	92.7
40%	80.9	83.0	83.3	82.0	82.2	59.2	89.5	90.0	90.4	90.7	88.7
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.3	98.4	97.6	98.4	98.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.3	94.2	95.2	93.9	95.2	84.7	97.4	98.4	98.6	98.0	98.7
20%	93.2	93.0	94.1	94.7	95.6	84.6	97.5	97.0	97.5	98.0	98.7
30%	95.6	96.0	96.2	96.0	96.6	88.0	98.2	98.0	99.4	98.4	99.1
40%	95.0	95.9	96.1	96.2	95.8	88.1	98.5	98.7	98.4	98.6	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.7	93.1	92.2	93.3	80.8	96.1	97.1	97.7	97.7	96.9
20%	88.3	88.3	88.9	87.8	90.3	73.9	94.1	94.1	94.0	94.6	95.8
30%	85.2	85.3	86.9	84.7	85.3	66.9	92.0	93.0	93.1	92.7	93.1
40%	80.3	79.0	78.5	78.1	79.5	55.7	87.8	87.0	86.8	88.2	87.7
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.2	92.4	92.4	92.1	92.5	79.8	96.8	97.8	97.6	97.5	97.2
20%	88.0	87.4	87.9	88.3	90.1	72.4	94.0	93.9	93.8	94.1	96.1
30%	84.7	85.8	86.7	85.5	87.6	66.3	91.8	92.9	93.5	92.8	93.6
40%	80.6	79.1	80.4	81.0	80.4	58.2	89.2	87.3	88.1	88.7	88.3

table2_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.9
Imputation using linear regression	NA
10%	93.5
20%	95.3
30%	95.7
40%	96.3
Imputation using predictive mean matching method	NA
10%	91.8
20%	85.7
30%	81.8
40%	73.1
Imputation using propensity−score method	NA
10%	91.0
20%	86.8
30%	81.2
40%	73.9
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.8
Imputation using linear regression	NA

table2_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
10%	94.0
20%	93.2
30%	95.6
40%	95.4
Imputation using predictive mean matching method	NA
10%	91.1
20%	86.4
30%	80.9
40%	71.8
Imputation using propensity−score method	NA
10%	91.3
20%	85.9
30%	80.7
40%	72.7

table2_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	99.2	99.0	98.5	98.2	97.3
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.1	94.1	94.6	94.0	94.8	81.5	98.8	98.6	98.0	97.5	98.0
20%	95.2	95.4	95.7	95.1	93.8	83.9	99.5	98.8	99.1	98.6	97.7
30%	95.4	95.9	93.9	95.8	94.4	83.4	99.0	99.5	98.1	99.2	97.8
40%	95.6	96.0	96.6	97.0	97.0	86.6	99.0	99.2	99.3	99.6	99.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.6	92.4	92.2	94.6	78.3	98.4	97.8	97.9	96.6	97.3
20%	91.8	91.3	92.2	92.6	91.7	76.1	97.9	96.9	97.5	96.6	95.8
30%	90.3	90.0	87.4	90.3	88.6	69.6	96.5	96.5	94.5	95.3	93.9
40%	88.2	89.5	88.6	89.5	88.4	66.3	95.4	95.3	95.3	95.3	94.7
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.5	92.0	92.6	92.6	93.6	77.5	98.3	97.7	97.7	96.7	97.3
20%	91.4	90.8	91.6	91.8	91.8	74.8	97.9	97.1	96.9	96.2	95.5
30%	90.7	89.5	87.3	90.0	88.0	68.1	96.5	96.0	94.1	95.7	92.9
40%	87.3	90.8	87.8	89.8	87.5	64.7	95.5	96.2	94.9	96.0	93.1
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.9	98.7	98.0	98.5	97.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.7	94.3	95.0	94.2	94.9	81.9	98.2	98.1	98.6	97.4	98.1
20%	94.3	93.9	94.0	94.6	96.2	82.5	98.7	98.7	98.3	98.2	98.5
30%	95.5	95.8	96.6	95.3	96.0	84.8	99.3	99.0	99.3	98.8	99.0
40%	96.0	96.3	96.1	96.4	95.7	84.7	98.9	99.5	99.5	99.6	98.3
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	92.7	94.4	92.6	93.6	79.1	97.9	97.5	97.9	96.1	96.5
20%	90.7	90.8	91.0	91.7	93.4	75.2	97.7	96.7	96.1	96.1	97.1
30%	89.9	90.9	91.5	89.9	91.4	71.6	96.6	96.7	96.7	95.2	95.6
40%	88.1	87.8	87.8	86.6	88.2	64.3	95.3	94.9	93.7	92.9	93.6
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.2	93.5	94.1	92.8	93.0	78.4	98.0	97.5	97.4	96.0	96.5
20%	91.5	90.0	90.8	91.7	92.5	74.2	97.6	96.1	95.8	96.0	96.7
30%	89.6	91.1	91.1	89.4	89.7	68.6	96.2	96.6	95.7	94.4	94.1
40%	88.2	87.5	87.2	87.0	85.4	62.5	94.9	94.3	92.8	92.9	91.7

table2_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.4
Imputation using linear regression	NA
10%	93.6
20%	94.7
30%	95.2
40%	97.1
Imputation using predictive mean matching method	NA
10%	91.3
20%	89.3
30%	85.6
40%	84.0
Imputation using propensity−score method	NA
10%	91.0
20%	88.3
30%	84.0
40%	82.5
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.3
Imputation using linear regression	NA

table2_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
10%	92.9
20%	94.5
30%	96.2
40%	96.1
Imputation using predictive mean matching method	NA
10%	90.1
20%	89.3
30%	87.3
40%	81.5
Imputation using propensity−score method	NA
10%	89.6
20%	87.8
30%	84.4
40%	78.4

table2_pooled_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	99.2	99.0	98.5	98.2	97.3
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.9	95.5	93.9	93.5	95.0	81.6	98.8	98.8	98.3	97.1	97.8
20%	95.4	95.0	96.5	94.6	94.3	83.6	99.1	99.2	99.2	98.7	98.9
30%	95.6	96.5	94.8	96.0	94.9	85.3	98.8	99.5	98.5	98.9	98.5
40%	96.1	95.7	96.3	96.1	96.5	85.7	99.2	99.0	99.1	99.1	98.5
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.8	93.0	90.5	93.7	77.6	98.7	98.1	97.5	95.9	96.8
20%	89.6	89.6	90.0	88.9	89.0	70.3	97.0	96.2	95.6	94.9	94.5
30%	84.9	85.5	82.9	86.3	85.3	62.2	94.2	94.0	91.5	93.8	91.9
40%	78.3	79.9	80.1	80.6	80.1	51.8	90.2	89.7	90.5	89.2	88.1
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.6	92.5	91.5	93.5	76.7	98.3	98.1	97.5	95.7	96.8
20%	90.4	88.6	89.9	88.8	88.9	68.7	96.9	95.6	96.0	95.2	93.7
30%	85.7	87.0	84.6	86.4	85.8	62.8	94.5	93.9	92.2	92.9	91.7
40%	80.4	83.0	83.3	82.5	81.9	53.8	90.9	91.7	90.5	90.6	88.5
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.9	98.7	98.0	98.5	97.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.8	94.7	96.1	94.6	95.1	83.5	98.2	98.9	98.7	97.9	98.1
20%	94.4	93.0	94.2	94.8	95.5	82.4	98.6	97.7	97.8	97.9	98.6
30%	95.0	95.4	96.1	96.0	96.5	86.2	99.1	99.1	99.4	98.5	98.9
40%	96.0	95.8	96.3	96.4	95.7	86.7	98.6	99.1	98.7	98.8	98.5
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.2	93.3	92.5	93.4	78.6	97.6	98.0	97.7	96.6	96.1
20%	89.1	88.1	88.5	88.3	90.2	70.2	96.4	94.3	95.2	94.2	94.8
30%	84.9	86.0	87.3	85.0	85.5	63.3	93.1	93.2	93.7	91.9	91.5
40%	79.5	78.0	78.2	78.3	79.1	50.8	89.9	88.8	87.7	87.0	87.2
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.6	93.6	92.9	92.2	92.4	77.9	97.6	98.0	97.6	96.5	95.7
20%	89.5	86.7	88.5	88.6	89.3	68.9	96.2	94.1	94.5	93.1	95.0
30%	85.5	86.3	86.3	85.5	86.0	62.2	93.1	93.0	93.1	92.2	92.9
40%	79.0	80.0	79.5	79.9	79.4	53.1	90.1	89.0	88.6	87.4	86.0

table2_pooled_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.4
Imputation using linear regression	NA
10%	93.3
20%	95.9
30%	95.5
40%	96.4
Imputation using predictive mean matching method	NA
10%	90.4
20%	86.1
30%	80.3
40%	71.3
Imputation using propensity−score method	NA
10%	89.7
20%	84.8
30%	79.0
40%	72.6
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.3
Imputation using linear regression	NA

table2_pooled_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
10%	93.2
20%	93.8
30%	96.0
40%	95.3
Imputation using predictive mean matching method	NA
10%	89.8
20%	86.0
30%	78.6
40%	69.0
Imputation using propensity−score method	NA
10%	88.9
20%	83.3
30%	79.5
40%	69.3

table2_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	99.1	98.5	98.2	97.9	96.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.2	94.3	94.6	93.6	94.7	82.7	98.5	98.3	97.6	96.2	96.8
20%	94.3	94.8	95.6	94.8	93.8	84.8	99.4	98.5	98.9	98.1	96.3
30%	95.3	95.2	93.8	95.7	94.4	84.2	98.8	99.0	97.7	98.6	97.3
40%	96.1	96.3	96.5	97.3	97.0	88.3	98.9	98.9	99.2	99.1	98.9
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.7	93.0	92.6	92.0	94.6	80.3	98.4	97.8	96.5	95.6	95.8
20%	91.2	91.2	91.9	92.9	91.8	77.8	97.6	96.4	96.7	95.7	94.4
30%	90.9	90.5	87.9	90.3	88.6	71.9	96.5	96.1	93.4	94.3	92.7
40%	89.1	88.4	88.9	89.8	88.6	69.1	96.0	95.2	95.1	94.7	93.3
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.4	92.5	92.6	93.0	93.8	79.2	98.3	97.8	96.8	95.6	95.8
20%	91.4	91.3	91.6	91.7	91.7	77.1	97.4	96.6	96.4	95.1	94.0
30%	89.7	90.2	87.4	90.7	88.2	70.8	96.9	95.7	92.4	94.3	91.4
40%	89.2	91.1	88.4	90.0	87.8	68.5	96.6	96.3	94.8	95.0	91.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	99.1	98.5	97.7	97.2	96.3
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.0	94.1	95.0	93.6	94.9	83.0	98.5	97.9	97.7	96.6	96.9
20%	94.5	93.6	94.0	94.6	96.1	84.1	98.7	98.2	98.2	97.7	98.0
30%	95.5	95.8	96.9	95.3	95.9	86.1	99.2	99.0	99.0	98.0	98.7
40%	95.5	96.5	96.2	96.5	95.7	85.6	98.9	99.4	99.2	99.3	97.8
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.9	92.8	94.0	92.4	93.6	80.4	98.0	97.4	97.1	94.7	94.9
20%	91.2	91.1	91.7	91.9	93.3	77.8	97.9	96.6	95.4	94.7	95.5
30%	90.5	91.0	91.7	90.1	91.2	74.1	96.7	96.4	96.2	94.3	94.1
40%	88.1	87.6	87.7	86.8	88.2	66.7	95.0	94.1	92.9	91.9	92.1
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.2	94.0	92.7	92.9	79.0	98.0	97.2	96.9	95.2	94.8
20%	91.0	90.1	91.1	92.5	92.4	76.4	97.5	95.6	95.3	94.8	94.8
30%	90.6	91.3	91.0	90.1	90.0	72.3	96.5	96.3	95.7	93.6	93.1
40%	88.5	88.3	88.0	87.3	86.6	66.5	94.6	94.0	92.4	91.9	90.5

table2_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	91.9
Imputation using linear regression	NA
10%	91.5
20%	92.5
30%	94.0
40%	96.3
Imputation using predictive mean matching method	NA
10%	89.2
20%	87.3
30%	84.1
40%	83.5
Imputation using propensity−score method	NA
10%	89.3
20%	86.2
30%	82.0
40%	82.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.0
Imputation using linear regression	NA

table2_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
10%	91.7
20%	93.8
30%	95.0
40%	95.3
Imputation using predictive mean matching method	NA
10%	88.1
20%	87.6
30%	86.1
40%	80.3
Imputation using propensity−score method	NA
10%	87.4
20%	86.2
30%	83.7
40%	77.6

table2_pooled_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	99.1	98.5	98.2	97.9	96.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.7	95.1	94.3	93.7	95.1	82.7	98.9	98.7	98.0	96.1	97.1
20%	95.3	94.6	96.2	94.6	94.2	84.1	99.1	98.9	99.0	98.3	97.8
30%	95.8	96.2	94.9	96.0	95.0	86.5	98.8	99.0	98.2	98.8	97.5
40%	96.3	96.0	96.4	96.0	96.5	87.0	99.2	98.9	99.1	98.8	98.4
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.1	93.2	90.5	93.6	78.8	98.4	97.9	96.7	94.8	95.5
20%	89.7	90.0	90.1	89.0	89.0	72.5	96.7	95.9	94.8	94.0	92.7
30%	85.8	86.1	83.2	86.5	85.4	65.1	94.0	93.6	90.4	91.9	89.9
40%	78.9	79.3	80.5	80.5	79.8	55.1	90.6	89.1	88.9	87.7	85.1
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.3	92.6	91.6	93.5	78.1	98.1	97.8	96.4	94.4	95.7
20%	89.6	89.1	90.0	89.1	89.0	70.7	96.8	95.0	95.1	93.9	91.8
30%	85.3	87.2	85.3	86.2	85.9	64.6	94.1	93.3	91.3	91.1	89.9
40%	80.2	82.9	83.4	82.3	82.0	57.0	91.1	90.8	89.2	88.8	86.3
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	99.1	98.5	97.7	97.2	96.3
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.5	94.3	95.8	94.1	95.2	84.2	98.2	98.4	98.4	96.9	96.6
20%	93.7	92.9	94.3	94.7	95.5	83.4	98.7	97.6	97.7	97.6	97.9
30%	95.6	95.9	96.2	95.9	96.6	87.3	99.0	98.7	99.2	98.2	98.4
40%	95.6	96.0	96.2	96.2	95.7	87.5	98.6	98.9	98.2	98.5	98.3
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.0	93.2	92.3	93.4	80.2	97.4	97.5	96.9	95.0	94.5
20%	88.6	87.9	88.8	88.1	90.2	72.4	96.1	93.8	93.7	92.2	93.8
30%	85.6	85.3	86.8	85.0	85.4	65.6	93.7	92.8	92.8	90.3	90.5
40%	79.7	78.7	78.2	78.0	79.2	53.5	89.8	88.1	85.7	85.8	85.0
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.3	92.8	92.8	92.1	92.5	79.1	97.5	97.8	96.5	95.7	94.5
20%	88.5	87.4	88.0	88.4	90.0	71.1	95.8	94.3	93.7	91.4	93.2
30%	85.3	85.8	86.3	85.5	86.6	64.1	93.6	93.1	92.5	90.7	91.3
40%	79.3	79.5	80.3	80.6	80.0	55.8	90.6	88.4	87.5	86.6	84.5

table2_pooled_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	91.9
Imputation using linear regression	NA
10%	92.2
20%	94.4
30%	94.6
40%	96.2
Imputation using predictive mean matching method	NA
10%	88.8
20%	83.9
30%	78.4
40%	70.1
Imputation using propensity−score method	NA
10%	88.2
20%	82.5
30%	76.9
40%	70.9
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	92.0
Imputation using linear regression	NA

table2_pooled_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
10%	91.5
20%	93.4
30%	95.3
40%	95.0
Imputation using predictive mean matching method	NA
10%	87.7
20%	83.2
30%	77.4
40%	68.2
Imputation using propensity−score method	NA
10%	87.2
20%	82.3
30%	78.4
40%	69.1

table2_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	99.2	98.1	97.6	97.8	95.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.4	94.4	94.5	93.4	94.8	83.6	98.4	98.0	97.3	95.9	95.9
20%	93.6	94.4	95.4	94.9	93.8	85.0	99.0	98.4	98.4	97.5	95.8
30%	94.7	95.2	93.7	95.9	94.4	84.8	99.0	98.8	97.5	98.4	96.5
40%	95.7	95.8	96.4	97.0	97.0	88.2	99.1	98.7	99.1	99.0	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.6	93.2	92.8	92.0	94.5	81.6	98.6	97.4	95.9	94.6	94.6
20%	91.3	91.4	91.7	92.9	91.9	78.8	97.9	95.9	96.1	95.0	93.5
30%	90.4	90.2	88.0	90.5	88.5	73.1	96.7	95.9	92.2	93.4	90.9
40%	90.3	89.3	89.8	89.8	88.8	71.7	96.6	95.7	94.6	93.5	92.3
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.8	92.6	93.1	93.6	80.5	98.4	97.4	96.2	94.7	94.5
20%	91.2	91.7	91.4	91.7	91.7	78.1	97.3	96.1	95.4	94.7	92.7
30%	90.5	89.8	87.8	90.6	88.1	72.1	96.6	95.2	91.8	93.6	89.7
40%	90.1	91.4	88.6	90.2	87.9	71.3	96.9	95.8	94.2	94.3	90.7
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.9	98.3	97.4	95.9	95.2
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.3	94.1	94.8	93.4	94.8	83.4	98.5	97.8	97.4	95.8	95.6
20%	94.2	94.0	94.2	94.4	96.1	85.4	98.5	98.1	97.7	96.8	97.6
30%	95.5	95.5	96.9	95.5	95.8	86.9	99.4	98.9	99.0	97.7	98.4
40%	95.1	95.8	96.0	96.6	95.7	85.9	99.0	99.2	99.0	98.8	97.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	91.8	92.6	94.0	92.5	93.8	81.1	97.9	97.4	96.6	94.1	93.8
20%	90.9	91.1	91.6	91.8	93.4	79.2	97.8	96.2	94.8	93.9	94.7
30%	90.6	91.7	91.8	90.2	91.2	75.8	97.3	95.8	95.9	93.2	93.0
40%	88.2	88.6	88.0	87.1	88.2	68.8	94.7	93.8	92.4	90.7	91.0
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.9	93.9	93.1	92.7	80.3	97.6	97.2	96.4	94.5	93.7
20%	91.2	90.1	91.0	92.6	92.8	77.7	97.0	95.7	94.5	94.0	94.2
30%	90.7	91.3	91.4	90.4	90.5	74.6	96.2	96.1	95.4	92.9	92.1
40%	88.5	88.3	88.5	87.6	86.7	68.6	94.6	94.0	92.2	91.1	90.4

table2_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	90.7
Imputation using linear regression	NA
10%	90.7
20%	91.4
30%	93.2
40%	96.0
Imputation using predictive mean matching method	NA
10%	87.6
20%	86.6
30%	82.4
40%	83.5
Imputation using propensity−score method	NA
10%	87.9
20%	84.7
30%	80.7
40%	82.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	90.3
Imputation using linear regression	NA

table2_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
10%	89.7
20%	93.1
30%	94.3
40%	95.0
Imputation using predictive mean matching method	NA
10%	87.3
20%	86.7
30%	84.9
40%	78.8
Imputation using propensity−score method	NA
10%	86.8
20%	85.8
30%	83.0
40%	77.4

table2_pooled_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	99.2	98.1	97.6	97.8	95.1
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.6	95.4	94.1	94.0	95.1	83.7	98.9	98.7	97.6	95.4	96.0
20%	94.5	94.2	96.0	94.5	94.3	84.2	99.1	98.7	98.6	98.0	96.9
30%	95.3	96.2	94.8	95.9	95.0	87.2	98.5	98.9	98.0	98.6	96.9
40%	96.4	96.2	96.4	96.1	96.5	88.3	99.1	98.8	98.9	98.2	98.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.7	92.5	90.7	93.5	79.3	98.3	97.3	96.1	93.4	94.4
20%	89.3	89.7	90.1	89.2	89.2	74.0	96.7	95.4	94.3	92.8	91.6
30%	85.5	85.8	83.4	86.8	85.6	66.8	94.6	92.8	88.9	90.6	88.5
40%	79.3	79.3	80.6	80.0	79.8	57.2	91.5	88.8	88.2	86.3	84.2
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.3	92.7	91.8	93.5	79.3	98.5	97.4	95.7	93.9	94.6
20%	89.5	89.1	89.9	89.4	89.0	72.1	96.8	95.0	94.6	92.8	90.8
30%	85.4	86.6	85.0	86.2	85.9	65.9	94.2	93.6	90.5	90.0	88.5
40%	80.9	83.0	83.3	82.0	82.2	59.2	92.2	90.6	88.8	87.7	85.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.9	98.3	97.4	95.9	95.2
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_pooled_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.3	94.2	95.2	93.9	95.2	84.7	98.5	98.4	98.0	96.4	95.8
20%	93.2	93.0	94.1	94.7	95.6	84.6	98.7	97.4	97.5	97.0	96.9
30%	95.6	96.0	96.2	96.0	96.6	88.0	98.8	98.5	98.8	98.1	98.0
40%	95.0	95.9	96.1	96.2	95.8	88.1	98.8	98.8	97.9	98.1	97.9
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.7	93.1	92.2	93.3	80.8	97.3	96.9	96.3	94.2	93.3
20%	88.3	88.3	88.9	87.8	90.3	73.9	96.2	93.6	93.0	91.2	92.3
30%	85.2	85.3	86.9	84.7	85.3	66.9	94.5	92.3	92.1	89.1	89.0
40%	80.3	79.0	78.5	78.1	79.5	55.7	90.0	87.6	85.0	84.6	83.3
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.2	92.4	92.4	92.1	92.5	79.8	97.6	97.3	95.9	94.6	93.7
20%	88.0	87.4	87.9	88.3	90.1	72.4	95.7	93.6	93.3	90.4	91.8
30%	84.7	85.8	86.7	85.5	87.6	66.3	93.8	93.0	92.1	89.7	90.2
40%	80.6	79.1	80.4	81.0	80.4	58.2	90.6	88.2	86.6	85.2	83.3

table2_pooled_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	90.7
Imputation using linear regression	NA
10%	91.0
20%	93.2
30%	94.2
40%	95.3
Imputation using predictive mean matching method	NA
10%	87.0
20%	82.7
30%	77.0
40%	70.2
Imputation using propensity−score method	NA
10%	87.2
20%	81.0
30%	76.4
40%	70.9
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	90.3
Imputation using linear regression	NA

table2_pooled_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
10%	90.3
20%	92.5
30%	95.0
40%	94.8
Imputation using predictive mean matching method	NA
10%	86.0
20%	82.0
30%	76.6
40%	67.1
Imputation using propensity−score method	NA
10%	86.3
20%	81.4
30%	78.2
40%	68.4

table2_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	99.0	98.8	98.6	98.5	97.6
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.1	94.1	94.6	94.0	94.8	81.5	98.6	98.6	98.3	97.6	98.2
20%	95.2	95.4	95.7	95.1	93.8	83.9	99.5	98.8	99.1	98.5	98.3
30%	95.4	95.9	93.9	95.8	94.4	83.4	98.7	99.3	98.1	99.1	98.0
40%	95.6	96.0	96.6	97.0	97.0	86.6	99.0	99.1	99.2	99.6	99.2
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.6	92.4	92.2	94.6	78.3	98.3	98.0	97.8	97.2	97.5
20%	91.8	91.3	92.2	92.6	91.7	76.1	97.6	97.0	97.4	97.1	96.3
30%	90.3	90.0	87.4	90.3	88.6	69.6	96.7	96.4	94.8	95.7	94.1
40%	88.2	89.5	88.6	89.5	88.4	66.3	95.0	95.3	95.1	95.5	95.2
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.5	92.0	92.6	92.6	93.6	77.5	98.2	98.0	97.8	97.1	97.4
20%	91.4	90.8	91.6	91.8	91.8	74.8	97.9	97.2	96.9	96.6	96.1
30%	90.7	89.5	87.3	90.0	88.0	68.1	96.4	96.5	94.3	96.2	93.5
40%	87.3	90.8	87.8	89.8	87.5	64.7	95.1	96.1	94.9	96.2	93.5
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.8	98.7	98.0	98.4	97.8
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.7	94.3	95.0	94.2	94.9	81.9	98.3	98.5	98.5	97.8	98.6
20%	94.3	93.9	94.0	94.6	96.2	82.5	98.7	98.6	98.1	98.2	98.6
30%	95.5	95.8	96.6	95.3	96.0	84.8	99.2	99.0	99.4	98.8	99.1
40%	96.0	96.3	96.1	96.4	95.7	84.7	98.7	99.3	99.2	99.6	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	92.7	94.4	92.6	93.6	79.1	97.9	97.7	98.2	96.7	97.4
20%	90.7	90.8	91.0	91.7	93.4	75.2	97.9	96.6	96.3	96.3	97.4
30%	89.9	90.9	91.5	89.9	91.4	71.6	95.9	96.8	96.8	95.5	95.8
40%	88.1	87.8	87.8	86.6	88.2	64.3	95.3	94.7	93.9	93.8	94.5
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.2	93.5	94.1	92.8	93.0	78.4	98.1	97.7	97.7	96.4	97.4
20%	91.5	90.0	90.8	91.7	92.5	74.2	97.5	95.8	96.1	96.5	97.2
30%	89.6	91.1	91.1	89.4	89.7	68.6	96.2	96.6	96.3	94.6	94.7
40%	88.2	87.5	87.2	87.0	85.4	62.5	94.4	94.5	93.8	93.4	92.5

table2_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.7
Imputation using linear regression	NA
10%	93.7
20%	95.0
30%	95.0
40%	96.9
Imputation using predictive mean matching method	NA
10%	91.6
20%	89.4
30%	86.0
40%	84.0
Imputation using propensity−score method	NA
10%	91.7
20%	89.2
30%	84.6
40%	82.2
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.5
Imputation using linear regression	NA

table2_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.0
20%	94.4
30%	96.4
40%	95.9
Imputation using predictive mean matching method	NA
10%	91.5
20%	89.7
30%	87.5
40%	81.8
Imputation using propensity−score method	NA
10%	91.2
20%	88.3
30%	85.6
40%	79.6

table2_shrink_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.6	95.5	95.2	96.6	95.4	83.7	99.0	98.8	98.6	98.5	97.6
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.9	95.5	93.9	93.5	95.0	81.6	98.8	99.0	98.4	97.2	98.2
20%	95.4	95.0	96.5	94.6	94.3	83.6	99.1	99.1	99.2	99.0	99.0
30%	95.6	96.5	94.8	96.0	94.9	85.3	99.0	99.6	98.8	99.0	98.8
40%	96.1	95.7	96.3	96.1	96.5	85.7	99.1	99.2	99.1	99.2	98.8
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.8	93.0	90.5	93.7	77.6	98.5	98.0	97.8	96.6	97.1
20%	89.6	89.6	90.0	88.9	89.0	70.3	96.6	96.2	95.4	95.2	94.9
30%	84.9	85.5	82.9	86.3	85.3	62.2	93.6	94.0	91.7	94.2	92.2
40%	78.3	79.9	80.1	80.6	80.1	51.8	89.8	89.5	90.5	89.8	88.8
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.6	92.5	91.5	93.5	76.7	98.3	98.0	97.9	96.0	97.2
20%	90.4	88.6	89.9	88.8	88.9	68.7	96.8	95.8	96.2	95.5	94.7
30%	85.7	87.0	84.6	86.4	85.8	62.8	94.0	93.8	92.7	93.7	92.4
40%	80.4	83.0	83.3	82.5	81.9	53.8	90.6	91.4	91.2	91.2	89.1
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.2	95.4	95.5	95.0	95.2	84.2	98.8	98.7	98.0	98.4	97.8
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R2_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.8	94.7	96.1	94.6	95.1	83.5	98.0	98.9	98.6	98.2	98.3
20%	94.4	93.0	94.2	94.8	95.5	82.4	98.5	97.7	97.8	98.0	98.8
30%	95.0	95.4	96.1	96.0	96.5	86.2	98.7	98.7	99.6	98.5	99.2
40%	96.0	95.8	96.3	96.4	95.7	86.7	98.5	98.8	98.6	98.7	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.2	93.3	92.5	93.4	78.6	97.4	98.0	97.8	97.3	96.7
20%	89.1	88.1	88.5	88.3	90.2	70.2	96.2	94.4	94.9	94.7	95.2
30%	84.9	86.0	87.3	85.0	85.5	63.3	92.7	93.4	93.9	92.7	92.8
40%	79.5	78.0	78.2	78.3	79.1	50.8	90.0	88.0	87.9	87.8	87.7
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.6	93.6	92.9	92.2	92.4	77.9	97.5	98.1	97.8	96.9	96.6
20%	89.5	86.7	88.5	88.6	89.3	68.9	95.5	94.1	94.4	94.1	95.8
30%	85.5	86.3	86.3	85.5	86.0	62.2	92.8	92.9	93.5	92.9	93.3
40%	79.0	80.0	79.5	79.9	79.4	53.1	89.4	88.2	88.7	88.0	87.1

table2_shrink_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.7
Imputation using linear regression	NA
10%	93.6
20%	95.8
30%	96.1
40%	96.7
Imputation using predictive mean matching method	NA
10%	91.1
20%	86.2
30%	80.2
40%	71.2
Imputation using propensity−score method	NA
10%	90.6
20%	85.6
30%	79.4
40%	73.0
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.5
Imputation using linear regression	NA

table2_shrink_5_treatments_R2_opposite_sign_MAR

MR	Joint_many
10%	93.3
20%	93.9
30%	95.8
40%	95.1
Imputation using predictive mean matching method	NA
10%	90.6
20%	86.4
30%	79.2
40%	69.3
Imputation using propensity−score method	NA
10%	90.0
20%	84.1
30%	79.7
40%	69.6

table2_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	98.8	98.7	98.3	98.4	97.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.2	94.3	94.6	93.6	94.7	82.7	98.5	98.4	98.0	96.9	98.0
20%	94.3	94.8	95.6	94.8	93.8	84.8	99.1	98.7	98.8	98.5	97.9
30%	95.3	95.2	93.8	95.7	94.4	84.2	98.8	98.9	97.8	99.0	97.9
40%	96.1	96.3	96.5	97.3	97.0	88.3	98.8	98.8	99.2	99.4	99.0
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.7	93.0	92.6	92.0	94.6	80.3	98.2	97.9	96.9	97.0	97.3
20%	91.2	91.2	91.9	92.9	91.8	77.8	97.5	96.6	97.1	96.8	95.9
30%	90.9	90.5	87.9	90.3	88.6	71.9	96.5	96.7	94.5	95.4	93.8
40%	89.1	88.4	88.9	89.8	88.6	69.1	95.3	95.1	94.8	95.5	95.0
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.4	92.5	92.6	93.0	93.8	79.2	98.0	98.0	97.0	96.5	97.2
20%	91.4	91.3	91.6	91.7	91.7	77.1	97.1	96.8	96.8	96.7	96.0
30%	89.7	90.2	87.4	90.7	88.2	70.8	97.1	96.0	93.6	95.8	93.2
40%	89.2	91.1	88.4	90.0	87.8	68.5	95.9	96.3	95.4	96.3	93.3
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	99.0	98.5	97.9	97.9	97.6
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.0	94.1	95.0	93.6	94.9	83.0	98.3	98.4	98.2	97.4	98.4
20%	94.5	93.6	94.0	94.6	96.1	84.1	98.5	98.6	98.0	98.0	98.3
30%	95.5	95.8	96.9	95.3	95.9	86.1	99.2	98.9	99.2	98.6	98.9
40%	95.5	96.5	96.2	96.5	95.7	85.6	98.8	99.3	99.1	99.5	98.3
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.9	92.8	94.0	92.4	93.6	80.4	98.0	97.2	97.5	96.0	97.3
20%	91.2	91.1	91.7	91.9	93.3	77.8	97.4	96.9	95.8	96.1	96.9
30%	90.5	91.0	91.7	90.1	91.2	74.1	96.3	96.6	96.9	95.5	95.6
40%	88.1	87.6	87.7	86.8	88.2	66.7	94.8	94.1	93.6	93.7	94.2
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.2	94.0	92.7	92.9	79.0	97.9	97.5	97.5	95.9	97.3
20%	91.0	90.1	91.1	92.5	92.4	76.4	97.3	96.2	95.8	96.0	96.7
30%	90.6	91.3	91.0	90.1	90.0	72.3	96.3	96.5	96.5	94.8	94.3
40%	88.5	88.3	88.0	87.3	86.6	66.5	94.3	94.3	93.8	93.6	92.6

table2_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.5
Imputation using linear regression	NA
10%	93.0
20%	94.2
30%	94.9
40%	96.5
Imputation using predictive mean matching method	NA
10%	91.0
20%	88.5
30%	85.8
40%	84.9
Imputation using propensity−score method	NA
10%	91.0
20%	88.8
30%	84.4
40%	84.3
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.4
Imputation using linear regression	NA

table2_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.0
20%	94.0
30%	95.8
40%	95.6
Imputation using predictive mean matching method	NA
10%	91.4
20%	89.5
30%	87.8
40%	82.2
Imputation using propensity−score method	NA
10%	91.1
20%	88.7
30%	86.1
40%	80.8

table2_shrink_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.9	96.0	95.0	96.6	95.5	85.4	98.8	98.7	98.3	98.4	97.4
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.7	95.1	94.3	93.7	95.1	82.7	98.6	98.8	98.2	96.8	98.0
20%	95.3	94.6	96.2	94.6	94.2	84.1	99.1	99.0	99.0	98.7	98.9
30%	95.8	96.2	94.9	96.0	95.0	86.5	98.8	99.1	98.6	99.0	98.3
40%	96.3	96.0	96.4	96.0	96.5	87.0	99.0	99.2	99.1	99.0	98.7
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.8	93.1	93.2	90.5	93.6	78.8	98.0	98.2	96.9	96.3	96.9
20%	89.7	90.0	90.1	89.0	89.0	72.5	96.4	95.6	95.1	94.8	94.5
30%	85.8	86.1	83.2	86.5	85.4	65.1	93.3	93.4	90.6	93.7	91.7
40%	78.9	79.3	80.5	80.5	79.8	55.1	89.5	89.1	89.7	89.0	88.0
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.3	92.6	91.6	93.5	78.1	97.9	98.2	96.9	95.6	97.1
20%	89.6	89.1	90.0	89.1	89.0	70.7	96.5	95.5	96.1	95.1	94.5
30%	85.3	87.2	85.3	86.2	85.9	64.6	93.3	93.5	91.8	93.4	92.1
40%	80.2	82.9	83.4	82.3	82.0	57.0	91.0	90.7	90.2	90.6	88.0
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.7	95.9	95.0	95.3	85.8	99.0	98.5	97.9	97.9	97.6
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R4_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.5	94.3	95.8	94.1	95.2	84.2	98.1	98.6	98.5	97.8	98.0
20%	93.7	92.9	94.3	94.7	95.5	83.4	98.4	97.3	97.9	97.7	98.7
30%	95.6	95.9	96.2	95.9	96.6	87.3	98.6	98.1	99.4	98.4	99.1
40%	95.6	96.0	96.2	96.2	95.7	87.5	98.5	99.0	98.5	98.7	98.5
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	93.0	93.2	92.3	93.4	80.2	97.0	97.5	97.6	96.8	96.0
20%	88.6	87.9	88.8	88.1	90.2	72.4	95.1	94.4	93.9	94.0	95.0
30%	85.6	85.3	86.8	85.0	85.4	65.6	93.1	93.1	93.3	92.1	92.6
40%	79.7	78.7	78.2	78.0	79.2	53.5	89.4	87.8	87.0	87.3	86.9
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.3	92.8	92.8	92.1	92.5	79.1	97.2	97.9	97.4	96.8	96.8
20%	88.5	87.4	88.0	88.4	90.0	71.1	95.3	94.3	93.9	93.6	95.4
30%	85.3	85.8	86.3	85.5	86.6	64.1	92.8	93.1	93.5	92.4	93.1
40%	79.3	79.5	80.3	80.6	80.0	55.8	90.3	87.5	87.9	88.1	87.5

table2_shrink_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.5
Imputation using linear regression	NA
10%	93.1
20%	95.5
30%	96.0
40%	96.5
Imputation using predictive mean matching method	NA
10%	90.7
20%	85.3
30%	80.1
40%	71.8
Imputation using propensity−score method	NA
10%	90.6
20%	85.8
30%	79.2
40%	73.1
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.4
Imputation using linear regression	NA

table2_shrink_5_treatments_R4_opposite_sign_MAR

MR	Joint_many
10%	93.2
20%	93.8
30%	95.5
40%	95.5
Imputation using predictive mean matching method	NA
10%	90.3
20%	85.6
30%	79.6
40%	70.6
Imputation using propensity−score method	NA
10%	90.3
20%	85.2
30%	80.2
40%	71.3

table2_shrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	98.7	98.6	98.3	98.4	97.2
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.4	94.4	94.5	93.4	94.8	83.6	98.6	97.9	97.7	96.5	97.8
20%	93.6	94.4	95.4	94.9	93.8	85.0	98.5	98.5	98.8	98.4	97.9
30%	94.7	95.2	93.7	95.9	94.4	84.8	98.3	98.8	97.9	99.0	97.7
40%	95.7	95.8	96.4	97.0	97.0	88.2	99.0	98.6	99.1	99.3	98.9
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.6	93.2	92.8	92.0	94.5	81.6	97.7	97.6	96.9	96.8	97.2
20%	91.3	91.4	91.7	92.9	91.9	78.8	97.2	96.5	96.8	96.6	95.9
30%	90.4	90.2	88.0	90.5	88.5	73.1	96.0	96.2	94.6	95.5	93.8
40%	90.3	89.3	89.8	89.8	88.8	71.7	96.0	95.1	94.7	95.2	94.5
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.3	92.8	92.6	93.1	93.6	80.5	97.8	97.5	97.0	96.6	97.3
20%	91.2	91.7	91.4	91.7	91.7	78.1	97.1	96.2	96.5	96.6	95.8
30%	90.5	89.8	87.8	90.6	88.1	72.1	96.4	95.7	93.5	95.8	93.2
40%	90.1	91.4	88.6	90.2	87.9	71.3	96.2	96.0	95.4	95.9	93.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.7	98.3	97.6	98.0	97.7
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	93.3	94.1	94.8	93.4	94.8	83.4	98.1	98.2	98.1	97.4	98.3
20%	94.2	94.0	94.2	94.4	96.1	85.4	98.1	98.3	97.9	97.8	98.3
30%	95.5	95.5	96.9	95.5	95.8	86.9	98.9	98.5	99.2	98.5	98.8
40%	95.1	95.8	96.0	96.6	95.7	85.9	98.7	99.1	99.0	99.5	98.3
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	91.8	92.6	94.0	92.5	93.8	81.1	97.6	97.3	97.5	95.9	97.1
20%	90.9	91.1	91.6	91.8	93.4	79.2	97.0	96.7	95.7	95.8	96.9
30%	90.6	91.7	91.8	90.2	91.2	75.8	96.3	96.4	96.8	95.5	95.6
40%	88.2	88.6	88.0	87.1	88.2	68.8	94.6	94.0	93.9	93.6	93.9
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.9	93.9	93.1	92.7	80.3	97.6	97.8	97.4	96.2	97.3
20%	91.2	90.1	91.0	92.6	92.8	77.7	96.8	96.2	95.6	95.8	96.7
30%	90.7	91.3	91.4	90.4	90.5	74.6	96.1	95.8	96.6	94.8	94.2
40%	88.5	88.3	88.5	87.6	86.7	68.6	94.5	94.4	93.5	93.6	92.4

table2_shrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.5
Imputation using linear regression	NA
10%	92.8
20%	94.0
30%	94.6
40%	96.3
Imputation using predictive mean matching method	NA
10%	91.0
20%	89.2
30%	86.1
40%	85.2
Imputation using propensity−score method	NA
10%	91.2
20%	88.4
30%	84.6
40%	84.8
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.5
Imputation using linear regression	NA

table2_shrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	Joint_many
10%	94.1
20%	93.9
30%	95.4
40%	95.7
Imputation using predictive mean matching method	NA
10%	91.6
20%	89.7
30%	88.5
40%	82.6
Imputation using propensity−score method	NA
10%	91.9
20%	89.2
30%	86.6
40%	81.4

table2_shrink_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
Scenario I: all means are 0 for T0,T1,...,T5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	95.7	96.0	94.9	96.5	95.4	86.1	98.7	98.6	98.3	98.4	97.2
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	94.6	95.4	94.1	94.0	95.1	83.7	98.4	98.7	97.9	96.5	98.1
20%	94.5	94.2	96.0	94.5	94.3	84.2	99.0	98.9	99.0	98.7	98.8
30%	95.3	96.2	94.8	95.9	95.0	87.2	98.6	99.0	98.5	98.9	98.3
40%	96.4	96.2	96.4	96.1	96.5	88.3	98.8	99.1	99.0	98.8	98.6
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.7	92.7	92.5	90.7	93.5	79.3	97.8	97.8	96.8	95.8	96.8
20%	89.3	89.7	90.1	89.2	89.2	74.0	95.8	95.5	94.8	94.5	94.4
30%	85.5	85.8	83.4	86.8	85.6	66.8	93.5	93.3	90.3	93.3	91.6
40%	79.3	79.3	80.6	80.0	79.8	57.2	89.8	88.2	89.2	88.6	87.7
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	93.0	92.3	92.7	91.8	93.5	79.3	97.7	97.8	96.7	95.6	96.9
20%	89.5	89.1	89.9	89.4	89.0	72.1	96.5	94.9	95.5	95.0	94.1
30%	85.4	86.6	85.0	86.2	85.9	65.9	93.7	93.5	91.8	92.9	91.7
40%	80.9	83.0	83.3	82.0	82.2	59.2	90.6	90.2	89.9	90.4	87.8
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
0%	94.9	95.8	95.8	94.8	95.3	86.6	98.7	98.3	97.6	98.0	97.7
Imputation using linear regression	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

table2_shrink_5_treatments_R9_opposite_sign_MAR

MR	T1vs.T0_none	T2vs.T0_none	T3vs.T0_none	T4vs.T0_none	T5vs.T0_none	Joint_none	T1vs.T0_many	T2vs.T0_many	T3vs.T0_many	T4vs.T0_many	T5vs.T0_many
10%	94.3	94.2	95.2	93.9	95.2	84.7	97.8	98.3	98.4	97.8	98.1
20%	93.2	93.0	94.1	94.7	95.6	84.6	98.2	97.1	97.9	97.7	98.6
30%	95.6	96.0	96.2	96.0	96.6	88.0	98.4	98.1	99.3	98.4	98.9
40%	95.0	95.9	96.1	96.2	95.8	88.1	98.6	98.7	98.2	98.6	98.5
Imputation using predictive mean matching method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.0	92.7	93.1	92.2	93.3	80.8	96.7	97.3	97.6	96.9	95.9
20%	88.3	88.3	88.9	87.8	90.3	73.9	94.9	94.1	93.9	93.8	94.9
30%	85.2	85.3	86.9	84.7	85.3	66.9	92.9	92.9	92.6	91.8	92.6
40%	80.3	79.0	78.5	78.1	79.5	55.7	88.8	87.3	86.3	87.3	86.8
Imputation using propensity−score method	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
10%	92.2	92.4	92.4	92.1	92.5	79.8	97.1	97.6	97.3	96.8	96.6
20%	88.0	87.4	87.9	88.3	90.1	72.4	94.7	94.2	93.4	93.3	95.4
30%	84.7	85.8	86.7	85.5	87.6	66.3	93.0	93.2	93.2	92.3	93.1
40%	80.6	79.1	80.4	81.0	80.4	58.2	89.7	87.6	87.8	88.2	87.5

table2_shrink_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
Scenario I: all means are 0 for T0,T1,...,T5	NA
0%	93.5
Imputation using linear regression	NA
10%	93.0
20%	95.5
30%	95.8
40%	96.2
Imputation using predictive mean matching method	NA
10%	90.7
20%	85.3
30%	80.7
40%	72.8
Imputation using propensity−score method	NA
10%	90.3
20%	85.7
30%	80.0
40%	73.4
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5	NA
0%	93.5
Imputation using linear regression	NA

table2_shrink_5_treatments_R9_opposite_sign_MAR

MR	Joint_many
10%	93.6
20%	93.8
30%	95.5
40%	95.2
Imputation using predictive mean matching method	NA
10%	90.5
20%	85.8
30%	80.0
40%	71.4
Imputation using propensity−score method	NA
10%	90.4
20%	85.4
30%	81.0
40%	72.5

table3_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.793	6.001	6.166	6.367	6.537
Imputation using linear regression										
10%	4.908 (4.68)	5.063 (4.40)	5.220 (4.84)	5.339 (3.90)	5.558 (5.41)	6.205 (7.12)	6.402 (6.68)	6.602 (7.06)	6.752 (6.04)	7.030 (7.54)
20%	5.332 (13.73)	5.541 (14.25)	5.696 (14.40)	5.945 (15.69)	6.030 (14.35)	6.766 (16.80)	7.030 (17.15)	7.229 (17.23)	7.543 (18.47)	7.652 (17.06)
30%	5.967 (27.27)	6.230 (28.46)	6.452 (29.58)	6.670 (29.80)	6.845 (29.81)	7.597 (31.15)	7.931 (32.16)	8.215 (33.22)	8.492 (33.37)	8.716 (33.34)
40%	7.000 (49.31)	7.264 (49.79)	7.559 (51.81)	7.799 (51.77)	8.125 (54.09)	8.950 (54.51)	9.289 (54.79)	9.666 (56.75)	9.972 (56.60)	10.391 (58.97)
Imputation using predictive mean matching method										
10%	4.674 (−0.30)	4.817 (−0.67)	4.966 (−0.27)	5.069 (−1.35)	5.264 (−0.16)	5.893 (1.74)	6.074 (1.22)	6.263 (1.57)	6.393 (0.41)	6.640 (1.58)
20%	4.770 (1.73)	4.939 (1.83)	5.070 (1.83)	5.255 (2.27)	5.352 (1.49)	6.011 (3.78)	6.225 (3.73)	6.392 (3.65)	6.625 (4.04)	6.747 (3.22)
30%	4.891 (4.31)	5.094 (5.03)	5.217 (4.78)	5.383 (4.76)	5.515 (4.59)	6.162 (6.38)	6.419 (6.97)	6.576 (6.64)	6.786 (6.57)	6.952 (6.36)
40%	5.088 (8.53)	5.236 (7.97)	5.390 (8.24)	5.553 (8.07)	5.706 (8.21)	6.406 (10.58)	6.592 (9.85)	6.787 (10.06)	6.994 (9.84)	7.187 (9.95)
Imputation using propensity−score method										
10%	4.676 (−0.27)	4.812 (−0.77)	4.958 (−0.43)	5.068 (−1.37)	5.256 (−0.31)	5.891 (1.70)	6.064 (1.04)	6.249 (1.33)	6.388 (0.32)	6.625 (1.35)
20%	4.777 (1.88)	4.954 (2.14)	5.083 (2.07)	5.264 (2.44)	5.376 (1.96)	6.013 (3.81)	6.236 (3.92)	6.400 (3.78)	6.628 (4.10)	6.770 (3.57)
30%	4.896 (4.42)	5.085 (4.84)	5.224 (4.90)	5.371 (4.51)	5.500 (4.30)	6.152 (6.20)	6.390 (6.48)	6.566 (6.47)	6.751 (6.03)	6.914 (5.77)
40%	5.065 (8.04)	5.240 (8.05)	5.355 (7.54)	5.517 (7.36)	5.678 (7.69)	6.351 (9.64)	6.569 (9.46)	6.715 (8.90)	6.920 (8.67)	7.123 (8.97)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.800	6.012	6.181	6.341	6.553
Imputation using linear regression										

table3_nonshrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.859 (3.40)	5.028 (3.40)	5.202 (4.16)	5.333 (4.13)	5.468 (3.37)	6.145 (5.96)	6.360 (5.79)	6.581 (6.47)	6.748 (6.41)	6.919 (5.59)
20%	5.307 (12.94)	5.480 (12.70)	5.659 (13.32)	5.887 (14.95)	6.026 (13.92)	6.735 (16.12)	6.956 (15.69)	7.184 (16.22)	7.473 (17.84)	7.650 (16.74)
30%	5.942 (26.45)	6.234 (28.19)	6.401 (28.16)	6.633 (29.52)	6.885 (30.16)	7.568 (30.47)	7.939 (32.04)	8.153 (31.91)	8.449 (33.23)	8.768 (33.81)
40%	6.904 (46.94)	7.274 (49.59)	7.576 (51.68)	7.814 (52.58)	8.216 (55.32)	8.837 (52.35)	9.307 (54.80)	9.692 (56.81)	9.999 (57.68)	10.514 (60.46)
Imputation using predictive mean matching method										
10%	4.632 (−1.43)	4.785 (−1.60)	4.942 (−1.05)	5.077 (−0.86)	5.190 (−1.89)	5.842 (0.72)	6.036 (0.39)	6.235 (0.87)	6.407 (1.03)	6.548 (−0.06)
20%	4.772 (1.56)	4.917 (1.13)	5.056 (1.23)	5.234 (2.20)	5.336 (0.87)	6.016 (3.72)	6.200 (3.12)	6.375 (3.14)	6.601 (4.09)	6.729 (2.69)
30%	4.897 (4.22)	5.058 (4.01)	5.196 (4.03)	5.344 (4.35)	5.495 (3.87)	6.171 (6.40)	6.375 (6.04)	6.549 (5.96)	6.737 (6.24)	6.927 (5.72)
40%	5.045 (7.37)	5.230 (7.56)	5.401 (8.13)	5.538 (8.13)	5.716 (8.07)	6.353 (9.53)	6.587 (9.55)	6.802 (10.04)	6.976 (10.00)	7.201 (9.89)
Imputation using propensity−score method										
10%	4.626 (−1.56)	4.783 (−1.63)	4.933 (−1.22)	5.076 (−0.88)	5.187 (−1.94)	5.830 (0.51)	6.029 (0.28)	6.219 (0.61)	6.400 (0.93)	6.539 (−0.20)
20%	4.782 (1.78)	4.932 (1.43)	5.059 (1.30)	5.238 (2.28)	5.341 (0.97)	6.021 (3.81)	6.211 (3.30)	6.371 (3.08)	6.598 (4.04)	6.727 (2.67)
30%	4.941 (5.15)	5.074 (4.34)	5.237 (4.86)	5.360 (4.65)	5.523 (4.40)	6.207 (7.02)	6.376 (6.05)	6.582 (6.48)	6.736 (6.22)	6.941 (5.93)
40%	5.017 (6.77)	5.203 (7.00)	5.386 (7.84)	5.523 (7.85)	5.694 (7.65)	6.289 (8.43)	6.523 (8.50)	6.753 (9.26)	6.926 (9.22)	7.141 (8.98)

table3_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.793	6.001	6.166	6.367	6.537
Imputation using linear regression										
10%	4.973 (6.08)	5.133 (5.84)	5.286 (6.16)	5.406 (5.19)	5.616 (6.50)	6.276 (8.35)	6.478 (7.95)	6.673 (8.22)	6.824 (7.17)	7.090 (8.47)
20%	5.534 (18.03)	5.735 (18.26)	5.874 (17.97)	6.122 (19.14)	6.216 (17.88)	6.985 (20.58)	7.241 (20.66)	7.418 (20.30)	7.731 (21.42)	7.851 (20.10)
30%	6.508 (38.81)	6.721 (38.59)	6.947 (39.51)	7.150 (39.14)	7.335 (39.10)	8.193 (41.44)	8.464 (41.03)	8.749 (41.88)	9.008 (41.46)	9.243 (41.41)
40%	7.865 (67.75)	8.111 (67.24)	8.407 (68.83)	8.623 (67.80)	8.945 (69.63)	9.903 (70.96)	10.218 (70.26)	10.592 (71.77)	10.866 (70.65)	11.275 (72.50)
Imputation using predictive mean matching method										
10%	4.610 (−1.68)	4.755 (−1.96)	4.902 (−1.55)	5.006 (−2.59)	5.208 (−1.23)	5.821 (0.49)	6.004 (0.05)	6.191 (0.41)	6.322 (−0.71)	6.579 (0.64)
20%	4.675 (−0.28)	4.848 (−0.03)	4.984 (0.10)	5.169 (0.58)	5.268 (−0.09)	5.904 (1.92)	6.123 (2.03)	6.296 (2.10)	6.529 (2.54)	6.656 (1.82)
30%	4.814 (2.67)	5.014 (3.39)	5.148 (3.39)	5.313 (3.40)	5.443 (3.23)	6.073 (4.85)	6.328 (5.44)	6.498 (5.37)	6.708 (5.34)	6.872 (5.12)
40%	4.970 (6.00)	5.125 (5.67)	5.282 (6.07)	5.452 (6.10)	5.595 (6.11)	6.271 (8.27)	6.468 (7.78)	6.667 (8.12)	6.883 (8.09)	7.065 (8.08)
Imputation using propensity−score method										
10%	4.599 (−1.90)	4.741 (−2.23)	4.890 (−1.80)	4.998 (−2.74)	5.191 (−1.55)	5.804 (0.19)	5.984 (−0.29)	6.172 (0.09)	6.309 (−0.92)	6.553 (0.25)
20%	4.673 (−0.33)	4.856 (0.12)	4.990 (0.21)	5.170 (0.62)	5.285 (0.23)	5.894 (1.74)	6.124 (2.05)	6.295 (2.08)	6.523 (2.44)	6.668 (2.01)
30%	4.738 (1.06)	4.934 (1.74)	5.073 (1.87)	5.232 (1.82)	5.350 (1.46)	5.972 (3.11)	6.221 (3.66)	6.396 (3.73)	6.599 (3.63)	6.748 (3.24)
40%	4.871 (3.88)	5.051 (4.16)	5.166 (3.75)	5.322 (3.57)	5.515 (4.58)	6.127 (5.77)	6.355 (5.90)	6.501 (5.42)	6.699 (5.20)	6.943 (6.21)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.800	6.012	6.181	6.341	6.553
Imputation using linear regression										

table3_nonshrink_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.931 (4.93)	5.103 (4.94)	5.273 (5.59)	5.403 (5.50)	5.535 (4.64)	6.225 (7.32)	6.443 (7.17)	6.660 (7.74)	6.824 (7.62)	6.992 (6.70)
20%	5.566 (18.45)	5.732 (17.87)	5.898 (18.09)	6.110 (19.31)	6.265 (18.43)	7.020 (21.03)	7.230 (20.26)	7.441 (20.39)	7.711 (21.59)	7.905 (20.65)
30%	6.421 (36.65)	6.682 (37.40)	6.869 (37.53)	7.103 (38.69)	7.323 (38.45)	8.093 (39.53)	8.423 (40.09)	8.661 (40.12)	8.959 (41.27)	9.238 (40.98)
40%	7.909 (68.31)	8.231 (69.26)	8.505 (70.29)	8.635 (68.61)	9.464 (78.91)	9.944 (71.45)	10.354 (72.22)	10.701 (73.13)	10.868 (71.38)	11.931 (82.08)
Imputation using predictive mean matching method										
10%	4.573 (−2.69)	4.727 (−2.79)	4.889 (−2.11)	5.023 (−1.91)	5.138 (−2.87)	5.775 (−0.43)	5.971 (−0.69)	6.176 (−0.07)	6.347 (0.08)	6.491 (−0.93)
20%	4.688 (−0.23)	4.840 (−0.47)	4.985 (−0.19)	5.160 (0.76)	5.260 (−0.56)	5.920 (2.07)	6.113 (1.67)	6.296 (1.86)	6.518 (2.79)	6.646 (1.42)
30%	4.800 (2.14)	4.972 (2.25)	5.103 (2.17)	5.252 (2.54)	5.405 (2.17)	6.060 (4.48)	6.279 (4.43)	6.445 (4.27)	6.634 (4.61)	6.827 (4.19)
40%	4.958 (5.51)	5.154 (5.99)	5.324 (6.59)	5.449 (6.39)	5.616 (6.18)	6.253 (7.81)	6.501 (8.13)	6.716 (8.66)	6.876 (8.43)	7.088 (8.16)
Imputation using propensity−score method										
10%	4.559 (−2.99)	4.716 (−3.02)	4.870 (−2.49)	5.010 (−2.18)	5.129 (−3.05)	5.754 (−0.80)	5.953 (−0.98)	6.149 (−0.52)	6.325 (−0.25)	6.476 (−1.17)
20%	4.706 (0.15)	4.855 (−0.17)	4.982 (−0.25)	5.176 (1.07)	5.276 (−0.25)	5.931 (2.26)	6.120 (1.79)	6.281 (1.62)	6.528 (2.93)	6.654 (1.55)
30%	4.765 (1.40)	4.896 (0.69)	5.076 (1.63)	5.199 (1.52)	5.372 (1.56)	6.006 (3.55)	6.173 (2.68)	6.401 (3.56)	6.556 (3.39)	6.775 (3.39)
40%	4.826 (2.71)	5.028 (3.40)	5.196 (4.04)	5.354 (4.53)	5.521 (4.37)	6.070 (4.65)	6.325 (5.20)	6.539 (5.79)	6.737 (6.24)	6.949 (6.05)

table3_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.381	7.839	8.233	8.653	9.021
Imputation using linear regression										
10%	6.307 (3.60)	6.684 (3.97)	7.042 (4.59)	7.338 (3.87)	7.754 (5.37)	7.887 (6.85)	8.360 (6.65)	8.810 (7.01)	9.181 (6.11)	9.703 (7.56)
20%	6.811 (11.87)	7.276 (13.19)	7.665 (13.85)	8.148 (15.34)	8.408 (14.26)	8.551 (15.85)	9.138 (16.58)	9.628 (16.95)	10.235 (18.29)	10.563 (17.09)
30%	7.570 (24.34)	8.144 (26.68)	8.650 (28.47)	9.130 (29.25)	9.537 (29.61)	9.541 (29.27)	10.266 (30.97)	10.906 (32.47)	11.512 (33.05)	12.028 (33.33)
40%	8.750 (43.73)	9.427 (46.64)	10.092 (49.89)	10.653 (50.80)	11.309 (53.69)	11.082 (50.14)	11.944 (52.37)	12.788 (55.33)	13.498 (56.00)	14.335 (58.90)
Imputation using predictive mean matching method										
10%	6.042 (−0.75)	6.383 (−0.72)	6.714 (−0.29)	6.976 (−1.25)	7.348 (−0.14)	7.532 (2.04)	7.958 (1.53)	8.373 (1.70)	8.701 (0.56)	9.167 (1.61)
20%	6.176 (1.44)	6.542 (1.77)	6.857 (1.84)	7.225 (2.28)	7.471 (1.54)	7.695 (4.25)	8.155 (4.03)	8.549 (3.84)	9.009 (4.12)	9.318 (3.29)
30%	6.337 (4.09)	6.743 (4.89)	7.056 (4.79)	7.404 (4.81)	7.699 (4.63)	7.894 (6.94)	8.403 (7.20)	8.794 (6.82)	9.230 (6.68)	9.600 (6.42)
40%	6.577 (8.03)	6.941 (7.96)	7.289 (8.26)	7.639 (8.14)	7.966 (8.26)	8.183 (10.87)	8.639 (10.21)	9.077 (10.25)	9.515 (9.96)	9.925 (10.02)
Imputation using propensity−score method										
10%	6.049 (−0.64)	6.381 (−0.74)	6.706 (−0.41)	6.975 (−1.26)	7.337 (−0.29)	7.534 (2.07)	7.950 (1.42)	8.356 (1.50)	8.694 (0.47)	9.146 (1.38)
20%	6.194 (1.73)	6.565 (2.13)	6.877 (2.14)	7.240 (2.49)	7.506 (2.01)	7.708 (4.42)	8.173 (4.27)	8.563 (4.01)	9.017 (4.21)	9.349 (3.64)
30%	6.362 (4.51)	6.746 (4.94)	7.072 (5.03)	7.391 (4.63)	7.683 (4.41)	7.901 (7.05)	8.381 (6.92)	8.788 (6.75)	9.187 (6.18)	9.552 (5.89)
40%	6.581 (8.10)	6.958 (8.23)	7.260 (7.83)	7.592 (7.48)	7.928 (7.74)	8.153 (10.45)	8.621 (9.99)	9.001 (9.33)	9.415 (8.81)	9.835 (9.02)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.400	7.858	8.254	8.622	9.043
Imputation using linear regression										

table3_nonshrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.244 (2.20)	6.632 (2.82)	7.011 (3.79)	7.324 (3.96)	7.629 (3.34)	7.812 (5.57)	8.300 (5.62)	8.776 (6.33)	9.170 (6.36)	9.553 (5.63)
20%	6.773 (10.86)	7.202 (11.65)	7.614 (12.71)	8.070 (14.54)	8.401 (13.80)	8.507 (14.96)	9.048 (15.14)	9.566 (15.90)	10.140 (17.61)	10.558 (16.75)
30%	7.539 (23.39)	8.140 (26.20)	8.584 (27.07)	9.081 (28.91)	9.591 (29.91)	9.506 (28.46)	10.266 (30.65)	10.829 (31.20)	11.457 (32.89)	12.097 (33.77)
40%	8.652 (41.62)	9.427 (46.14)	10.104 (49.57)	10.669 (51.44)	11.430 (54.83)	10.972 (48.26)	11.953 (52.12)	12.811 (55.21)	13.532 (56.95)	14.499 (60.33)
Imputation using predictive mean matching method										
10%	5.989 (−1.98)	6.337 (−1.76)	6.677 (−1.15)	6.982 (−0.90)	7.245 (−1.86)	7.469 (0.93)	7.905 (0.60)	8.331 (0.94)	8.713 (1.07)	9.043 (−0.01)
20%	6.169 (0.96)	6.515 (1.00)	6.837 (1.21)	7.197 (2.15)	7.449 (0.90)	7.688 (3.89)	8.123 (3.37)	8.526 (3.29)	8.976 (4.11)	9.292 (2.75)
30%	6.333 (3.66)	6.700 (3.86)	7.026 (4.01)	7.352 (4.35)	7.670 (3.90)	7.891 (6.63)	8.350 (6.27)	8.759 (6.12)	9.167 (6.32)	9.565 (5.77)
40%	6.539 (7.03)	6.933 (7.48)	7.301 (8.09)	7.619 (8.14)	7.980 (8.09)	8.140 (10.00)	8.633 (9.87)	9.094 (10.18)	9.492 (10.09)	9.943 (9.95)
Imputation using propensity−score method										
10%	5.989 (−1.97)	6.339 (−1.74)	6.669 (−1.27)	6.981 (−0.91)	7.239 (−1.95)	7.462 (0.84)	7.900 (0.53)	8.314 (0.73)	8.704 (0.96)	9.026 (−0.19)
20%	6.190 (1.31)	6.539 (1.38)	6.846 (1.35)	7.203 (2.24)	7.451 (0.93)	7.704 (4.11)	8.141 (3.61)	8.525 (3.29)	8.972 (4.06)	9.282 (2.64)
30%	6.411 (4.93)	6.742 (4.52)	7.088 (4.93)	7.376 (4.69)	7.705 (4.36)	7.959 (7.56)	8.374 (6.57)	8.806 (6.69)	9.165 (6.30)	9.575 (5.88)
40%	6.545 (7.12)	6.924 (7.33)	7.294 (7.98)	7.597 (7.83)	7.933 (7.46)	8.104 (9.52)	8.578 (9.16)	9.040 (9.52)	9.417 (9.23)	9.836 (8.77)

table3_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.381	7.839	8.233	8.653	9.021
Imputation using linear regression										
10%	6.409 (5.28)	6.787 (5.57)	7.139 (6.02)	7.433 (5.23)	7.836 (6.49)	7.998 (8.35)	8.471 (8.07)	8.913 (8.27)	9.283 (7.28)	9.788 (8.50)
20%	7.120 (16.95)	7.568 (17.73)	7.930 (17.77)	8.406 (19.00)	8.674 (17.88)	8.886 (20.39)	9.451 (20.56)	9.906 (20.32)	10.504 (21.40)	10.841 (20.17)
30%	8.382 (37.69)	8.884 (38.19)	9.377 (39.27)	9.826 (39.11)	10.237 (39.12)	10.431 (41.31)	11.062 (41.12)	11.681 (41.88)	12.247 (41.55)	12.765 (41.51)
40%	10.089 (65.72)	10.704 (66.50)	11.338 (68.40)	11.845 (67.68)	12.479 (69.60)	12.554 (70.08)	13.333 (70.09)	14.130 (71.63)	14.769 (70.68)	15.568 (72.57)
Imputation using predictive mean matching method										
10%	5.942 (−2.39)	6.288 (−2.18)	6.620 (−1.68)	6.884 (−2.55)	7.267 (−1.24)	7.419 (0.52)	7.854 (0.20)	8.270 (0.45)	8.601 (−0.60)	9.081 (0.67)
20%	6.030 (−0.96)	6.405 (−0.37)	6.730 (−0.05)	7.100 (0.50)	7.352 (−0.08)	7.530 (2.01)	8.001 (2.08)	8.409 (2.14)	8.873 (2.55)	9.190 (1.87)
30%	6.219 (2.16)	6.625 (3.05)	6.953 (3.26)	7.302 (3.37)	7.596 (3.23)	7.759 (5.12)	8.268 (5.48)	8.680 (5.43)	9.119 (5.39)	9.488 (5.18)
40%	6.394 (5.03)	6.771 (5.33)	7.130 (5.89)	7.491 (6.05)	7.808 (6.11)	7.977 (8.07)	8.452 (7.83)	8.903 (8.14)	9.356 (8.13)	9.754 (8.13)
Imputation using propensity−score method										
10%	5.933 (−2.54)	6.274 (−2.40)	6.605 (−1.90)	6.874 (−2.70)	7.245 (−1.55)	7.403 (0.29)	7.831 (−0.10)	8.246 (0.16)	8.582 (−0.82)	9.047 (0.29)
20%	6.034 (−0.89)	6.417 (−0.18)	6.740 (0.10)	7.102 (0.54)	7.375 (0.22)	7.524 (1.94)	8.006 (2.13)	8.411 (2.16)	8.865 (2.45)	9.206 (2.05)
30%	6.115 (0.45)	6.516 (1.36)	6.848 (1.71)	7.187 (1.75)	7.467 (1.48)	7.621 (3.25)	8.125 (3.65)	8.542 (3.75)	8.968 (3.65)	9.319 (3.31)
40%	6.277 (3.11)	6.671 (3.77)	6.982 (3.70)	7.315 (3.55)	7.693 (4.55)	7.804 (5.73)	8.298 (5.86)	8.689 (5.54)	9.107 (5.26)	9.583 (6.23)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.400	7.858	8.254	8.622	9.043
Imputation using linear regression										

table3_nonshrink_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.355 (4.02)	6.743 (4.54)	7.116 (5.34)	7.426 (5.40)	7.725 (4.64)	7.934 (7.22)	8.421 (7.17)	8.889 (7.70)	9.279 (7.62)	9.654 (6.75)
20%	7.165 (17.27)	7.577 (17.46)	7.965 (17.91)	8.395 (19.16)	8.742 (18.42)	8.934 (20.73)	9.452 (20.29)	9.940 (20.43)	10.480 (21.56)	10.915 (20.70)
30%	8.271 (35.38)	8.820 (36.73)	9.269 (37.22)	9.759 (38.52)	10.217 (38.39)	10.306 (39.26)	10.996 (39.94)	11.562 (40.09)	12.178 (41.25)	12.754 (41.03)
40%	10.174 (66.53)	10.859 (68.34)	11.471 (69.81)	11.869 (68.48)	13.195 (78.73)	12.643 (70.86)	13.508 (71.91)	14.278 (72.98)	14.781 (71.44)	16.465 (82.07)
Imputation using predictive mean matching method										
10%	5.899 (−3.45)	6.250 (−3.10)	6.599 (−2.31)	6.903 (−2.01)	7.170 (−2.87)	7.367 (−0.44)	7.809 (−0.63)	8.246 (−0.09)	8.628 (0.08)	8.963 (−0.89)
20%	6.040 (−1.14)	6.397 (−0.83)	6.730 (−0.37)	7.089 (0.62)	7.341 (−0.56)	7.543 (1.93)	7.992 (1.70)	8.409 (1.89)	8.859 (2.75)	9.176 (1.47)
30%	6.181 (1.17)	6.567 (1.81)	6.888 (1.97)	7.218 (2.45)	7.542 (2.16)	7.717 (4.29)	8.202 (4.38)	8.606 (4.27)	9.020 (4.62)	9.426 (4.23)
40%	6.400 (4.76)	6.813 (5.62)	7.185 (6.36)	7.489 (6.31)	7.837 (6.16)	7.981 (7.86)	8.499 (8.16)	8.967 (8.64)	9.350 (8.45)	9.786 (8.21)
Imputation using propensity−score method										
10%	5.885 (−3.67)	6.238 (−3.29)	6.575 (−2.66)	6.884 (−2.29)	7.155 (−3.08)	7.345 (−0.74)	7.787 (−0.90)	8.211 (−0.52)	8.598 (−0.27)	8.937 (−1.17)
20%	6.069 (−0.66)	6.423 (−0.43)	6.733 (−0.32)	7.112 (0.95)	7.361 (−0.29)	7.563 (2.20)	8.007 (1.90)	8.396 (1.73)	8.871 (2.90)	9.184 (1.55)
30%	6.136 (0.42)	6.477 (0.41)	6.851 (1.41)	7.141 (1.36)	7.485 (1.38)	7.645 (3.30)	8.074 (2.75)	8.544 (3.51)	8.907 (3.31)	9.338 (3.26)
40%	6.249 (2.28)	6.656 (3.19)	7.016 (3.86)	7.353 (4.38)	7.685 (4.10)	7.767 (4.96)	8.278 (5.35)	8.731 (5.78)	9.153 (6.17)	9.569 (5.81)

table3_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.316	11.158	11.894	12.640	13.300
Imputation using linear regression										
10%	8.882 (2.44)	9.603 (3.54)	10.267 (4.38)	10.824 (3.84)	11.540 (5.35)	10.999 (6.62)	11.897 (6.62)	12.724 (6.98)	13.418 (6.16)	14.308 (7.57)
20%	9.541 (10.04)	10.415 (12.30)	11.157 (13.44)	12.000 (15.12)	12.511 (14.21)	11.867 (15.03)	12.959 (16.14)	13.888 (16.76)	14.938 (18.18)	15.577 (17.12)
30%	10.541 (21.59)	11.616 (25.25)	12.559 (27.69)	13.436 (28.90)	14.186 (29.50)	13.166 (27.62)	14.512 (30.06)	15.695 (31.96)	16.793 (32.86)	17.734 (33.34)
40%	12.021 (38.65)	13.371 (44.17)	14.611 (48.55)	15.655 (50.18)	16.809 (53.45)	15.090 (46.28)	16.794 (50.51)	18.357 (54.34)	19.670 (55.61)	21.130 (58.87)
Imputation using predictive mean matching method										
10%	8.548 (−1.41)	9.194 (−0.87)	9.802 (−0.34)	10.299 (−1.20)	10.940 (−0.13)	10.550 (2.27)	11.353 (1.75)	12.108 (1.80)	12.724 (0.67)	13.519 (1.64)
20%	8.745 (0.87)	9.422 (1.60)	10.014 (1.81)	10.661 (2.28)	11.126 (1.56)	10.791 (4.60)	11.632 (4.25)	12.366 (3.97)	13.169 (4.18)	13.745 (3.34)
30%	8.979 (3.56)	9.706 (4.66)	10.303 (4.75)	10.926 (4.82)	11.464 (4.65)	11.075 (7.35)	11.979 (7.35)	12.720 (6.95)	13.494 (6.75)	14.161 (6.47)
40%	9.297 (7.24)	9.999 (7.81)	10.644 (8.22)	11.275 (8.16)	11.862 (8.28)	11.454 (11.02)	12.325 (10.46)	13.128 (10.37)	13.910 (10.05)	14.640 (10.07)
Imputation using propensity−score method										
10%	8.563 (−1.24)	9.196 (−0.84)	9.793 (−0.43)	10.299 (−1.20)	10.924 (−0.28)	10.560 (2.36)	11.346 (1.69)	12.087 (1.62)	12.714 (0.58)	13.488 (1.41)
20%	8.778 (1.25)	9.459 (1.99)	10.043 (2.11)	10.684 (2.50)	11.174 (2.01)	10.818 (4.86)	11.662 (4.52)	12.388 (4.15)	13.182 (4.28)	13.789 (3.67)
30%	9.033 (4.19)	9.725 (4.86)	10.332 (5.05)	10.912 (4.68)	11.442 (4.45)	11.106 (7.66)	11.965 (7.23)	12.717 (6.92)	13.435 (6.29)	14.091 (5.94)
40%	9.337 (7.69)	10.034 (8.19)	10.614 (7.91)	11.211 (7.55)	11.811 (7.81)	11.449 (10.98)	12.311 (10.33)	13.031 (9.56)	13.770 (8.94)	14.511 (9.10)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.353	11.190	11.926	12.599	13.333
Imputation using linear regression										

table3_nonshrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.793 (0.96)	9.523 (2.29)	10.216 (3.51)	10.800 (3.85)	11.356 (3.32)	10.895 (5.24)	11.805 (5.50)	12.669 (6.23)	13.397 (6.33)	14.088 (5.66)
20%	9.484 (8.89)	10.315 (10.79)	11.081 (12.27)	11.885 (14.28)	12.500 (13.73)	11.799 (13.97)	12.838 (14.73)	13.795 (15.68)	14.800 (17.47)	15.568 (16.76)
30%	10.499 (20.55)	11.602 (24.62)	12.466 (26.30)	13.366 (28.51)	14.262 (29.76)	13.120 (26.73)	14.503 (29.60)	15.589 (30.72)	16.715 (32.67)	17.833 (33.75)
40%	11.909 (36.74)	13.357 (43.46)	14.617 (48.10)	15.675 (50.72)	16.986 (54.54)	14.971 (44.60)	16.795 (50.09)	18.382 (54.13)	19.718 (56.51)	21.371 (60.28)
Imputation using predictive mean matching method										
10%	8.474 (−2.70)	9.126 (−1.98)	9.745 (−1.26)	10.303 (−0.94)	10.788 (−1.85)	10.465 (1.09)	11.274 (0.75)	12.044 (0.99)	12.737 (1.10)	13.338 (0.03)
20%	8.725 (0.17)	9.384 (0.79)	9.983 (1.15)	10.619 (2.10)	11.092 (0.92)	10.767 (4.00)	11.586 (3.54)	12.331 (3.40)	13.120 (4.13)	13.707 (2.80)
30%	8.960 (2.88)	9.648 (3.63)	10.261 (3.96)	10.852 (4.35)	11.421 (3.91)	11.054 (6.77)	11.909 (6.42)	12.669 (6.23)	13.403 (6.38)	14.108 (5.81)
40%	9.266 (6.39)	9.988 (7.28)	10.660 (8.00)	11.246 (8.13)	11.881 (8.10)	11.420 (10.30)	12.318 (10.08)	13.152 (10.28)	13.879 (10.16)	14.667 (10.00)
Imputation using propensity−score method										
10%	8.483 (−2.60)	9.131 (−1.92)	9.736 (−1.36)	10.301 (−0.95)	10.777 (−1.95)	10.465 (1.09)	11.270 (0.71)	12.021 (0.80)	12.722 (0.98)	13.310 (−0.17)
20%	8.763 (0.61)	9.423 (1.21)	10.000 (1.32)	10.627 (2.18)	11.097 (0.97)	10.798 (4.30)	11.617 (3.81)	12.333 (3.41)	13.111 (4.06)	13.693 (2.69)
30%	9.090 (4.37)	9.728 (4.49)	10.356 (4.92)	10.886 (4.67)	11.464 (4.30)	11.172 (7.91)	11.964 (6.92)	12.740 (6.83)	13.395 (6.32)	14.110 (5.82)
40%	9.317 (6.97)	10.001 (7.42)	10.664 (8.04)	11.209 (7.77)	11.803 (7.38)	11.419 (10.30)	12.267 (9.62)	13.086 (9.73)	13.761 (9.22)	14.493 (8.70)

table3_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.316	11.158	11.894	12.640	13.300
Imputation using linear regression										
10%	9.044 (4.32)	9.761 (5.25)	10.415 (5.89)	10.969 (5.23)	11.665 (6.48)	11.175 (8.32)	12.067 (8.15)	12.881 (8.30)	13.570 (7.36)	14.434 (8.52)
20%	10.028 (15.67)	10.869 (17.19)	11.565 (17.59)	12.394 (18.90)	12.913 (17.88)	12.394 (20.14)	13.443 (20.47)	14.313 (20.33)	15.344 (21.39)	15.991 (20.23)
30%	11.811 (36.23)	12.770 (37.69)	13.675 (39.04)	14.496 (39.06)	15.241 (39.13)	14.551 (41.04)	15.748 (41.13)	16.875 (41.88)	17.900 (41.61)	18.831 (41.58)
40%	14.161 (63.34)	15.366 (65.68)	16.525 (68.01)	17.468 (67.58)	18.576 (69.57)	17.441 (69.06)	18.953 (69.86)	20.401 (71.52)	21.579 (70.72)	22.963 (72.65)
Imputation using predictive mean matching method										
10%	8.388 (−3.25)	9.047 (−2.45)	9.658 (−1.81)	10.159 (−2.54)	10.818 (−1.25)	10.371 (0.53)	11.191 (0.29)	11.951 (0.48)	12.573 (−0.53)	13.392 (0.69)
20%	8.513 (−1.81)	9.207 (−0.72)	9.817 (−0.19)	10.470 (0.44)	10.945 (−0.08)	10.528 (2.05)	11.393 (2.10)	12.152 (2.17)	12.964 (2.56)	13.555 (1.92)
30%	8.792 (1.41)	9.523 (2.68)	10.144 (3.14)	10.772 (3.34)	11.309 (3.24)	10.862 (5.29)	11.772 (5.50)	12.545 (5.48)	13.327 (5.44)	13.995 (5.22)
40%	9.008 (3.90)	9.733 (4.95)	10.399 (5.73)	11.049 (6.00)	11.623 (6.10)	11.127 (7.86)	12.032 (7.84)	12.862 (8.14)	13.670 (8.15)	14.387 (8.17)
Imputation using propensity−score method										
10%	8.379 (−3.35)	9.029 (−2.64)	9.638 (−2.01)	10.144 (−2.69)	10.784 (−1.55)	10.353 (0.35)	11.161 (0.03)	11.918 (0.20)	12.546 (−0.74)	13.341 (0.31)
20%	8.525 (−1.68)	9.226 (−0.52)	9.834 (−0.02)	10.476 (0.50)	10.980 (0.23)	10.527 (2.05)	11.401 (2.17)	12.157 (2.21)	12.955 (2.49)	13.581 (2.11)
30%	8.636 (−0.39)	9.366 (0.98)	9.988 (1.54)	10.603 (1.71)	11.122 (1.53)	10.658 (3.31)	11.566 (3.66)	12.341 (3.76)	13.107 (3.70)	13.753 (3.40)
40%	8.849 (2.06)	9.585 (3.36)	10.185 (3.55)	10.796 (3.56)	11.445 (4.48)	10.890 (5.56)	11.809 (5.83)	12.556 (5.56)	13.316 (5.35)	14.126 (6.21)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.353	11.190	11.926	12.599	13.333
Imputation using linear regression										

table3_nonshrink_5_treatments_R9_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.970 (2.99)	9.695 (4.13)	10.377 (5.14)	10.954 (5.33)	11.501 (4.64)	11.090 (7.12)	11.992 (7.16)	12.841 (7.67)	13.560 (7.63)	14.239 (6.79)
20%	10.097 (15.93)	10.894 (17.02)	11.620 (17.73)	12.381 (19.05)	13.014 (18.41)	12.466 (20.41)	13.460 (20.29)	14.365 (20.45)	15.313 (21.54)	16.100 (20.75)
30%	11.658 (33.86)	12.667 (36.06)	13.517 (36.95)	14.393 (38.39)	15.207 (38.36)	14.382 (38.92)	15.642 (39.78)	16.703 (40.06)	17.796 (41.25)	18.811 (41.08)
40%	14.310 (64.31)	15.584 (67.39)	16.719 (69.39)	17.511 (68.37)	19.632 (78.62)	17.603 (70.03)	19.200 (71.58)	20.617 (72.87)	21.608 (71.51)	24.282 (82.11)
Imputation using predictive mean matching method										
10%	8.332 (−4.34)	8.991 (−3.43)	9.625 (−2.48)	10.183 (−2.08)	10.675 (−2.87)	10.305 (−0.46)	11.125 (−0.58)	11.914 (−0.10)	12.609 (0.08)	13.220 (−0.85)
20%	8.521 (−2.16)	9.199 (−1.19)	9.818 (−0.52)	10.454 (0.52)	10.929 (−0.56)	10.538 (1.78)	11.382 (1.72)	12.153 (1.90)	12.944 (2.74)	13.534 (1.51)
30%	8.717 (0.09)	9.438 (1.37)	10.048 (1.80)	10.648 (2.38)	11.227 (2.14)	10.777 (4.09)	11.675 (4.33)	12.435 (4.27)	13.182 (4.63)	13.901 (4.26)
40%	9.039 (3.79)	9.795 (5.21)	10.478 (6.16)	11.049 (6.24)	11.667 (6.15)	11.162 (7.82)	12.104 (8.17)	12.955 (8.63)	13.667 (8.48)	14.435 (8.26)
Imputation using propensity−score method										
10%	8.317 (−4.50)	8.975 (−3.60)	9.592 (−2.82)	10.154 (−2.37)	10.650 (−3.11)	10.279 (−0.71)	11.097 (−0.83)	11.865 (−0.51)	12.563 (−0.28)	13.178 (−1.16)
20%	8.567 (−1.63)	9.240 (−0.75)	9.828 (−0.42)	10.487 (0.84)	10.957 (−0.31)	10.570 (2.10)	11.408 (1.95)	12.140 (1.79)	12.959 (2.86)	13.542 (1.57)
30%	8.649 (−0.69)	9.316 (0.07)	9.991 (1.23)	10.533 (1.28)	11.134 (1.30)	10.669 (3.05)	11.501 (2.78)	12.341 (3.48)	13.014 (3.30)	13.761 (3.21)
40%	8.846 (1.56)	9.580 (2.89)	10.236 (3.71)	10.840 (4.23)	11.411 (3.82)	10.883 (5.12)	11.797 (5.42)	12.615 (5.78)	13.366 (6.09)	14.077 (5.57)

table3_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.921	5.921	5.921	5.921	5.921
Imputation using linear regression										
10%	4.908 (4.68)	5.063 (4.40)	5.220 (4.84)	5.339 (3.90)	5.558 (5.41)	6.302 (6.45)	6.331 (6.92)	6.357 (7.36)	6.370 (7.59)	6.419 (8.42)
20%	5.332 (13.73)	5.541 (14.25)	5.696 (14.40)	5.945 (15.69)	6.030 (14.35)	6.928 (17.01)	6.993 (18.12)	7.042 (18.93)	7.165 (21.03)	7.161 (20.95)
30%	5.967 (27.27)	6.230 (28.46)	6.452 (29.58)	6.670 (29.80)	6.845 (29.81)	7.788 (31.55)	7.929 (33.91)	8.058 (36.10)	8.172 (38.03)	8.258 (39.48)
40%	7.000 (49.31)	7.264 (49.79)	7.559 (51.81)	7.799 (51.77)	8.125 (54.09)	9.164 (54.78)	9.347 (57.88)	9.587 (61.93)	9.740 (64.52)	10.037 (69.52)
Imputation using predictive mean matching method										
10%	4.674 (−0.30)	4.817 (−0.67)	4.966 (−0.27)	5.069 (−1.35)	5.264 (−0.16)	5.940 (0.32)	5.954 (0.57)	5.974 (0.90)	5.974 (0.90)	5.999 (1.33)
20%	4.770 (1.73)	4.939 (1.83)	5.070 (1.83)	5.255 (2.27)	5.352 (1.49)	6.080 (2.68)	6.107 (3.16)	6.128 (3.50)	6.185 (4.46)	6.195 (4.64)
30%	4.891 (4.31)	5.094 (5.03)	5.217 (4.78)	5.383 (4.76)	5.515 (4.59)	6.234 (5.30)	6.296 (6.35)	6.335 (7.00)	6.390 (7.92)	6.426 (8.54)
40%	5.088 (8.53)	5.236 (7.97)	5.390 (8.24)	5.553 (8.07)	5.706 (8.21)	6.451 (8.97)	6.506 (9.89)	6.577 (11.08)	6.638 (12.11)	6.713 (13.38)
Imputation using propensity−score method										
10%	4.676 (−0.27)	4.812 (−0.77)	4.958 (−0.43)	5.068 (−1.37)	5.256 (−0.31)	5.923 (0.04)	5.932 (0.19)	5.950 (0.50)	5.957 (0.61)	5.975 (0.92)
20%	4.777 (1.88)	4.954 (2.14)	5.083 (2.07)	5.264 (2.44)	5.376 (1.96)	6.074 (2.59)	6.109 (3.18)	6.122 (3.40)	6.183 (4.42)	6.207 (4.84)
30%	4.896 (4.42)	5.085 (4.84)	5.224 (4.90)	5.371 (4.51)	5.500 (4.30)	6.186 (4.49)	6.238 (5.35)	6.300 (6.40)	6.326 (6.85)	6.370 (7.59)
40%	5.065 (8.04)	5.240 (8.05)	5.355 (7.54)	5.517 (7.36)	5.678 (7.69)	6.335 (7.00)	6.430 (8.60)	6.465 (9.19)	6.530 (10.29)	6.615 (11.73)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.906	5.906	5.906	5.906	5.906
Imputation using linear regression										

table3_pooled_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.859 (3.40)	5.028 (3.40)	5.202 (4.16)	5.333 (4.13)	5.468 (3.37)	6.268 (6.12)	6.298 (6.62)	6.334 (7.24)	6.336 (7.27)	6.361 (7.69)
20%	5.307 (12.94)	5.480 (12.70)	5.659 (13.32)	5.887 (14.95)	6.026 (13.92)	6.883 (16.53)	6.936 (17.43)	7.001 (18.54)	7.089 (20.02)	7.156 (21.16)
30%	5.942 (26.45)	6.234 (28.19)	6.401 (28.16)	6.633 (29.52)	6.885 (30.16)	7.758 (31.35)	7.947 (34.55)	8.009 (35.61)	8.160 (38.16)	8.311 (40.71)
40%	6.904 (46.94)	7.274 (49.59)	7.576 (51.68)	7.814 (52.58)	8.216 (55.32)	9.104 (54.15)	9.401 (59.17)	9.629 (63.03)	9.783 (65.63)	10.142 (71.71)
Imputation using predictive mean matching method										
10%	4.632 (−1.43)	4.785 (−1.60)	4.942 (−1.05)	5.077 (−0.86)	5.190 (−1.89)	5.906 (0.00)	5.926 (0.33)	5.943 (0.63)	5.955 (0.82)	5.958 (0.88)
20%	4.772 (1.56)	4.917 (1.13)	5.056 (1.23)	5.234 (2.20)	5.336 (0.87)	6.067 (2.73)	6.092 (3.14)	6.115 (3.54)	6.160 (4.29)	6.189 (4.78)
30%	4.897 (4.22)	5.058 (4.01)	5.196 (4.03)	5.344 (4.35)	5.495 (3.87)	6.211 (5.15)	6.264 (6.06)	6.307 (6.78)	6.359 (7.67)	6.397 (8.31)
40%	5.045 (7.37)	5.230 (7.56)	5.401 (8.13)	5.538 (8.13)	5.716 (8.07)	6.415 (8.61)	6.498 (10.02)	6.568 (11.21)	6.619 (12.06)	6.711 (13.62)
Imputation using propensity−score method										
10%	4.626 (−1.56)	4.783 (−1.63)	4.933 (−1.22)	5.076 (−0.88)	5.187 (−1.94)	5.881 (−0.43)	5.904 (−0.04)	5.917 (0.18)	5.938 (0.54)	5.939 (0.55)
20%	4.782 (1.78)	4.932 (1.43)	5.059 (1.30)	5.238 (2.28)	5.341 (0.97)	6.050 (2.43)	6.082 (2.97)	6.098 (3.24)	6.139 (3.94)	6.173 (4.52)
30%	4.941 (5.15)	5.074 (4.34)	5.237 (4.86)	5.360 (4.65)	5.523 (4.40)	6.207 (5.10)	6.233 (5.53)	6.300 (6.66)	6.332 (7.21)	6.386 (8.12)
40%	5.017 (6.77)	5.203 (7.00)	5.386 (7.84)	5.523 (7.85)	5.694 (7.65)	6.287 (6.44)	6.373 (7.89)	6.466 (9.48)	6.520 (10.40)	6.609 (11.90)

table3_pooled_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.921	5.921	5.921	5.921	5.921
Imputation using linear regression										
10%	4.973 (6.08)	5.133 (5.84)	5.286 (6.16)	5.406 (5.19)	5.616 (6.50)	6.370 (7.59)	6.401 (8.11)	6.425 (8.51)	6.443 (8.83)	6.481 (9.46)
20%	5.534 (18.03)	5.735 (18.26)	5.874 (17.97)	6.122 (19.14)	6.216 (17.88)	7.138 (20.56)	7.197 (21.55)	7.231 (22.12)	7.356 (24.25)	7.366 (24.41)
30%	6.508 (38.81)	6.721 (38.59)	6.947 (39.51)	7.150 (39.14)	7.335 (39.10)	8.343 (40.91)	8.440 (42.55)	8.577 (44.86)	8.684 (46.67)	8.785 (48.37)
40%	7.865 (67.75)	8.111 (67.24)	8.407 (68.83)	8.623 (67.80)	8.945 (69.63)	10.066 (70.02)	10.238 (72.92)	10.490 (77.18)	10.624 (79.45)	10.919 (84.43)
Imputation using predictive mean matching method										
10%	4.610 (−1.68)	4.755 (−1.96)	4.902 (−1.55)	5.006 (−2.59)	5.208 (−1.23)	5.909 (−0.20)	5.923 (0.04)	5.939 (0.31)	5.940 (0.32)	5.969 (0.82)
20%	4.675 (−0.28)	4.848 (−0.03)	4.984 (0.10)	5.169 (0.58)	5.268 (−0.09)	6.036 (1.94)	6.066 (2.45)	6.089 (2.84)	6.144 (3.77)	6.154 (3.95)
30%	4.814 (2.67)	5.014 (3.39)	5.148 (3.39)	5.313 (3.40)	5.443 (3.23)	6.201 (4.74)	6.259 (5.71)	6.307 (6.52)	6.354 (7.32)	6.389 (7.91)
40%	4.970 (6.00)	5.125 (5.67)	5.282 (6.07)	5.452 (6.10)	5.595 (6.11)	6.386 (7.86)	6.440 (8.77)	6.515 (10.05)	6.581 (11.16)	6.640 (12.16)
Imputation using propensity−score method										
10%	4.599 (−1.90)	4.741 (−2.23)	4.890 (−1.80)	4.998 (−2.74)	5.191 (−1.55)	5.880 (−0.69)	5.891 (−0.50)	5.909 (−0.20)	5.915 (−0.10)	5.933 (0.22)
20%	4.673 (−0.33)	4.856 (0.12)	4.990 (0.21)	5.170 (0.62)	5.285 (0.23)	6.026 (1.78)	6.062 (2.39)	6.079 (2.67)	6.135 (3.62)	6.160 (4.04)
30%	4.738 (1.06)	4.934 (1.74)	5.073 (1.87)	5.232 (1.82)	5.350 (1.46)	6.087 (2.81)	6.143 (3.75)	6.200 (4.72)	6.235 (5.31)	6.261 (5.75)
40%	4.871 (3.88)	5.051 (4.16)	5.166 (3.75)	5.322 (3.57)	5.515 (4.58)	6.204 (4.78)	6.294 (6.31)	6.329 (6.90)	6.379 (7.74)	6.494 (9.69)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.906	5.906	5.906	5.906	5.906
Imputation using linear regression										

table3_pooled_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.931 (4.93)	5.103 (4.94)	5.273 (5.59)	5.403 (5.50)	5.535 (4.64)	6.336 (7.28)	6.371 (7.88)	6.405 (8.44)	6.407 (8.48)	6.429 (8.86)
20%	5.566 (18.45)	5.732 (17.87)	5.898 (18.09)	6.110 (19.31)	6.265 (18.43)	7.144 (20.96)	7.193 (21.79)	7.250 (22.75)	7.328 (24.07)	7.413 (25.51)
30%	6.421 (36.65)	6.682 (37.40)	6.869 (37.53)	7.103 (38.69)	7.323 (38.45)	8.254 (39.75)	8.415 (42.48)	8.510 (44.08)	8.657 (46.58)	8.776 (48.59)
40%	7.909 (68.31)	8.231 (69.26)	8.505 (70.29)	8.635 (68.61)	9.464 (78.91)	10.179 (72.34)	10.431 (76.61)	10.637 (80.09)	10.677 (80.77)	11.544 (95.46)
Imputation using predictive mean matching method										
10%	4.573 (−2.69)	4.727 (−2.79)	4.889 (−2.11)	5.023 (−1.91)	5.138 (−2.87)	5.877 (−0.50)	5.894 (−0.21)	5.915 (0.15)	5.925 (0.32)	5.928 (0.37)
20%	4.688 (−0.23)	4.840 (−0.47)	4.985 (−0.19)	5.160 (0.76)	5.260 (−0.56)	6.026 (2.02)	6.055 (2.52)	6.085 (3.03)	6.124 (3.69)	6.147 (4.08)
30%	4.800 (2.14)	4.972 (2.25)	5.103 (2.17)	5.252 (2.54)	5.405 (2.17)	6.165 (4.39)	6.229 (5.46)	6.254 (5.89)	6.307 (6.79)	6.345 (7.43)
40%	4.958 (5.51)	5.154 (5.99)	5.324 (6.59)	5.449 (6.39)	5.616 (6.18)	6.379 (8.00)	6.471 (9.57)	6.537 (10.68)	6.566 (11.18)	6.644 (12.48)
Imputation using propensity−score method										
10%	4.559 (−2.99)	4.716 (−3.02)	4.870 (−2.49)	5.010 (−2.18)	5.129 (−3.05)	5.847 (−1.01)	5.865 (−0.69)	5.881 (−0.42)	5.897 (−0.15)	5.904 (−0.03)
20%	4.706 (0.15)	4.855 (−0.17)	4.982 (−0.25)	5.176 (1.07)	5.276 (−0.25)	6.022 (1.96)	6.050 (2.43)	6.066 (2.70)	6.120 (3.61)	6.151 (4.15)
30%	4.765 (1.40)	4.896 (0.69)	5.076 (1.63)	5.199 (1.52)	5.372 (1.56)	6.097 (3.23)	6.117 (3.58)	6.190 (4.81)	6.221 (5.33)	6.282 (6.37)
40%	4.826 (2.71)	5.028 (3.40)	5.196 (4.04)	5.354 (4.53)	5.521 (4.37)	6.153 (4.18)	6.256 (5.92)	6.319 (6.99)	6.392 (8.22)	6.472 (9.57)

table3_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.641	7.641	7.641	7.641	7.641
Imputation using linear regression										
10%	6.307 (3.60)	6.684 (3.97)	7.042 (4.59)	7.338 (3.87)	7.754 (5.37)	8.108 (6.11)	8.172 (6.95)	8.233 (7.75)	8.276 (8.32)	8.368 (9.52)
20%	6.811 (11.87)	7.276 (13.19)	7.665 (13.85)	8.148 (15.34)	8.408 (14.26)	8.848 (15.80)	9.007 (17.88)	9.141 (19.63)	9.366 (22.58)	9.416 (23.24)
30%	7.570 (24.34)	8.144 (26.68)	8.650 (28.47)	9.130 (29.25)	9.537 (29.61)	9.873 (29.21)	10.185 (33.29)	10.475 (37.09)	10.742 (40.58)	10.961 (43.46)
40%	8.750 (43.73)	9.427 (46.64)	10.092 (49.89)	10.653 (50.80)	11.309 (53.69)	11.460 (49.99)	11.941 (56.27)	12.463 (63.11)	12.853 (68.22)	13.427 (75.73)
Imputation using predictive mean matching method										
10%	6.042 (−0.75)	6.383 (−0.72)	6.714 (−0.29)	6.976 (−1.25)	7.348 (−0.14)	7.671 (0.39)	7.704 (0.83)	7.744 (1.35)	7.759 (1.54)	7.806 (2.16)
20%	6.176 (1.44)	6.542 (1.77)	6.857 (1.84)	7.225 (2.28)	7.471 (1.54)	7.847 (2.70)	7.916 (3.60)	7.973 (4.35)	8.077 (5.71)	8.119 (6.26)
30%	6.337 (4.09)	6.743 (4.89)	7.056 (4.79)	7.404 (4.81)	7.699 (4.63)	8.043 (5.27)	8.168 (6.90)	8.267 (8.20)	8.384 (9.72)	8.475 (10.92)
40%	6.577 (8.03)	6.941 (7.96)	7.289 (8.26)	7.639 (8.14)	7.966 (8.26)	8.310 (8.76)	8.452 (10.61)	8.603 (12.59)	8.743 (14.43)	8.902 (16.51)
Imputation using propensity−score method										
10%	6.049 (−0.64)	6.381 (−0.74)	6.706 (−0.41)	6.975 (−1.26)	7.337 (−0.29)	7.651 (0.13)	7.677 (0.48)	7.713 (0.94)	7.735 (1.23)	7.773 (1.72)
20%	6.194 (1.73)	6.565 (2.13)	6.877 (2.14)	7.240 (2.49)	7.506 (2.01)	7.844 (2.66)	7.919 (3.63)	7.967 (4.27)	8.075 (5.68)	8.136 (6.48)
30%	6.362 (4.51)	6.746 (4.94)	7.072 (5.03)	7.391 (4.63)	7.683 (4.41)	7.992 (4.59)	8.100 (6.01)	8.223 (7.62)	8.300 (8.62)	8.403 (9.97)
40%	6.581 (8.10)	6.958 (8.23)	7.260 (7.83)	7.592 (7.48)	7.928 (7.74)	8.180 (7.06)	8.355 (9.35)	8.471 (10.87)	8.603 (12.59)	8.775 (14.85)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.625	7.625	7.625	7.625	7.625
Imputation using linear regression										

table3_pooled_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.244 (2.20)	6.632 (2.82)	7.011 (3.79)	7.324 (3.96)	7.629 (3.34)	8.059 (5.69)	8.125 (6.55)	8.200 (7.53)	8.229 (7.91)	8.285 (8.66)
20%	6.773 (10.86)	7.202 (11.65)	7.614 (12.71)	8.070 (14.54)	8.401 (13.80)	8.793 (15.31)	8.933 (17.15)	9.083 (19.12)	9.260 (21.44)	9.410 (23.40)
30%	7.539 (23.39)	8.140 (26.20)	8.584 (27.07)	9.081 (28.91)	9.591 (29.91)	9.843 (29.09)	10.199 (33.75)	10.413 (36.55)	10.724 (40.63)	11.034 (44.70)
40%	8.652 (41.62)	9.427 (46.14)	10.104 (49.57)	10.669 (51.44)	11.430 (54.83)	11.412 (49.65)	12.002 (57.40)	12.516 (64.14)	12.915 (69.37)	13.578 (78.06)
Imputation using predictive mean matching method										
10%	5.989 (−1.98)	6.337 (−1.76)	6.677 (−1.15)	6.982 (−0.90)	7.245 (−1.86)	7.627 (0.03)	7.666 (0.53)	7.702 (1.01)	7.732 (1.40)	7.750 (1.63)
20%	6.169 (0.96)	6.515 (1.00)	6.837 (1.21)	7.197 (2.15)	7.449 (0.90)	7.827 (2.64)	7.891 (3.48)	7.952 (4.28)	8.039 (5.42)	8.106 (6.31)
30%	6.333 (3.66)	6.700 (3.86)	7.026 (4.01)	7.352 (4.35)	7.670 (3.90)	8.008 (5.02)	8.122 (6.51)	8.226 (7.88)	8.340 (9.37)	8.432 (10.58)
40%	6.539 (7.03)	6.933 (7.48)	7.301 (8.09)	7.619 (8.14)	7.980 (8.09)	8.278 (8.56)	8.443 (10.72)	8.597 (12.74)	8.724 (14.40)	8.905 (16.78)
Imputation using propensity−score method										
10%	5.989 (−1.97)	6.339 (−1.74)	6.669 (−1.27)	6.981 (−0.91)	7.239 (−1.95)	7.598 (−0.36)	7.637 (0.16)	7.669 (0.57)	7.710 (1.11)	7.721 (1.25)
20%	6.190 (1.31)	6.539 (1.38)	6.846 (1.35)	7.203 (2.24)	7.451 (0.93)	7.806 (2.37)	7.878 (3.31)	7.929 (3.99)	8.010 (5.05)	8.078 (5.94)
30%	6.411 (4.93)	6.742 (4.52)	7.088 (4.93)	7.376 (4.69)	7.705 (4.36)	8.012 (5.06)	8.095 (6.16)	8.221 (7.81)	8.307 (8.93)	8.418 (10.40)
40%	6.545 (7.12)	6.924 (7.33)	7.294 (7.98)	7.597 (7.83)	7.933 (7.46)	8.137 (6.71)	8.298 (8.83)	8.475 (11.14)	8.596 (12.72)	8.760 (14.88)

table3_pooled_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.641	7.641	7.641	7.641	7.641
Imputation using linear regression										
10%	6.409 (5.28)	6.787 (5.57)	7.139 (6.02)	7.433 (5.23)	7.836 (6.49)	8.213 (7.48)	8.278 (8.34)	8.336 (9.10)	8.386 (9.76)	8.462 (10.74)
20%	7.120 (16.95)	7.568 (17.73)	7.930 (17.77)	8.406 (19.00)	8.674 (17.88)	9.171 (20.03)	9.318 (21.95)	9.430 (23.41)	9.655 (26.36)	9.722 (27.24)
30%	8.382 (37.69)	8.884 (38.19)	9.377 (39.27)	9.826 (39.11)	10.237 (39.12)	10.705 (40.10)	10.960 (43.45)	11.249 (47.23)	11.500 (50.50)	11.734 (53.56)
40%	10.089 (65.72)	10.704 (66.50)	11.338 (68.40)	11.845 (67.68)	12.479 (69.60)	12.856 (68.26)	13.294 (73.99)	13.805 (80.67)	14.153 (85.23)	14.711 (92.53)
Imputation using predictive mean matching method										
10%	5.942 (−2.39)	6.288 (−2.18)	6.620 (−1.68)	6.884 (−2.55)	7.267 (−1.24)	7.623 (−0.24)	7.657 (0.21)	7.691 (0.66)	7.706 (0.85)	7.760 (1.56)
20%	6.030 (−0.96)	6.405 (−0.37)	6.730 (−0.05)	7.100 (0.50)	7.352 (−0.08)	7.780 (1.82)	7.850 (2.74)	7.912 (3.55)	8.014 (4.88)	8.056 (5.43)
30%	6.219 (2.16)	6.625 (3.05)	6.953 (3.26)	7.302 (3.37)	7.596 (3.23)	7.993 (4.61)	8.111 (6.16)	8.223 (7.62)	8.330 (9.02)	8.419 (10.18)
40%	6.394 (5.03)	6.771 (5.33)	7.130 (5.89)	7.491 (6.05)	7.808 (6.11)	8.209 (7.44)	8.349 (9.27)	8.509 (11.36)	8.657 (13.30)	8.793 (15.08)
Imputation using propensity−score method										
10%	5.933 (−2.54)	6.274 (−2.40)	6.605 (−1.90)	6.874 (−2.70)	7.245 (−1.55)	7.587 (−0.71)	7.615 (−0.34)	7.651 (0.13)	7.671 (0.39)	7.710 (0.91)
20%	6.034 (−0.89)	6.417 (−0.18)	6.740 (0.10)	7.102 (0.54)	7.375 (0.22)	7.770 (1.69)	7.846 (2.69)	7.901 (3.40)	8.000 (4.70)	8.062 (5.51)
30%	6.115 (0.45)	6.516 (1.36)	6.848 (1.71)	7.187 (1.75)	7.467 (1.48)	7.834 (2.53)	7.949 (4.03)	8.070 (5.61)	8.157 (6.75)	8.235 (7.78)
40%	6.277 (3.11)	6.671 (3.77)	6.982 (3.70)	7.315 (3.55)	7.693 (4.55)	7.974 (4.36)	8.144 (6.58)	8.263 (8.14)	8.382 (9.70)	8.591 (12.44)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.625	7.625	7.625	7.625	7.625
Imputation using linear regression										

table3_pooled_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.355 (4.02)	6.743 (4.54)	7.116 (5.34)	7.426 (5.40)	7.725 (4.64)	8.166 (7.09)	8.237 (8.02)	8.307 (8.94)	8.337 (9.33)	8.390 (10.02)
20%	7.165 (17.27)	7.577 (17.46)	7.965 (17.91)	8.395 (19.16)	8.742 (18.42)	9.188 (20.50)	9.322 (22.25)	9.459 (24.04)	9.619 (26.15)	9.792 (28.41)
30%	8.271 (35.38)	8.820 (36.73)	9.269 (37.22)	9.759 (38.52)	10.217 (38.39)	10.601 (39.02)	10.918 (43.17)	11.158 (46.33)	11.459 (50.27)	11.719 (53.68)
40%	10.174 (66.53)	10.859 (68.34)	11.471 (69.81)	11.869 (68.48)	13.195 (78.73)	13.034 (70.92)	13.550 (77.69)	14.013 (83.77)	14.240 (86.75)	15.581 (104.33)
Imputation using predictive mean matching method										
10%	5.899 (−3.45)	6.250 (−3.10)	6.599 (−2.31)	6.903 (−2.01)	7.170 (−2.87)	7.582 (−0.57)	7.617 (−0.11)	7.660 (0.45)	7.687 (0.81)	7.704 (1.03)
20%	6.040 (−1.14)	6.397 (−0.83)	6.730 (−0.37)	7.089 (0.62)	7.341 (−0.56)	7.765 (1.83)	7.834 (2.73)	7.904 (3.65)	7.984 (4.70)	8.043 (5.48)
30%	6.181 (1.17)	6.567 (1.81)	6.888 (1.97)	7.218 (2.45)	7.542 (2.16)	7.937 (4.08)	8.064 (5.75)	8.147 (6.84)	8.261 (8.34)	8.354 (9.56)
40%	6.400 (4.76)	6.813 (5.62)	7.185 (6.36)	7.489 (6.31)	7.837 (6.16)	8.217 (7.76)	8.395 (10.10)	8.546 (12.08)	8.645 (13.37)	8.806 (15.48)
Imputation using propensity−score method										
10%	5.885 (−3.67)	6.238 (−3.29)	6.575 (−2.66)	6.884 (−2.29)	7.155 (−3.08)	7.544 (−1.07)	7.579 (−0.60)	7.613 (−0.16)	7.647 (0.29)	7.668 (0.56)
20%	6.069 (−0.66)	6.423 (−0.43)	6.733 (−0.32)	7.112 (0.95)	7.361 (−0.29)	7.763 (1.81)	7.830 (2.68)	7.882 (3.36)	7.978 (4.62)	8.047 (5.53)
30%	6.136 (0.42)	6.477 (0.41)	6.851 (1.41)	7.141 (1.36)	7.485 (1.38)	7.838 (2.79)	7.919 (3.85)	8.055 (5.63)	8.134 (6.67)	8.254 (8.25)
40%	6.249 (2.28)	6.656 (3.19)	7.016 (3.86)	7.353 (4.38)	7.685 (4.10)	7.929 (3.98)	8.112 (6.38)	8.253 (8.23)	8.406 (10.23)	8.553 (12.17)

table3_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.802	10.802	10.802	10.802	10.802
Imputation using linear regression										
10%	8.882 (2.44)	9.603 (3.54)	10.267 (4.38)	10.824 (3.84)	11.540 (5.35)	11.433 (5.84)	11.554 (6.96)	11.670 (8.03)	11.759 (8.86)	11.917 (10.33)
20%	9.541 (10.04)	10.415 (12.30)	11.157 (13.44)	12.000 (15.12)	12.511 (14.21)	12.404 (14.83)	12.711 (17.67)	12.976 (20.13)	13.363 (23.71)	13.492 (24.90)
30%	10.541 (21.59)	11.616 (25.25)	12.559 (27.69)	13.436 (28.90)	14.186 (29.50)	13.750 (27.30)	14.340 (32.75)	14.883 (37.78)	15.384 (42.42)	15.805 (46.31)
40%	12.021 (38.65)	13.371 (44.17)	14.611 (48.55)	15.655 (50.18)	16.809 (53.45)	15.767 (45.97)	16.734 (54.92)	17.702 (63.88)	18.455 (70.85)	19.458 (80.14)
Imputation using predictive mean matching method										
10%	8.548 (−1.41)	9.194 (−0.87)	9.802 (−0.34)	10.299 (−1.20)	10.940 (−0.13)	10.849 (0.44)	10.913 (1.03)	10.984 (1.69)	11.020 (2.02)	11.103 (2.79)
20%	8.745 (0.87)	9.422 (1.60)	10.014 (1.81)	10.661 (2.28)	11.126 (1.56)	11.094 (2.70)	11.226 (3.92)	11.339 (4.97)	11.518 (6.63)	11.606 (7.44)
30%	8.979 (3.56)	9.706 (4.66)	10.303 (4.75)	10.926 (4.82)	11.464 (4.65)	11.365 (5.22)	11.590 (7.30)	11.783 (9.08)	11.995 (11.04)	12.169 (12.66)
40%	9.297 (7.24)	9.999 (7.81)	10.644 (8.22)	11.275 (8.16)	11.862 (8.28)	11.726 (8.56)	12.003 (11.11)	12.279 (13.68)	12.542 (16.11)	12.830 (18.77)
Imputation using propensity−score method										
10%	8.563 (−1.24)	9.196 (−0.84)	9.793 (−0.43)	10.299 (−1.20)	10.924 (−0.28)	10.824 (0.20)	10.877 (0.70)	10.940 (1.28)	10.985 (1.70)	11.053 (2.32)
20%	8.778 (1.25)	9.459 (1.99)	10.043 (2.11)	10.684 (2.50)	11.174 (2.01)	11.093 (2.69)	11.230 (3.96)	11.329 (4.88)	11.514 (6.60)	11.628 (7.64)
30%	9.033 (4.19)	9.725 (4.86)	10.332 (5.05)	10.912 (4.68)	11.442 (4.45)	11.303 (4.63)	11.500 (6.47)	11.719 (8.49)	11.875 (9.93)	12.063 (11.67)
40%	9.337 (7.69)	10.034 (8.19)	10.614 (7.91)	11.211 (7.55)	11.811 (7.81)	11.562 (7.03)	11.867 (9.86)	12.099 (12.01)	12.345 (14.29)	12.656 (17.16)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.783	10.783	10.783	10.783	10.783
Imputation using linear regression										

table3_pooled_5_treatments_R9_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.793 (0.96)	9.523 (2.29)	10.216 (3.51)	10.800 (3.85)	11.356 (3.32)	11.360 (5.35)	11.483 (6.49)	11.618 (7.74)	11.687 (8.38)	11.793 (9.37)
20%	9.484 (8.89)	10.315 (10.79)	11.081 (12.27)	11.885 (14.28)	12.500 (13.73)	12.330 (14.34)	12.608 (16.92)	12.890 (19.54)	13.206 (22.47)	13.482 (25.03)
30%	10.499 (20.55)	11.602 (24.62)	12.466 (26.30)	13.366 (28.51)	14.262 (29.76)	13.720 (27.23)	14.349 (33.07)	14.796 (37.21)	15.357 (42.42)	15.913 (47.57)
40%	11.909 (36.74)	13.357 (43.46)	14.617 (48.10)	15.675 (50.72)	16.986 (54.54)	15.729 (45.86)	16.813 (55.92)	17.778 (64.87)	18.549 (72.02)	19.686 (82.56)
Imputation using predictive mean matching method										
10%	8.474 (−2.70)	9.126 (−1.98)	9.745 (−1.26)	10.303 (−0.94)	10.788 (−1.85)	10.787 (0.04)	10.855 (0.67)	10.923 (1.29)	10.981 (1.83)	11.020 (2.19)
20%	8.725 (0.17)	9.384 (0.79)	9.983 (1.15)	10.619 (2.10)	11.092 (0.92)	11.060 (2.56)	11.185 (3.73)	11.305 (4.83)	11.458 (6.26)	11.584 (7.43)
30%	8.960 (2.88)	9.648 (3.63)	10.261 (3.96)	10.852 (4.35)	11.421 (3.91)	11.311 (4.89)	11.520 (6.83)	11.719 (8.68)	11.929 (10.62)	12.103 (12.24)
40%	9.266 (6.39)	9.988 (7.28)	10.660 (8.00)	11.246 (8.13)	11.881 (8.10)	11.699 (8.49)	11.993 (11.22)	12.277 (13.85)	12.519 (16.10)	12.839 (19.06)
Imputation using propensity−score method										
10%	8.483 (−2.60)	9.131 (−1.92)	9.736 (−1.36)	10.301 (−0.95)	10.777 (−1.95)	10.750 (−0.31)	10.816 (0.31)	10.874 (0.84)	10.947 (1.52)	10.974 (1.77)
20%	8.763 (0.61)	9.423 (1.21)	10.000 (1.32)	10.627 (2.18)	11.097 (0.97)	11.033 (2.32)	11.167 (3.55)	11.272 (4.53)	11.412 (5.83)	11.545 (7.07)
30%	9.090 (4.37)	9.728 (4.49)	10.356 (4.92)	10.886 (4.67)	11.464 (4.30)	11.321 (4.99)	11.496 (6.61)	11.715 (8.64)	11.876 (10.13)	12.076 (11.99)
40%	9.317 (6.97)	10.001 (7.42)	10.664 (8.04)	11.209 (7.77)	11.803 (7.38)	11.526 (6.88)	11.808 (9.50)	12.118 (12.37)	12.333 (14.37)	12.621 (17.04)

table3_pooled_5_treatments_R9_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.802	10.802	10.802	10.802	10.802
Imputation using linear regression										
10%	9.044 (4.32)	9.761 (5.25)	10.415 (5.89)	10.969 (5.23)	11.665 (6.48)	11.600 (7.38)	11.720 (8.50)	11.832 (9.54)	11.930 (10.44)	12.064 (11.68)
20%	10.028 (15.67)	10.869 (17.19)	11.565 (17.59)	12.394 (18.90)	12.913 (17.88)	12.915 (19.56)	13.201 (22.21)	13.431 (24.34)	13.815 (27.89)	13.966 (29.30)
30%	11.811 (36.23)	12.770 (37.69)	13.675 (39.04)	14.496 (39.06)	15.241 (39.13)	15.049 (39.31)	15.557 (44.02)	16.085 (48.90)	16.553 (53.24)	16.988 (57.27)
40%	14.161 (63.34)	15.366 (65.68)	16.525 (68.01)	17.468 (67.58)	18.576 (69.57)	17.996 (66.60)	18.860 (74.60)	19.778 (83.09)	20.448 (89.30)	21.412 (98.22)
Imputation using predictive mean matching method										
10%	8.388 (−3.25)	9.047 (−2.45)	9.658 (−1.81)	10.159 (−2.54)	10.818 (−1.25)	10.773 (−0.27)	10.837 (0.33)	10.902 (0.92)	10.937 (1.25)	11.030 (2.11)
20%	8.513 (−1.81)	9.207 (−0.72)	9.817 (−0.19)	10.470 (0.44)	10.945 (−0.08)	10.987 (1.71)	11.120 (2.95)	11.241 (4.07)	11.418 (5.70)	11.506 (6.52)
30%	8.792 (1.41)	9.523 (2.68)	10.144 (3.14)	10.772 (3.34)	11.309 (3.24)	11.286 (4.48)	11.502 (6.48)	11.711 (8.42)	11.911 (10.27)	12.080 (11.83)
40%	9.008 (3.90)	9.733 (4.95)	10.399 (5.73)	11.049 (6.00)	11.623 (6.10)	11.566 (7.08)	11.841 (9.62)	12.132 (12.31)	12.406 (14.85)	12.660 (17.20)
Imputation using propensity−score method										
10%	8.379 (−3.35)	9.029 (−2.64)	9.638 (−2.01)	10.144 (−2.69)	10.784 (−1.55)	10.723 (−0.73)	10.778 (−0.22)	10.842 (0.37)	10.884 (0.76)	10.953 (1.40)
20%	8.525 (−1.68)	9.226 (−0.52)	9.834 (−0.02)	10.476 (0.50)	10.980 (0.23)	10.975 (1.61)	11.115 (2.89)	11.226 (3.92)	11.397 (5.51)	11.516 (6.61)
30%	8.636 (−0.39)	9.366 (0.98)	9.988 (1.54)	10.603 (1.71)	11.122 (1.53)	11.049 (2.29)	11.260 (4.24)	11.476 (6.24)	11.650 (7.85)	11.808 (9.32)
40%	8.849 (2.06)	9.585 (3.36)	10.185 (3.55)	10.796 (3.56)	11.445 (4.48)	11.228 (3.94)	11.533 (6.77)	11.769 (8.95)	12.006 (11.15)	12.353 (14.36)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.783	10.783	10.783	10.783	10.783
Imputation using linear regression										

table3_pooled_5_treatments_R9_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.970 (2.99)	9.695 (4.13)	10.377 (5.14)	10.954 (5.33)	11.501 (4.64)	11.530 (6.93)	11.658 (8.11)	11.787 (9.31)	11.857 (9.96)	11.956 (10.88)
20%	10.097 (15.93)	10.894 (17.02)	11.620 (17.73)	12.381 (19.05)	13.014 (18.41)	12.948 (20.08)	13.216 (22.56)	13.476 (24.97)	13.765 (27.65)	14.073 (30.50)
30%	11.658 (33.86)	12.667 (36.06)	13.517 (36.95)	14.393 (38.39)	15.207 (38.36)	14.917 (38.34)	15.485 (43.60)	15.951 (47.93)	16.489 (52.91)	16.964 (57.32)
40%	14.310 (64.31)	15.584 (67.39)	16.719 (69.39)	17.511 (68.37)	19.632 (78.62)	18.277 (69.49)	19.226 (78.29)	20.092 (86.32)	20.591 (90.96)	22.707 (110.57)
Imputation using predictive mean matching method										
10%	8.332 (−4.34)	8.991 (−3.43)	9.625 (−2.48)	10.183 (−2.08)	10.675 (−2.87)	10.715 (−0.63)	10.780 (−0.03)	10.855 (0.67)	10.910 (1.18)	10.947 (1.52)
20%	8.521 (−2.16)	9.199 (−1.19)	9.818 (−0.52)	10.454 (0.52)	10.929 (−0.56)	10.963 (1.67)	11.095 (2.89)	11.226 (4.11)	11.370 (5.44)	11.485 (6.50)
30%	8.717 (0.09)	9.438 (1.37)	10.048 (1.80)	10.648 (2.38)	11.227 (2.14)	11.195 (3.82)	11.424 (5.94)	11.595 (7.52)	11.805 (9.48)	11.981 (11.11)
40%	9.039 (3.79)	9.795 (5.21)	10.478 (6.16)	11.049 (6.24)	11.667 (6.15)	11.594 (7.52)	11.911 (10.46)	12.194 (13.08)	12.397 (14.96)	12.685 (17.64)
Imputation using propensity−score method										
10%	8.317 (−4.50)	8.975 (−3.60)	9.592 (−2.82)	10.154 (−2.37)	10.650 (−3.11)	10.664 (−1.11)	10.726 (−0.53)	10.787 (0.03)	10.849 (0.61)	10.891 (1.00)
20%	8.567 (−1.63)	9.240 (−0.75)	9.828 (−0.42)	10.487 (0.84)	10.957 (−0.31)	10.963 (1.66)	11.091 (2.85)	11.197 (3.83)	11.358 (5.33)	11.487 (6.53)
30%	8.649 (−0.69)	9.316 (0.07)	9.991 (1.23)	10.533 (1.28)	11.134 (1.30)	11.043 (2.41)	11.218 (4.03)	11.456 (6.24)	11.612 (7.68)	11.822 (9.63)
40%	8.846 (1.56)	9.580 (2.89)	10.236 (3.71)	10.840 (4.23)	11.411 (3.82)	11.192 (3.79)	11.505 (6.69)	11.769 (9.14)	12.037 (11.63)	12.277 (13.85)

table3_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.908	5.957	6.005	6.072	6.139
Imputation using linear regression										
10%	4.908 (4.68)	5.063 (4.40)	5.220 (4.84)	5.339 (3.90)	5.558 (5.41)	6.282 (6.33)	6.369 (6.92)	6.465 (7.66)	6.537 (7.66)	6.685 (8.89)
20%	5.332 (13.73)	5.541 (14.25)	5.696 (14.40)	5.945 (15.69)	6.030 (14.35)	6.862 (16.14)	7.014 (17.75)	7.135 (18.82)	7.357 (21.17)	7.413 (20.75)
30%	5.967 (27.27)	6.230 (28.46)	6.452 (29.58)	6.670 (29.80)	6.845 (29.81)	7.689 (30.14)	7.927 (33.07)	8.142 (35.58)	8.352 (37.56)	8.521 (38.80)
40%	7.000 (49.31)	7.264 (49.79)	7.559 (51.81)	7.799 (51.77)	8.125 (54.09)	9.026 (52.76)	9.305 (56.21)	9.631 (60.38)	9.880 (62.72)	10.259 (67.11)
Imputation using predictive mean matching method										
10%	4.674 (−0.30)	4.817 (−0.67)	4.966 (−0.27)	5.069 (−1.35)	5.264 (−0.16)	5.955 (0.78)	6.017 (1.01)	6.098 (1.55)	6.149 (1.27)	6.259 (1.96)
20%	4.770 (1.73)	4.939 (1.83)	5.070 (1.83)	5.255 (2.27)	5.352 (1.49)	6.085 (3.00)	6.176 (3.68)	6.256 (4.17)	6.391 (5.26)	6.456 (5.16)
30%	4.891 (4.31)	5.094 (5.03)	5.217 (4.78)	5.383 (4.76)	5.515 (4.59)	6.228 (5.42)	6.368 (6.91)	6.460 (7.57)	6.591 (8.55)	6.695 (9.05)
40%	5.088 (8.53)	5.236 (7.97)	5.390 (8.24)	5.553 (8.07)	5.706 (8.21)	6.452 (9.21)	6.566 (10.22)	6.703 (11.62)	6.838 (12.63)	6.986 (13.80)
Imputation using propensity−score method										
10%	4.676 (−0.27)	4.812 (−0.77)	4.958 (−0.43)	5.068 (−1.37)	5.256 (−0.31)	5.945 (0.63)	6.002 (0.77)	6.083 (1.30)	6.141 (1.14)	6.247 (1.75)
20%	4.777 (1.88)	4.954 (2.14)	5.083 (2.07)	5.264 (2.44)	5.376 (1.96)	6.080 (2.90)	6.186 (3.84)	6.264 (4.31)	6.406 (5.50)	6.489 (5.69)
30%	4.896 (4.42)	5.085 (4.84)	5.224 (4.90)	5.371 (4.51)	5.500 (4.30)	6.197 (4.89)	6.331 (6.28)	6.449 (7.39)	6.560 (8.04)	6.667 (8.60)
40%	5.065 (8.04)	5.240 (8.05)	5.355 (7.54)	5.517 (7.36)	5.678 (7.69)	6.372 (7.85)	6.525 (9.54)	6.628 (10.37)	6.770 (11.50)	6.929 (12.88)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.904	5.956	6.005	6.057	6.129
Imputation using linear regression										

table3_shrink_5_treatments_R2_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.859 (3.40)	5.028 (3.40)	5.202 (4.16)	5.333 (4.13)	5.468 (3.37)	6.244 (5.76)	6.336 (6.38)	6.440 (7.23)	6.516 (7.58)	6.606 (7.79)
20%	5.307 (12.94)	5.480 (12.70)	5.659 (13.32)	5.887 (14.95)	6.026 (13.92)	6.828 (15.65)	6.948 (16.65)	7.092 (18.09)	7.279 (20.18)	7.412 (20.94)
30%	5.942 (26.45)	6.234 (28.19)	6.401 (28.16)	6.633 (29.52)	6.885 (30.16)	7.655 (29.66)	7.936 (33.24)	8.082 (34.58)	8.320 (37.37)	8.570 (39.83)
40%	6.904 (46.94)	7.274 (49.59)	7.576 (51.68)	7.814 (52.58)	8.216 (55.32)	8.930 (51.25)	9.328 (56.61)	9.657 (60.81)	9.909 (63.59)	10.373 (69.26)
Imputation using predictive mean matching method										
10%	4.632 (−1.43)	4.785 (−1.60)	4.942 (−1.05)	5.077 (−0.86)	5.190 (−1.89)	5.918 (0.24)	5.991 (0.59)	6.067 (1.02)	6.140 (1.38)	6.202 (1.20)
20%	4.772 (1.56)	4.917 (1.13)	5.056 (1.23)	5.234 (2.20)	5.336 (0.87)	6.083 (3.03)	6.158 (3.39)	6.241 (3.93)	6.366 (5.10)	6.443 (5.13)
30%	4.897 (4.22)	5.058 (4.01)	5.196 (4.03)	5.344 (4.35)	5.495 (3.87)	6.217 (5.30)	6.328 (6.23)	6.430 (7.06)	6.550 (8.14)	6.660 (8.67)
40%	5.045 (7.37)	5.230 (7.56)	5.401 (8.13)	5.538 (8.13)	5.716 (8.07)	6.411 (8.58)	6.562 (10.17)	6.708 (11.70)	6.824 (12.67)	6.989 (14.05)
Imputation using propensity−score method										
10%	4.626 (−1.56)	4.783 (−1.63)	4.933 (−1.22)	5.076 (−0.88)	5.187 (−1.94)	5.898 (−0.10)	5.979 (0.38)	6.050 (0.74)	6.134 (1.27)	6.194 (1.07)
20%	4.782 (1.78)	4.932 (1.43)	5.059 (1.30)	5.238 (2.28)	5.341 (0.97)	6.075 (2.90)	6.159 (3.40)	6.239 (3.89)	6.364 (5.07)	6.447 (5.20)
30%	4.941 (5.15)	5.074 (4.34)	5.237 (4.86)	5.360 (4.65)	5.523 (4.40)	6.234 (5.58)	6.319 (6.08)	6.457 (7.52)	6.555 (8.22)	6.684 (9.06)
40%	5.017 (6.77)	5.203 (7.00)	5.386 (7.84)	5.523 (7.85)	5.694 (7.65)	6.320 (7.05)	6.481 (8.80)	6.652 (10.77)	6.774 (11.84)	6.939 (13.21)

table3_shrink_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	4.689	4.850	4.980	5.139	5.273	5.908	5.957	6.005	6.072	6.139
Imputation using linear regression										
10%	4.973 (6.08)	5.133 (5.84)	5.286 (6.16)	5.406 (5.19)	5.616 (6.50)	6.355 (7.57)	6.447 (8.24)	6.540 (8.90)	6.615 (8.95)	6.755 (10.03)
20%	5.534 (18.03)	5.735 (18.26)	5.874 (17.97)	6.122 (19.14)	6.216 (17.88)	7.084 (19.91)	7.230 (21.37)	7.334 (22.13)	7.555 (24.42)	7.625 (24.20)
30%	6.508 (38.81)	6.721 (38.59)	6.947 (39.51)	7.150 (39.14)	7.335 (39.10)	8.269 (39.96)	8.454 (41.92)	8.675 (44.46)	8.873 (46.13)	9.057 (47.52)
40%	7.865 (67.75)	8.111 (67.24)	8.407 (68.83)	8.623 (67.80)	8.945 (69.63)	9.961 (68.60)	10.223 (71.62)	10.554 (75.75)	10.774 (77.45)	11.150 (81.63)
Imputation using predictive mean matching method										
10%	4.610 (−1.68)	4.755 (−1.96)	4.902 (−1.55)	5.006 (−2.59)	5.208 (−1.23)	5.904 (−0.06)	5.966 (0.16)	6.043 (0.63)	6.093 (0.35)	6.208 (1.12)
20%	4.675 (−0.28)	4.848 (−0.03)	4.984 (0.10)	5.169 (0.58)	5.268 (−0.09)	6.011 (1.73)	6.102 (2.45)	6.184 (2.98)	6.316 (4.02)	6.383 (3.97)
30%	4.814 (2.67)	5.014 (3.39)	5.148 (3.39)	5.313 (3.40)	5.443 (3.23)	6.163 (4.31)	6.299 (5.75)	6.400 (6.58)	6.527 (7.50)	6.628 (7.97)
40%	4.970 (6.00)	5.125 (5.67)	5.282 (6.07)	5.452 (6.10)	5.595 (6.11)	6.346 (7.41)	6.462 (8.49)	6.604 (9.98)	6.745 (11.09)	6.878 (12.04)
Imputation using propensity−score method										
10%	4.599 (−1.90)	4.741 (−2.23)	4.890 (−1.80)	4.998 (−2.74)	5.191 (−1.55)	5.881 (−0.46)	5.942 (−0.25)	6.021 (0.26)	6.077 (0.09)	6.183 (0.71)
20%	4.673 (−0.33)	4.856 (0.12)	4.990 (0.21)	5.170 (0.62)	5.285 (0.23)	5.996 (1.49)	6.102 (2.45)	6.183 (2.96)	6.320 (4.09)	6.404 (4.32)
30%	4.738 (1.06)	4.934 (1.74)	5.073 (1.87)	5.232 (1.82)	5.350 (1.46)	6.053 (2.44)	6.191 (3.94)	6.306 (5.02)	6.428 (5.87)	6.520 (6.21)
40%	4.871 (3.88)	5.051 (4.16)	5.166 (3.75)	5.322 (3.57)	5.515 (4.58)	6.186 (4.71)	6.342 (6.47)	6.441 (7.27)	6.573 (8.26)	6.771 (10.29)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	4.699	4.863	4.994	5.121	5.290	5.904	5.956	6.005	6.057	6.129
Imputation using linear regression										

table3_shrink_5_treatments_R2_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	4.931 (4.93)	5.103 (4.94)	5.273 (5.59)	5.403 (5.50)	5.535 (4.64)	6.322 (7.08)	6.419 (7.77)	6.521 (8.59)	6.596 (8.90)	6.683 (9.04)
20%	5.566 (18.45)	5.732 (17.87)	5.898 (18.09)	6.110 (19.31)	6.265 (18.43)	7.105 (20.33)	7.219 (21.20)	7.355 (22.47)	7.526 (24.26)	7.677 (25.26)
30%	6.421 (36.65)	6.682 (37.40)	6.869 (37.53)	7.103 (38.69)	7.323 (38.45)	8.174 (38.45)	8.423 (41.41)	8.598 (43.17)	8.837 (45.91)	9.049 (47.66)
40%	7.909 (68.31)	8.231 (69.26)	8.505 (70.29)	8.635 (68.61)	9.464 (78.91)	10.030 (69.89)	10.373 (74.16)	10.669 (77.65)	10.790 (78.15)	11.798 (92.50)
Imputation using predictive mean matching method										
10%	4.573 (−2.69)	4.727 (−2.79)	4.889 (−2.11)	5.023 (−1.91)	5.138 (−2.87)	5.873 (−0.53)	5.942 (−0.25)	6.019 (0.22)	6.092 (0.58)	6.154 (0.41)
20%	4.688 (−0.23)	4.840 (−0.47)	4.985 (−0.19)	5.160 (0.76)	5.260 (−0.56)	6.014 (1.85)	6.093 (2.30)	6.186 (3.00)	6.300 (4.01)	6.375 (4.01)
30%	4.800 (2.14)	4.972 (2.25)	5.103 (2.17)	5.252 (2.54)	5.405 (2.17)	6.141 (4.02)	6.265 (5.18)	6.348 (5.71)	6.472 (6.85)	6.582 (7.40)
40%	4.958 (5.51)	5.154 (5.99)	5.324 (6.59)	5.449 (6.39)	5.616 (6.18)	6.341 (7.39)	6.501 (9.15)	6.642 (10.60)	6.741 (11.29)	6.891 (12.44)
Imputation using propensity−score method										
10%	4.559 (−2.99)	4.716 (−3.02)	4.870 (−2.49)	5.010 (−2.18)	5.129 (−3.05)	5.845 (−1.00)	5.920 (−0.62)	5.992 (−0.22)	6.072 (0.25)	6.140 (0.18)
20%	4.706 (0.15)	4.855 (−0.17)	4.982 (−0.25)	5.176 (1.07)	5.276 (−0.25)	6.017 (1.91)	6.095 (2.33)	6.180 (2.90)	6.316 (4.28)	6.395 (4.35)
30%	4.765 (1.40)	4.896 (0.69)	5.076 (1.63)	5.199 (1.52)	5.372 (1.56)	6.077 (2.93)	6.161 (3.44)	6.307 (5.02)	6.408 (5.80)	6.544 (6.78)
40%	4.826 (2.71)	5.028 (3.40)	5.196 (4.04)	5.354 (4.53)	5.521 (4.37)	6.136 (3.93)	6.313 (5.99)	6.460 (7.56)	6.603 (9.02)	6.765 (10.38)

table3_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.564	7.779	7.987	8.225	8.459
Imputation using linear regression										
10%	6.307 (3.60)	6.684 (3.97)	7.042 (4.59)	7.338 (3.87)	7.754 (5.37)	8.034 (6.22)	8.309 (6.82)	8.592 (7.57)	8.835 (7.41)	9.200 (8.76)
20%	6.811 (11.87)	7.276 (13.19)	7.665 (13.85)	8.148 (15.34)	8.408 (14.26)	8.711 (15.16)	9.103 (17.02)	9.452 (18.34)	9.922 (20.63)	10.159 (20.10)
30%	7.570 (24.34)	8.144 (26.68)	8.650 (28.47)	9.130 (29.25)	9.537 (29.61)	9.690 (28.11)	10.241 (31.66)	10.754 (34.65)	11.254 (36.83)	11.683 (38.11)
40%	8.750 (43.73)	9.427 (46.64)	10.092 (49.89)	10.653 (50.80)	11.309 (53.69)	11.217 (48.30)	11.939 (53.48)	12.680 (58.76)	13.295 (61.63)	14.065 (66.27)
Imputation using predictive mean matching method										
10%	6.042 (−0.75)	6.383 (−0.72)	6.714 (−0.29)	6.976 (−1.25)	7.348 (−0.14)	7.655 (1.21)	7.890 (1.43)	8.138 (1.90)	8.342 (1.42)	8.649 (2.25)
20%	6.176 (1.44)	6.542 (1.77)	6.857 (1.84)	7.225 (2.28)	7.471 (1.54)	7.821 (3.40)	8.093 (4.04)	8.345 (4.48)	8.670 (5.41)	8.886 (5.05)
30%	6.337 (4.09)	6.743 (4.89)	7.056 (4.79)	7.404 (4.81)	7.699 (4.63)	8.006 (5.85)	8.339 (7.21)	8.612 (7.83)	8.934 (8.62)	9.212 (8.90)
40%	6.577 (8.03)	6.941 (7.96)	7.289 (8.26)	7.639 (8.14)	7.966 (8.26)	8.279 (9.46)	8.597 (10.52)	8.928 (11.78)	9.262 (12.60)	9.602 (13.51)
Imputation using propensity−score method										
10%	6.049 (−0.64)	6.381 (−0.74)	6.706 (−0.41)	6.975 (−1.26)	7.337 (−0.29)	7.650 (1.14)	7.876 (1.26)	8.120 (1.67)	8.334 (1.32)	8.632 (2.04)
20%	6.194 (1.73)	6.565 (2.13)	6.877 (2.14)	7.240 (2.49)	7.506 (2.01)	7.825 (3.45)	8.109 (4.24)	8.360 (4.68)	8.688 (5.63)	8.928 (5.54)
30%	6.362 (4.51)	6.746 (4.94)	7.072 (5.03)	7.391 (4.63)	7.683 (4.41)	7.990 (5.64)	8.310 (6.83)	8.607 (7.77)	8.898 (8.18)	9.179 (8.51)
40%	6.581 (8.10)	6.958 (8.23)	7.260 (7.83)	7.592 (7.48)	7.928 (7.74)	8.215 (8.61)	8.562 (10.07)	8.852 (10.83)	9.175 (11.54)	9.528 (12.64)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.566	7.791	8.000	8.208	8.466
Imputation using linear regression										

table3_shrink_5_treatments_R4_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.244 (2.20)	6.632 (2.82)	7.011 (3.79)	7.324 (3.96)	7.629 (3.34)	7.975 (5.40)	8.256 (5.97)	8.554 (6.92)	8.806 (7.28)	9.076 (7.20)
20%	6.773 (10.86)	7.202 (11.65)	7.614 (12.71)	8.070 (14.54)	8.401 (13.80)	8.665 (14.52)	9.021 (15.78)	9.390 (17.37)	9.820 (19.63)	10.156 (19.96)
30%	7.539 (23.39)	8.140 (26.20)	8.584 (27.07)	9.081 (28.91)	9.591 (29.91)	9.654 (27.60)	10.242 (31.46)	10.680 (33.50)	11.208 (36.55)	11.750 (38.79)
40%	8.652 (41.62)	9.427 (46.14)	10.104 (49.57)	10.669 (51.44)	11.430 (54.83)	11.123 (47.00)	11.959 (53.50)	12.709 (58.86)	13.338 (62.50)	14.227 (68.05)
Imputation using predictive mean matching method										
10%	5.989 (−1.98)	6.337 (−1.76)	6.677 (−1.15)	6.982 (−0.90)	7.245 (−1.86)	7.603 (0.49)	7.845 (0.69)	8.094 (1.18)	8.332 (1.51)	8.551 (1.00)
20%	6.169 (0.96)	6.515 (1.00)	6.837 (1.21)	7.197 (2.15)	7.449 (0.90)	7.813 (3.26)	8.067 (3.54)	8.321 (4.01)	8.631 (5.15)	8.862 (4.67)
30%	6.333 (3.66)	6.700 (3.86)	7.026 (4.01)	7.352 (4.35)	7.670 (3.90)	7.990 (5.60)	8.288 (6.38)	8.575 (7.18)	8.878 (8.16)	9.172 (8.33)
40%	6.539 (7.03)	6.933 (7.48)	7.301 (8.09)	7.619 (8.14)	7.980 (8.09)	8.244 (8.96)	8.595 (10.32)	8.941 (11.76)	9.249 (12.68)	9.617 (13.60)
Imputation using propensity−score method										
10%	5.989 (−1.97)	6.339 (−1.74)	6.669 (−1.27)	6.981 (−0.91)	7.239 (−1.95)	7.588 (0.29)	7.834 (0.56)	8.077 (0.96)	8.325 (1.42)	8.539 (0.86)
20%	6.190 (1.31)	6.539 (1.38)	6.846 (1.35)	7.203 (2.24)	7.451 (0.93)	7.813 (3.26)	8.076 (3.66)	8.323 (4.03)	8.631 (5.15)	8.861 (4.66)
30%	6.411 (4.93)	6.742 (4.52)	7.088 (4.93)	7.376 (4.69)	7.705 (4.36)	8.032 (6.15)	8.302 (6.57)	8.621 (7.76)	8.888 (8.28)	9.201 (8.68)
40%	6.545 (7.12)	6.924 (7.33)	7.294 (7.98)	7.597 (7.83)	7.933 (7.46)	8.175 (8.04)	8.523 (9.39)	8.887 (11.08)	9.185 (11.90)	9.532 (12.59)

table3_shrink_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	6.088	6.429	6.733	7.064	7.358	7.564	7.779	7.987	8.225	8.459
Imputation using linear regression										
10%	6.409 (5.28)	6.787 (5.57)	7.139 (6.02)	7.433 (5.23)	7.836 (6.49)	8.147 (7.72)	8.424 (8.30)	8.701 (8.94)	8.947 (8.78)	9.297 (9.90)
20%	7.120 (16.95)	7.568 (17.73)	7.930 (17.77)	8.406 (19.00)	8.674 (17.88)	9.048 (19.62)	9.423 (21.14)	9.742 (21.98)	10.206 (24.08)	10.456 (23.60)
30%	8.382 (37.69)	8.884 (38.19)	9.377 (39.27)	9.826 (39.11)	10.237 (39.12)	10.559 (39.60)	11.036 (41.87)	11.537 (44.45)	12.005 (45.96)	12.442 (47.09)
40%	10.089 (65.72)	10.704 (66.50)	11.338 (68.40)	11.845 (67.68)	12.479 (69.60)	12.664 (67.43)	13.321 (71.25)	14.029 (75.65)	14.581 (77.28)	15.323 (81.14)
Imputation using predictive mean matching method										
10%	5.942 (−2.39)	6.288 (−2.18)	6.620 (−1.68)	6.884 (−2.55)	7.267 (−1.24)	7.570 (0.08)	7.804 (0.32)	8.047 (0.75)	8.250 (0.30)	8.565 (1.25)
20%	6.030 (−0.96)	6.405 (−0.37)	6.730 (−0.05)	7.100 (0.50)	7.352 (−0.08)	7.694 (1.72)	7.967 (2.43)	8.221 (2.94)	8.544 (3.87)	8.764 (3.61)
30%	6.219 (2.16)	6.625 (3.05)	6.953 (3.26)	7.302 (3.37)	7.596 (3.23)	7.900 (4.45)	8.227 (5.77)	8.514 (6.60)	8.833 (7.38)	9.109 (7.68)
40%	6.394 (5.03)	6.771 (5.33)	7.130 (5.89)	7.491 (6.05)	7.808 (6.11)	8.109 (7.21)	8.432 (8.40)	8.773 (9.84)	9.118 (10.86)	9.441 (11.60)
Imputation using propensity−score method										
10%	5.933 (−2.54)	6.274 (−2.40)	6.605 (−1.90)	6.874 (−2.70)	7.245 (−1.55)	7.545 (−0.25)	7.776 (−0.03)	8.020 (0.42)	8.231 (0.07)	8.534 (0.88)
20%	6.034 (−0.89)	6.417 (−0.18)	6.740 (0.10)	7.102 (0.54)	7.375 (0.22)	7.683 (1.58)	7.970 (2.46)	8.224 (2.98)	8.546 (3.90)	8.790 (3.91)
30%	6.115 (0.45)	6.516 (1.36)	6.848 (1.71)	7.187 (1.75)	7.467 (1.48)	7.755 (2.52)	8.083 (3.91)	8.381 (4.94)	8.689 (5.63)	8.952 (5.83)
40%	6.277 (3.11)	6.671 (3.77)	6.982 (3.70)	7.315 (3.55)	7.693 (4.55)	7.916 (4.66)	8.270 (6.32)	8.565 (7.23)	8.885 (8.02)	9.289 (9.81)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	6.110	6.451	6.755	7.045	7.382	7.566	7.791	8.000	8.208	8.466
Imputation using linear regression										

table3_shrink_5_treatments_R4_opposite_sign_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	6.355 (4.02)	6.743 (4.54)	7.116 (5.34)	7.426 (5.40)	7.725 (4.64)	8.095 (6.98)	8.379 (7.55)	8.672 (8.40)	8.920 (8.68)	9.184 (8.47)
20%	7.165 (17.27)	7.577 (17.46)	7.965 (17.91)	8.395 (19.16)	8.742 (18.42)	9.082 (20.03)	9.424 (20.97)	9.774 (22.18)	10.176 (23.98)	10.531 (24.39)
30%	8.271 (35.38)	8.820 (36.73)	9.269 (37.22)	9.759 (38.52)	10.217 (38.39)	10.443 (38.02)	10.980 (40.93)	11.429 (42.86)	11.949 (45.58)	12.430 (46.82)
40%	10.174 (66.53)	10.859 (68.34)	11.471 (69.81)	11.869 (68.48)	13.195 (78.73)	12.782 (68.94)	13.519 (73.52)	14.191 (77.39)	14.617 (78.09)	16.222 (91.61)
Imputation using predictive mean matching method										
10%	5.899 (−3.45)	6.250 (−3.10)	6.599 (−2.31)	6.903 (−2.01)	7.170 (−2.87)	7.525 (−0.54)	7.763 (−0.35)	8.016 (0.20)	8.252 (0.54)	8.472 (0.07)
20%	6.040 (−1.14)	6.397 (−0.83)	6.730 (−0.37)	7.089 (0.62)	7.341 (−0.56)	7.697 (1.73)	7.957 (2.13)	8.222 (2.78)	8.523 (3.83)	8.751 (3.36)
30%	6.181 (1.17)	6.567 (1.81)	6.888 (1.97)	7.218 (2.45)	7.542 (2.16)	7.858 (3.86)	8.174 (4.92)	8.442 (5.53)	8.749 (6.59)	9.046 (6.85)
40%	6.400 (4.76)	6.813 (5.62)	7.185 (6.36)	7.489 (6.31)	7.837 (6.16)	8.119 (7.31)	8.486 (8.92)	8.829 (10.36)	9.116 (11.07)	9.464 (11.79)
Imputation using propensity−score method										
10%	5.885 (−3.67)	6.238 (−3.29)	6.575 (−2.66)	6.884 (−2.29)	7.155 (−3.08)	7.496 (−0.93)	7.737 (−0.69)	7.982 (−0.23)	8.224 (0.20)	8.450 (−0.19)
20%	6.069 (−0.66)	6.423 (−0.43)	6.733 (−0.32)	7.112 (0.95)	7.361 (−0.29)	7.709 (1.89)	7.968 (2.27)	8.217 (2.71)	8.542 (4.06)	8.771 (3.60)
30%	6.136 (0.42)	6.477 (0.41)	6.851 (1.41)	7.141 (1.36)	7.485 (1.38)	7.773 (2.73)	8.046 (3.28)	8.383 (4.78)	8.650 (5.39)	8.977 (6.03)
40%	6.249 (2.28)	6.656 (3.19)	7.016 (3.86)	7.353 (4.38)	7.685 (4.10)	7.879 (4.14)	8.253 (5.93)	8.593 (7.40)	8.930 (8.79)	9.270 (9.49)

table3_shrink_5_treatments_R9_control_MCAR_treatment_MAR

MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.559	11.100	11.597	12.125	12.618
Imputation using linear regression										
10%	8.882 (2.44)	9.603 (3.54)	10.267 (4.38)	10.824 (3.84)	11.540 (5.35)	11.213 (6.20)	11.843 (6.69)	12.451 (7.36)	12.977 (7.03)	13.683 (8.44)
20%	9.541 (10.04)	10.415 (12.30)	11.157 (13.44)	12.000 (15.12)	12.511 (14.21)	12.098 (14.57)	12.919 (16.39)	13.648 (17.69)	14.522 (19.78)	15.039 (19.19)
30%	10.541 (21.59)	11.616 (25.25)	12.559 (27.69)	13.436 (28.90)	14.186 (29.50)	13.387 (26.78)	14.480 (30.45)	15.479 (33.48)	16.431 (35.52)	17.254 (36.74)
40%	12.021 (38.65)	13.371 (44.17)	14.611 (48.55)	15.655 (50.18)	16.809 (53.45)	15.309 (44.99)	16.779 (51.17)	18.184 (56.80)	19.362 (59.70)	20.730 (64.29)
Imputation using predictive mean matching method										
10%	8.548 (−1.41)	9.194 (−0.87)	9.802 (−0.34)	10.299 (−1.20)	10.940 (−0.13)	10.727 (1.59)	11.281 (1.63)	11.823 (1.95)	12.279 (1.27)	12.890 (2.16)
20%	8.745 (0.87)	9.422 (1.60)	10.014 (1.81)	10.661 (2.28)	11.126 (1.56)	10.965 (3.84)	11.565 (4.19)	12.109 (4.42)	12.741 (5.09)	13.198 (4.59)
30%	8.979 (3.56)	9.706 (4.66)	10.303 (4.75)	10.926 (4.82)	11.464 (4.65)	11.235 (6.41)	11.911 (7.31)	12.487 (7.68)	13.112 (8.15)	13.661 (8.27)
40%	9.297 (7.24)	9.999 (7.81)	10.644 (8.22)	11.275 (8.16)	11.862 (8.28)	11.600 (9.86)	12.276 (10.60)	12.927 (11.47)	13.575 (11.96)	14.208 (12.60)
Imputation using propensity−score method										
10%	8.563 (−1.24)	9.196 (−0.84)	9.793 (−0.43)	10.299 (−1.20)	10.924 (−0.28)	10.727 (1.60)	11.268 (1.51)	11.801 (1.76)	12.267 (1.18)	12.863 (1.94)
20%	8.778 (1.25)	9.459 (1.99)	10.043 (2.11)	10.684 (2.50)	11.174 (2.01)	10.983 (4.02)	11.592 (4.43)	12.130 (4.60)	12.763 (5.26)	13.250 (5.01)
30%	9.033 (4.19)	9.725 (4.86)	10.332 (5.05)	10.912 (4.68)	11.442 (4.45)	11.240 (6.45)	11.886 (7.08)	12.483 (7.65)	13.059 (7.71)	13.607 (7.84)
40%	9.337 (7.69)	10.034 (8.19)	10.614 (7.91)	11.211 (7.55)	11.811 (7.81)	11.553 (9.42)	12.242 (10.29)	12.829 (10.63)	13.447 (10.91)	14.099 (11.74)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.575	11.122	11.621	12.097	12.639
Imputation using linear regression										

table3_shrink_5_treatments_R9_control_MCAR_treatment_MAR

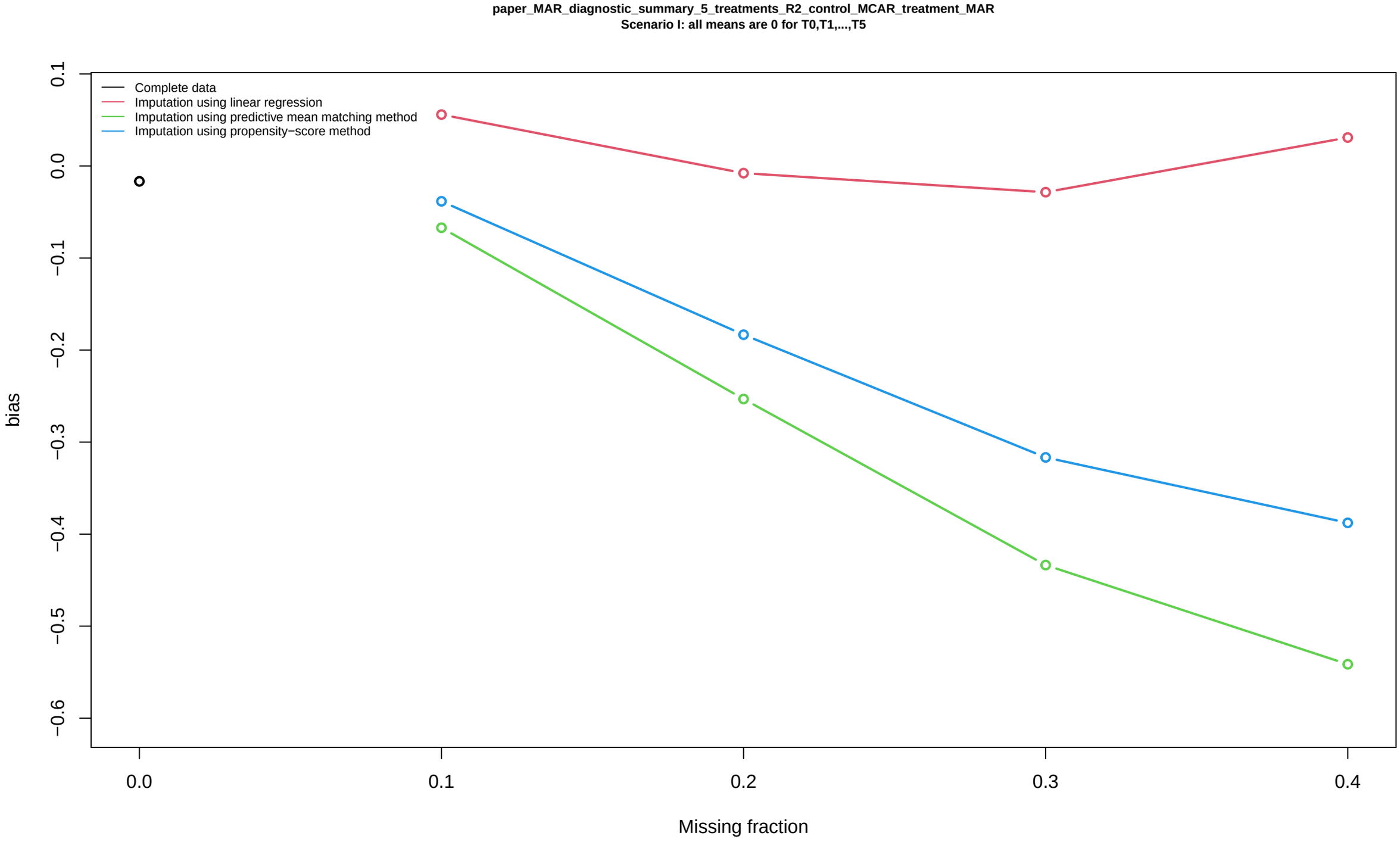
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.793 (0.96)	9.523 (2.29)	10.216 (3.51)	10.800 (3.85)	11.356 (3.32)	11.122 (5.17)	11.757 (5.72)	12.391 (6.62)	12.934 (6.92)	13.483 (6.68)
20%	9.484 (8.89)	10.315 (10.79)	11.081 (12.27)	11.885 (14.28)	12.500 (13.73)	12.031 (13.77)	12.806 (15.14)	13.557 (16.66)	14.377 (18.84)	15.032 (18.94)
30%	10.499 (20.55)	11.602 (24.62)	12.466 (26.30)	13.366 (28.51)	14.262 (29.76)	13.344 (26.18)	14.471 (30.12)	15.376 (32.31)	16.362 (35.25)	17.353 (37.30)
40%	11.909 (36.74)	13.357 (43.46)	14.617 (48.10)	15.675 (50.72)	16.986 (54.54)	15.205 (43.78)	16.796 (51.02)	18.218 (56.76)	19.424 (60.57)	20.970 (65.92)
Imputation using predictive mean matching method										
10%	8.474 (−2.70)	9.126 (−1.98)	9.745 (−1.26)	10.303 (−0.94)	10.788 (−1.85)	10.652 (0.73)	11.208 (0.78)	11.756 (1.16)	12.267 (1.40)	12.729 (0.71)
20%	8.725 (0.17)	9.384 (0.79)	9.983 (1.15)	10.619 (2.10)	11.092 (0.92)	10.942 (3.48)	11.523 (3.61)	12.073 (3.89)	12.686 (4.87)	13.160 (4.12)
30%	8.960 (2.88)	9.648 (3.63)	10.261 (3.96)	10.852 (4.35)	11.421 (3.91)	11.202 (5.93)	11.840 (6.46)	12.433 (6.98)	13.027 (7.69)	13.604 (7.63)
40%	9.266 (6.39)	9.988 (7.28)	10.660 (8.00)	11.246 (8.13)	11.881 (8.10)	11.572 (9.43)	12.273 (10.35)	12.948 (11.42)	13.553 (12.03)	14.233 (12.61)
Imputation using propensity−score method										
10%	8.483 (−2.60)	9.131 (−1.92)	9.736 (−1.36)	10.301 (−0.95)	10.777 (−1.95)	10.642 (0.63)	11.197 (0.68)	11.732 (0.95)	12.254 (1.29)	12.706 (0.53)
20%	8.763 (0.61)	9.423 (1.21)	10.000 (1.32)	10.627 (2.18)	11.097 (0.97)	10.956 (3.61)	11.543 (3.79)	12.075 (3.91)	12.680 (4.82)	13.156 (4.09)
30%	9.090 (4.37)	9.728 (4.49)	10.356 (4.92)	10.886 (4.67)	11.464 (4.30)	11.288 (6.75)	11.883 (6.84)	12.501 (7.57)	13.032 (7.72)	13.625 (7.81)
40%	9.317 (6.97)	10.001 (7.42)	10.664 (8.04)	11.209 (7.77)	11.803 (7.38)	11.528 (9.02)	12.201 (9.71)	12.883 (10.86)	13.448 (11.16)	14.084 (11.43)

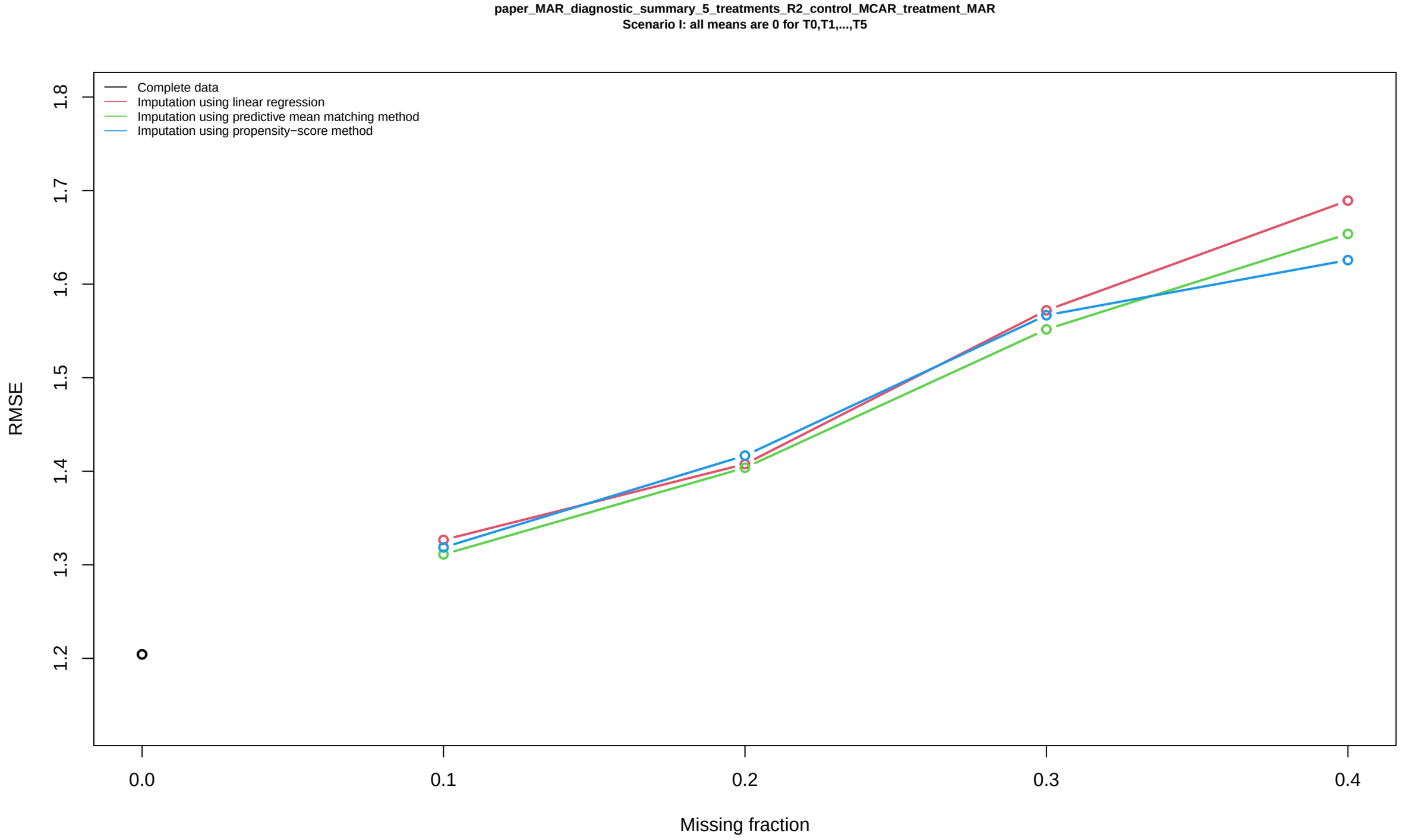
table3_shrink_5_treatments_R9_opposite_sign_MAR

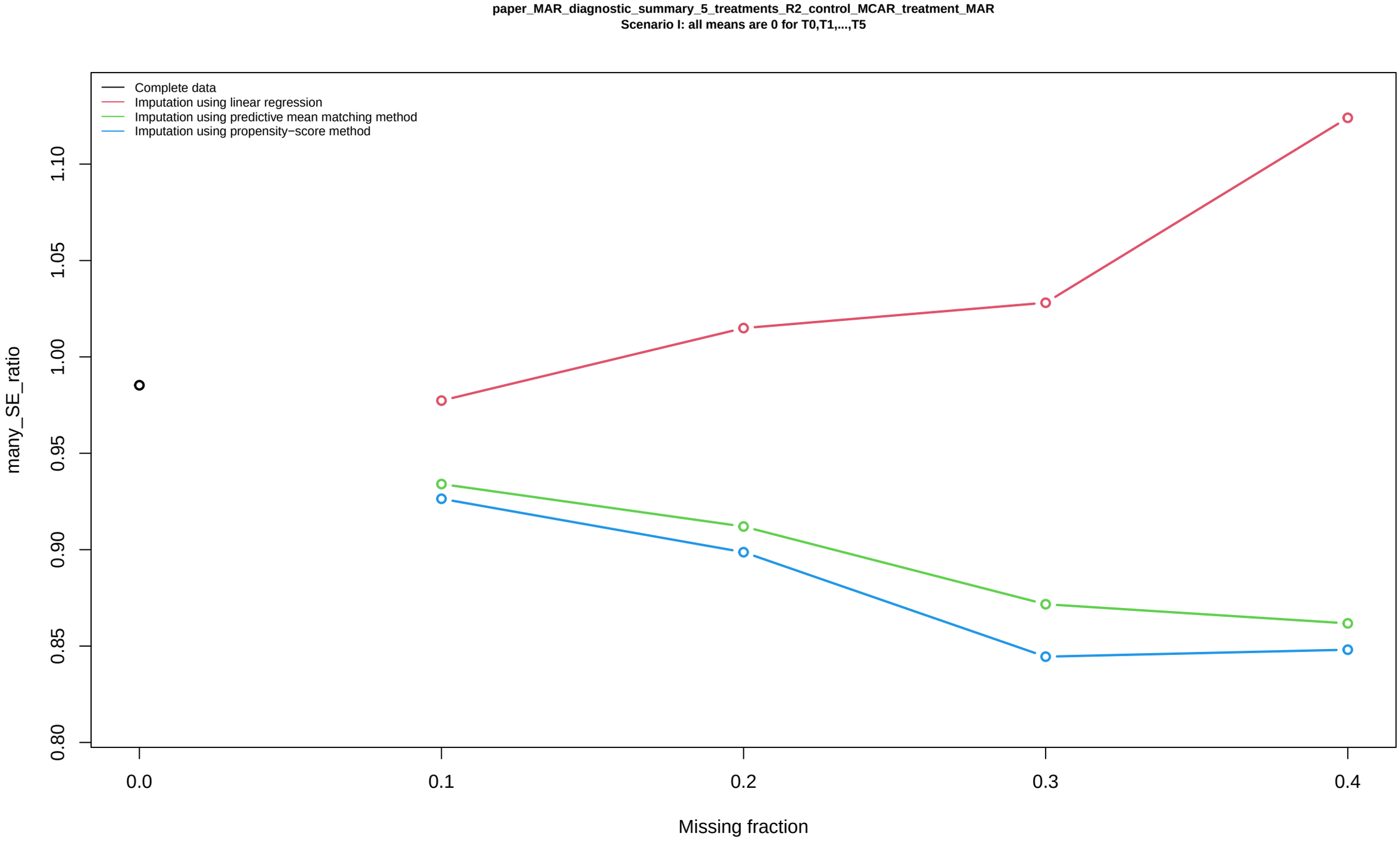
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
Scenario I: all means are 0 for T0,T1,...,T5										
0%	8.670	9.274	9.836	10.424	10.955	10.559	11.100	11.597	12.125	12.618
Imputation using linear regression										
10%	9.044 (4.32)	9.761 (5.25)	10.415 (5.89)	10.969 (5.23)	11.665 (6.48)	11.391 (7.88)	12.017 (8.26)	12.614 (8.77)	13.140 (8.37)	13.820 (9.53)
20%	10.028 (15.67)	10.869 (17.19)	11.565 (17.59)	12.394 (18.90)	12.913 (17.88)	12.626 (19.57)	13.412 (20.83)	14.087 (21.48)	14.946 (23.27)	15.475 (22.64)
30%	11.811 (36.23)	12.770 (37.69)	13.675 (39.04)	14.496 (39.06)	15.241 (39.13)	14.744 (39.64)	15.716 (41.58)	16.673 (43.77)	17.562 (44.85)	18.384 (45.70)
40%	14.161 (63.34)	15.366 (65.68)	16.525 (68.01)	17.468 (67.58)	18.576 (69.57)	17.622 (66.89)	18.933 (70.56)	20.241 (74.54)	21.299 (75.67)	22.600 (79.11)
Imputation using predictive mean matching method										
10%	8.388 (−3.25)	9.047 (−2.45)	9.658 (−1.81)	10.159 (−2.54)	10.818 (−1.25)	10.580 (0.20)	11.139 (0.35)	11.678 (0.70)	12.135 (0.08)	12.763 (1.15)
20%	8.513 (−1.81)	9.207 (−0.72)	9.817 (−0.19)	10.470 (0.44)	10.945 (−0.08)	10.749 (1.80)	11.357 (2.32)	11.912 (2.72)	12.546 (3.47)	13.011 (3.12)
30%	8.792 (1.41)	9.523 (2.68)	10.144 (3.14)	10.772 (3.34)	11.309 (3.24)	11.059 (4.74)	11.731 (5.68)	12.330 (6.32)	12.955 (6.85)	13.503 (7.01)
40%	9.008 (3.90)	9.733 (4.95)	10.399 (5.73)	11.049 (6.00)	11.623 (6.10)	11.321 (7.22)	12.012 (8.22)	12.683 (9.37)	13.351 (10.12)	13.962 (10.65)
Imputation using propensity−score method										
10%	8.379 (−3.35)	9.029 (−2.64)	9.638 (−2.01)	10.144 (−2.69)	10.784 (−1.55)	10.553 (−0.06)	11.103 (0.03)	11.642 (0.39)	12.107 (−0.14)	12.714 (0.76)
20%	8.525 (−1.68)	9.226 (−0.52)	9.834 (−0.02)	10.476 (0.50)	10.980 (0.23)	10.745 (1.76)	11.363 (2.37)	11.917 (2.76)	12.544 (3.46)	13.045 (3.39)
30%	8.636 (−0.39)	9.366 (0.98)	9.988 (1.54)	10.603 (1.71)	11.122 (1.53)	10.850 (2.76)	11.523 (3.81)	12.129 (4.59)	12.739 (5.07)	13.271 (5.17)
40%	8.849 (2.06)	9.585 (3.36)	10.185 (3.55)	10.796 (3.56)	11.445 (4.48)	11.061 (4.76)	11.777 (6.10)	12.379 (6.75)	13.009 (7.30)	13.721 (8.75)
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5										
0%	8.709	9.310	9.870	10.400	10.991	10.575	11.122	11.621	12.097	12.639
Imputation using linear regression										

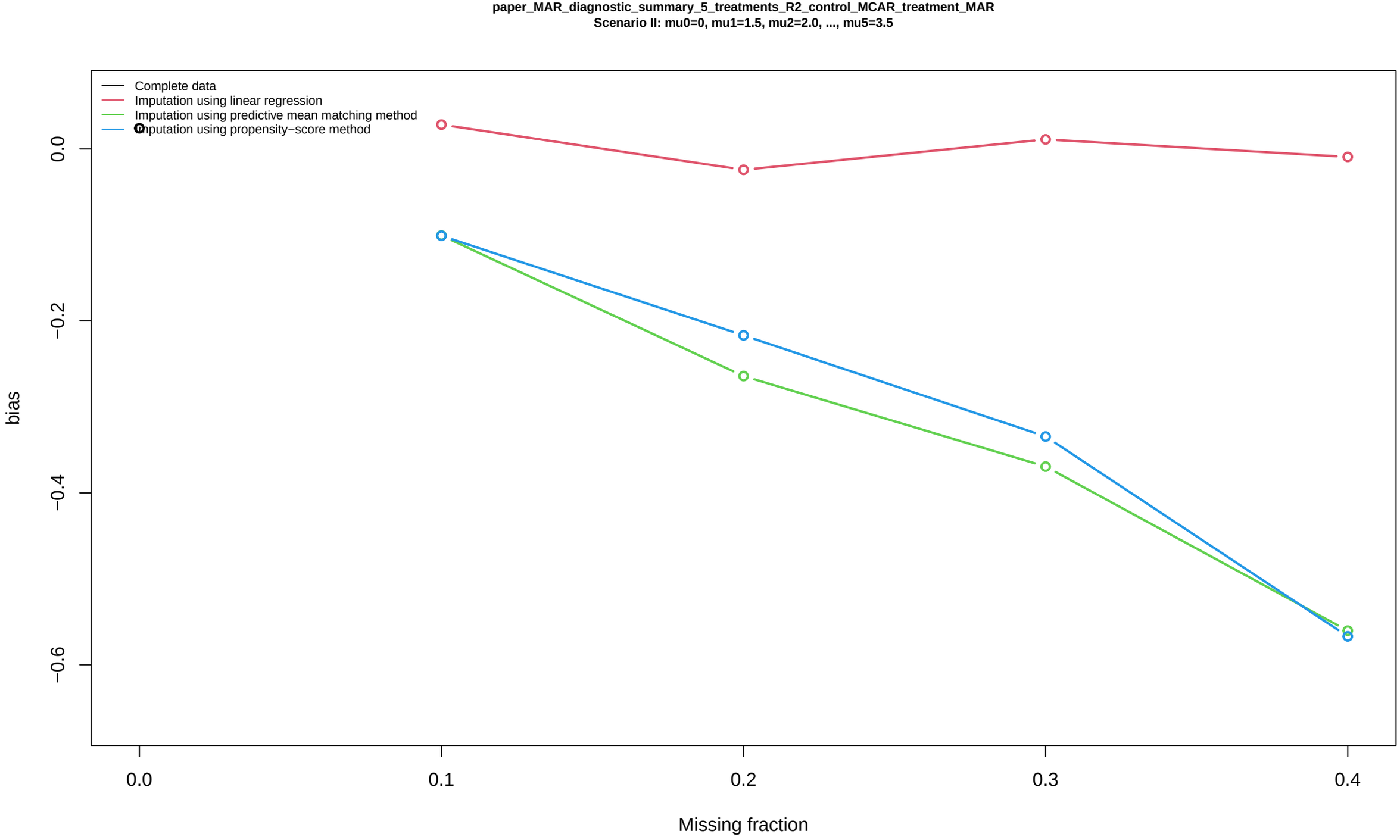
table3_shrink_5_treatments_R9_opposite_sign_MAR

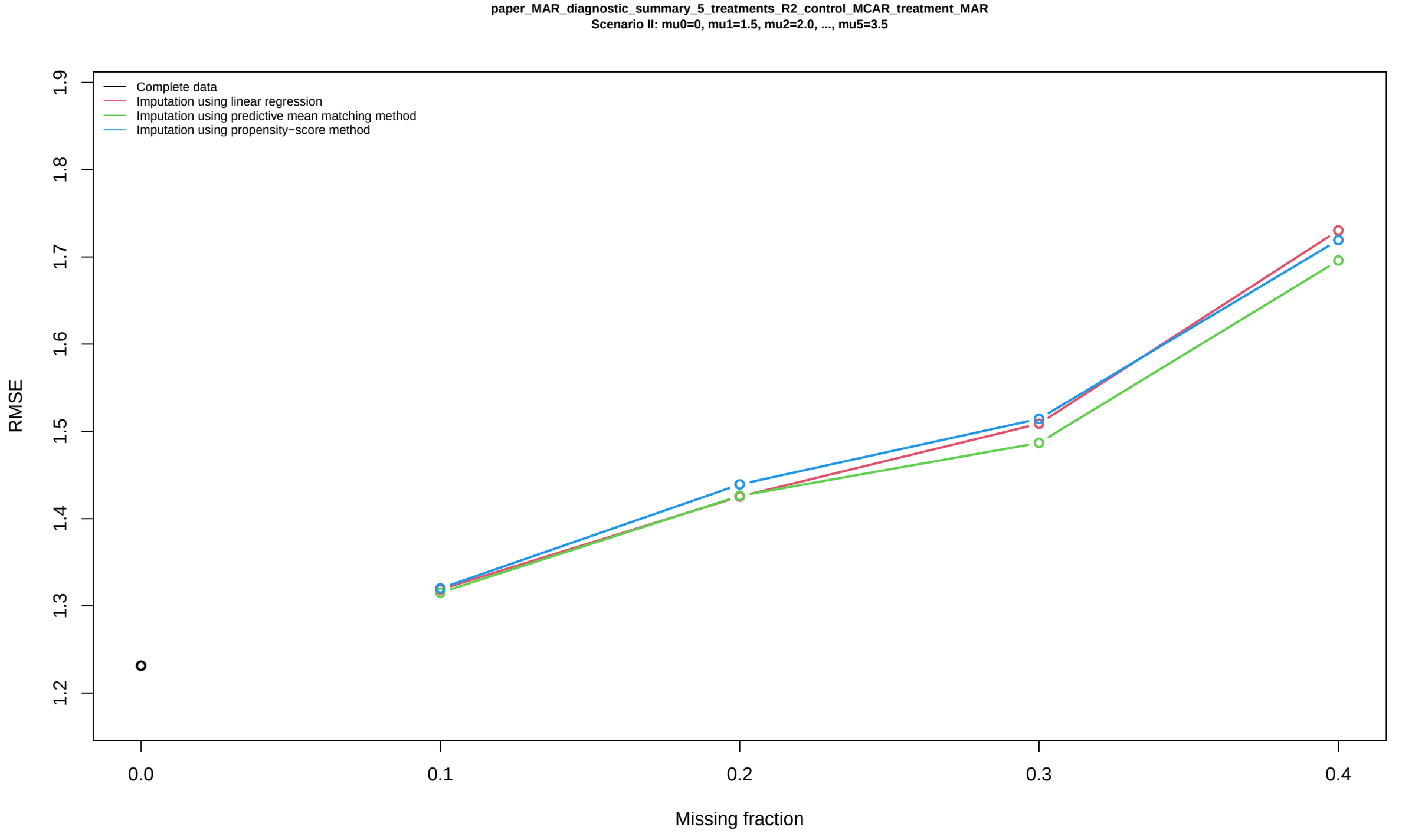
MR	None_T1vs.T0	None_T2vs.T0	None_T3vs.T0	None_T4vs.T0	None_T5vs.T0	Many_T1vs.T0	Many_T2vs.T0	Many_T3vs.T0	Many_T4vs.T0	Many_T5vs.T0
10%	8.970 (2.99)	9.695 (4.13)	10.377 (5.14)	10.954 (5.33)	11.501 (4.64)	11.312 (6.97)	11.946 (7.41)	12.568 (8.15)	13.105 (8.33)	13.642 (7.94)
20%	10.097 (15.93)	10.894 (17.02)	11.620 (17.73)	12.381 (19.05)	13.014 (18.41)	12.683 (19.94)	13.428 (20.74)	14.140 (21.67)	14.911 (23.25)	15.587 (23.32)
30%	11.658 (33.86)	12.667 (36.06)	13.517 (36.95)	14.393 (38.39)	15.207 (38.36)	14.589 (37.96)	15.619 (40.44)	16.512 (42.09)	17.471 (44.42)	18.362 (45.28)
40%	14.310 (64.31)	15.584 (67.39)	16.719 (69.39)	17.511 (68.37)	19.632 (78.62)	17.817 (68.49)	19.213 (72.75)	20.483 (76.26)	21.361 (76.58)	23.932 (89.35)
Imputation using predictive mean matching method										
10%	8.332 (−4.34)	8.991 (−3.43)	9.625 (−2.48)	10.183 (−2.08)	10.675 (−2.87)	10.519 (−0.52)	11.076 (−0.41)	11.634 (0.11)	12.145 (0.39)	12.611 (−0.22)
20%	8.521 (−2.16)	9.199 (−1.19)	9.818 (−0.52)	10.454 (0.52)	10.929 (−0.56)	10.751 (1.67)	11.344 (2.00)	11.912 (2.51)	12.519 (3.48)	12.991 (2.79)
30%	8.717 (0.09)	9.438 (1.37)	10.048 (1.80)	10.648 (2.38)	11.227 (2.14)	10.977 (3.80)	11.644 (4.70)	12.221 (5.16)	12.824 (6.01)	13.408 (6.09)
40%	9.039 (3.79)	9.795 (5.21)	10.478 (6.16)	11.049 (6.24)	11.667 (6.15)	11.357 (7.40)	12.087 (8.68)	12.768 (9.87)	13.349 (10.34)	14.002 (10.79)
Imputation using propensity−score method										
10%	8.317 (−4.50)	8.975 (−3.60)	9.592 (−2.82)	10.154 (−2.37)	10.650 (−3.11)	10.486 (−0.84)	11.042 (−0.71)	11.585 (−0.31)	12.101 (0.02)	12.574 (−0.52)
20%	8.567 (−1.63)	9.240 (−0.75)	9.828 (−0.42)	10.487 (0.84)	10.957 (−0.31)	10.776 (1.91)	11.365 (2.19)	11.905 (2.44)	12.540 (3.66)	13.013 (2.96)
30%	8.649 (−0.69)	9.316 (0.07)	9.991 (1.23)	10.533 (1.28)	11.134 (1.30)	10.856 (2.66)	11.468 (3.12)	12.128 (4.36)	12.669 (4.72)	13.285 (5.11)
40%	8.846 (1.56)	9.580 (2.89)	10.236 (3.71)	10.840 (4.23)	11.411 (3.82)	11.047 (4.47)	11.766 (5.80)	12.428 (6.94)	13.061 (7.97)	13.668 (8.14)

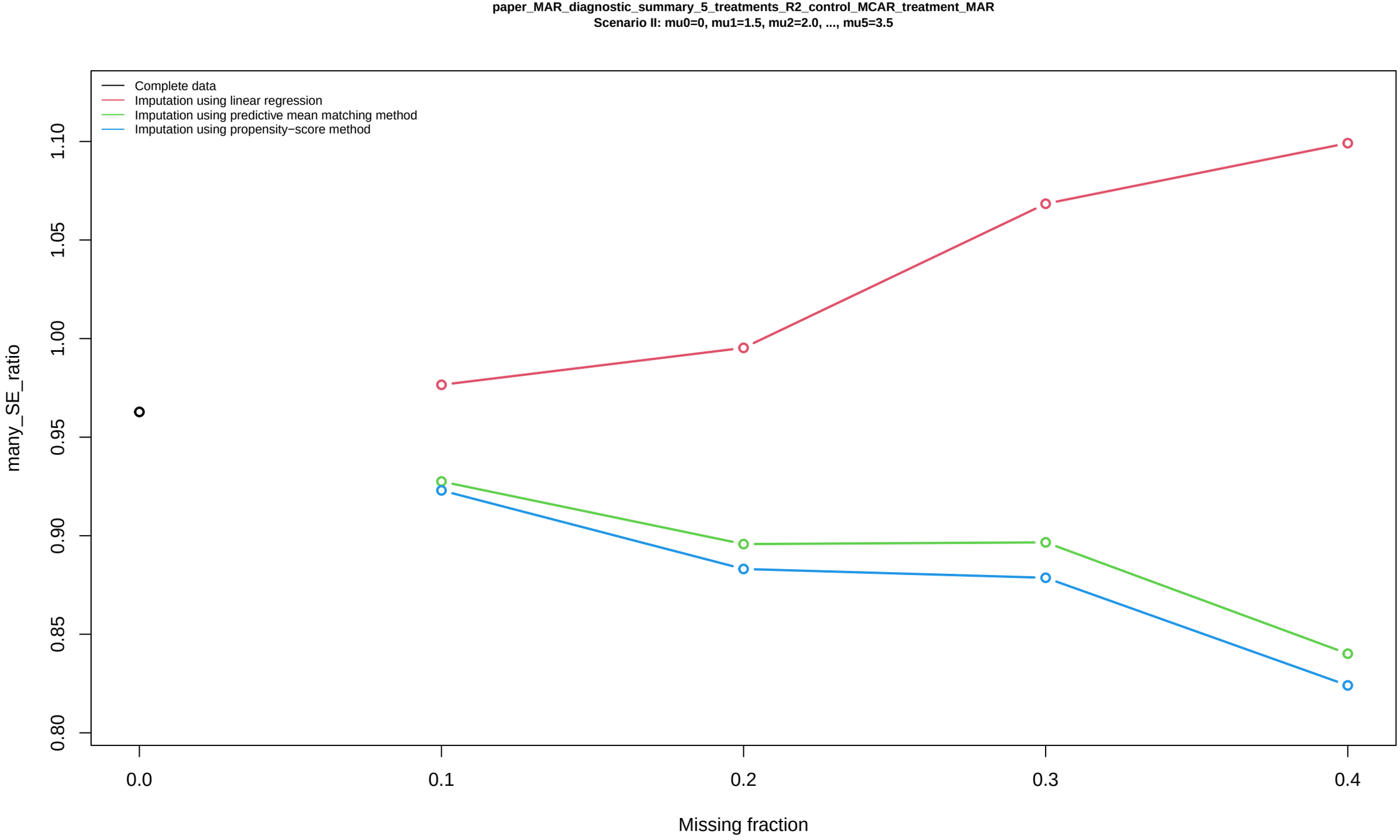

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

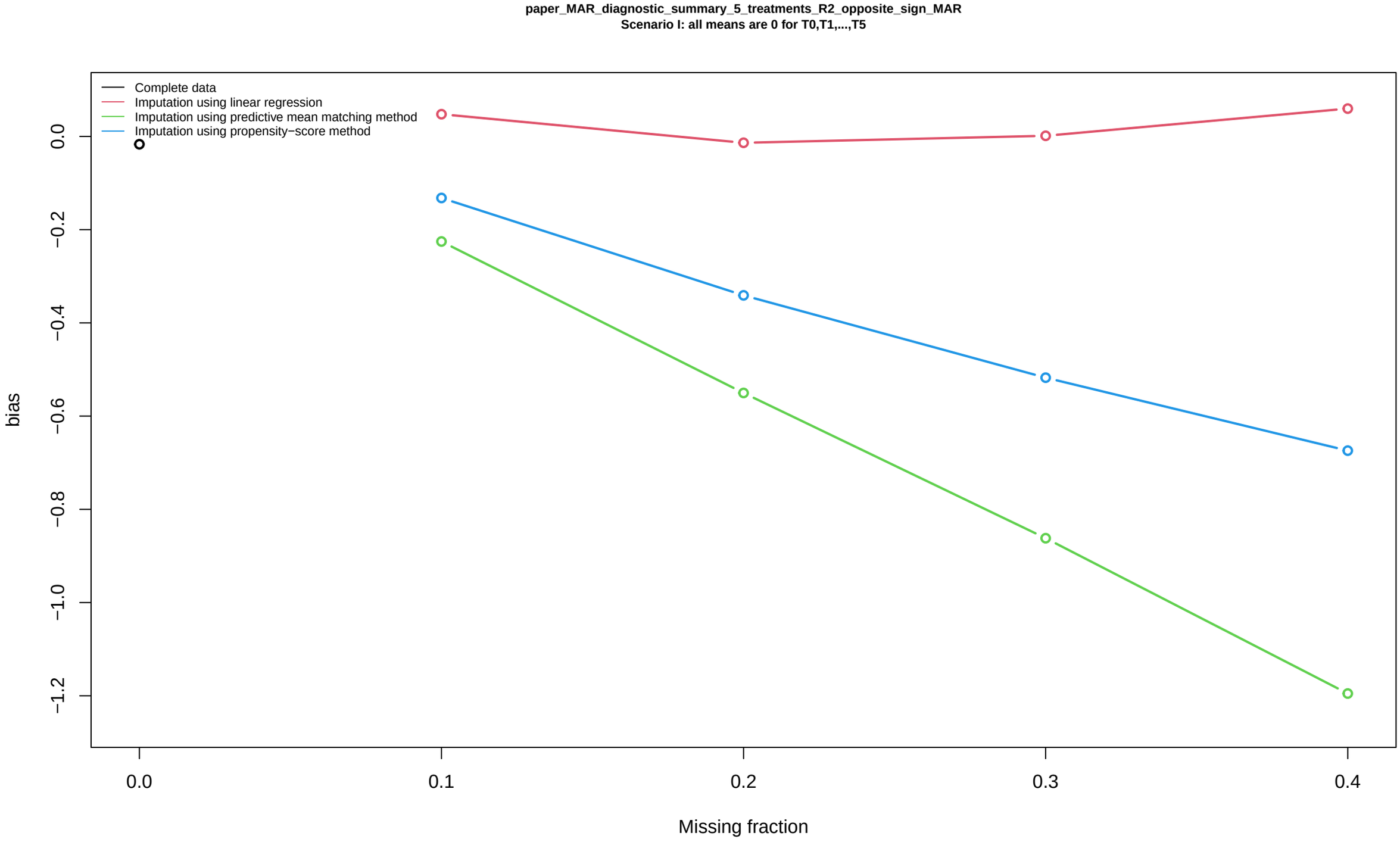

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.3
1.4
1.5
1.6
1.7
1.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

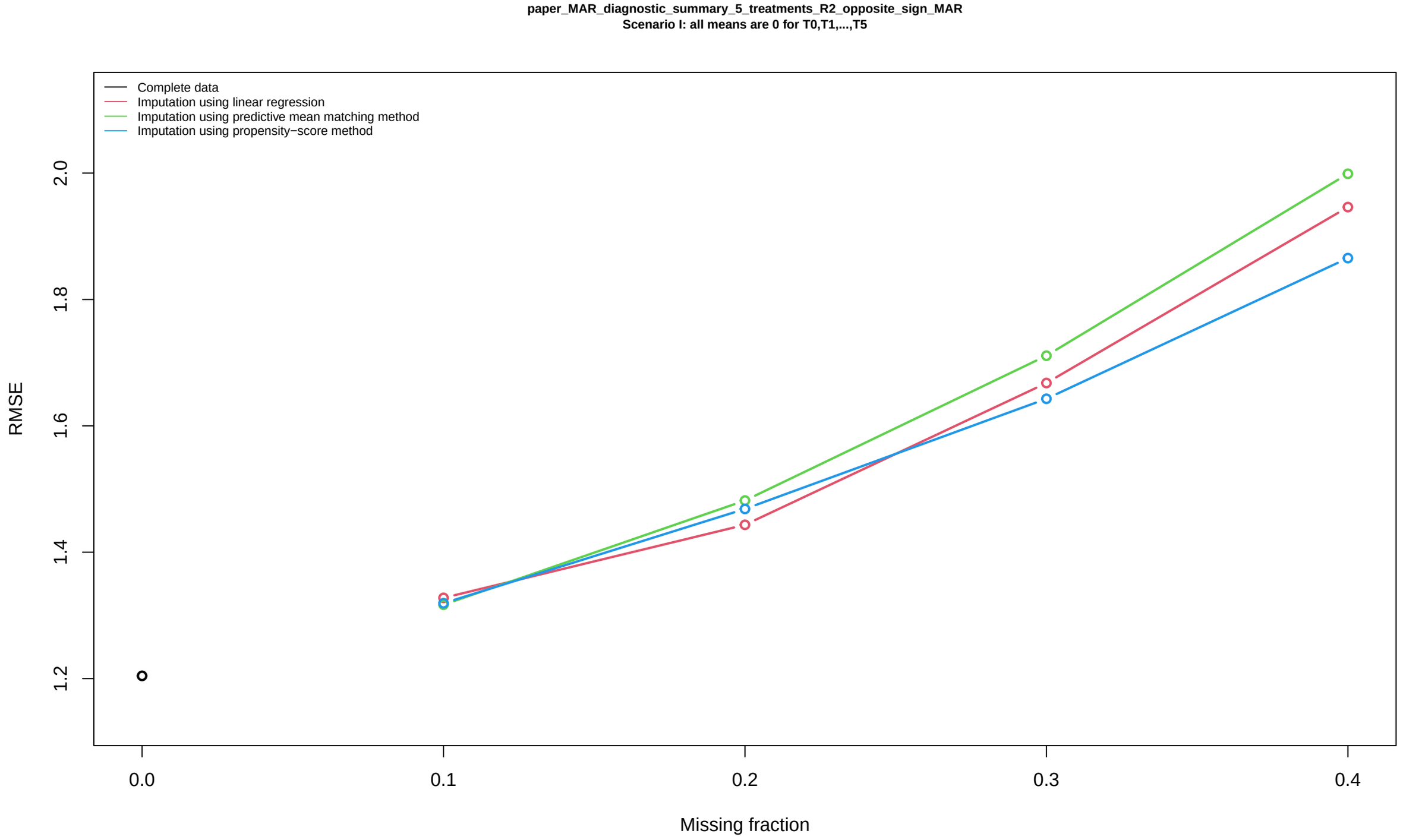

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

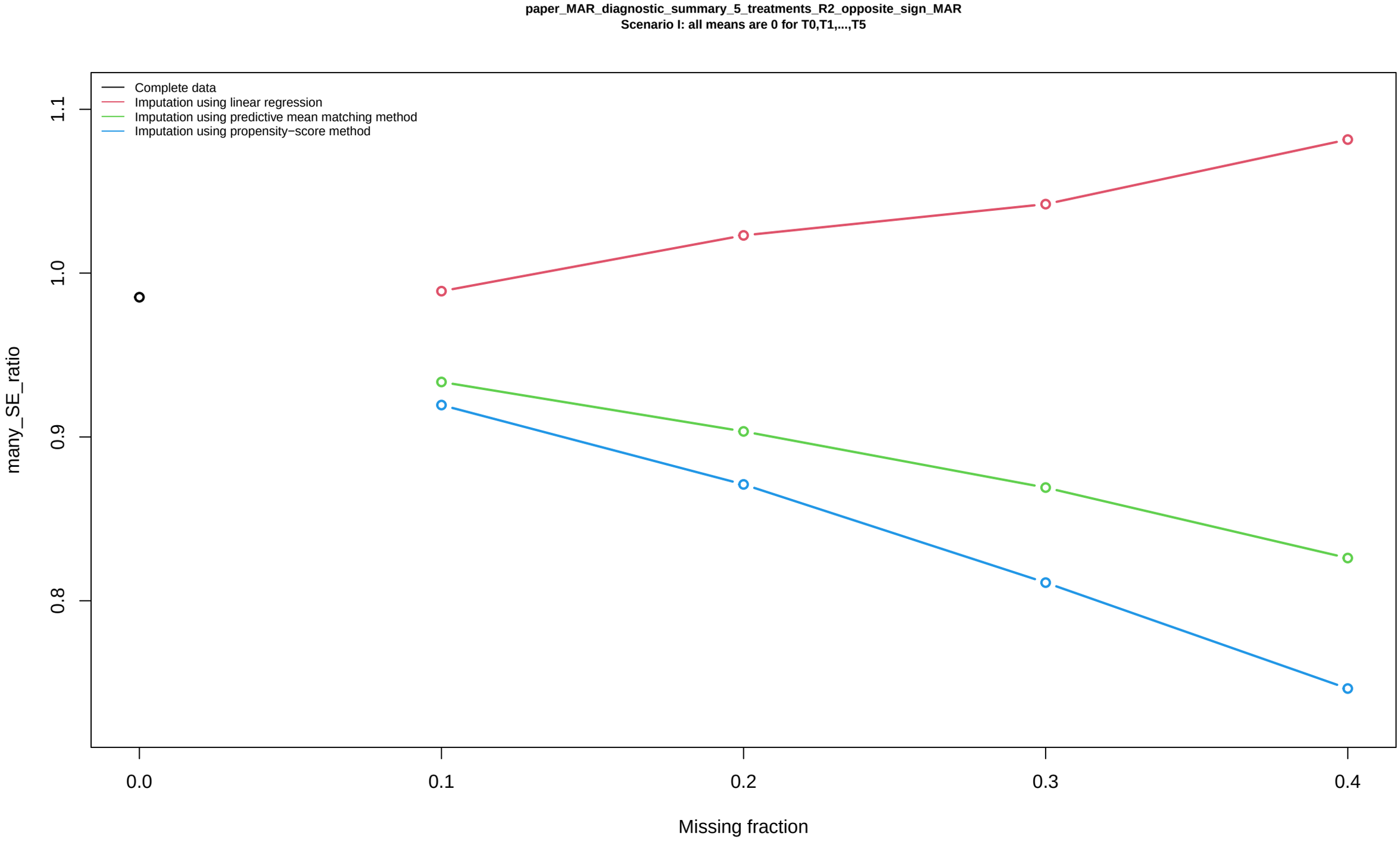

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

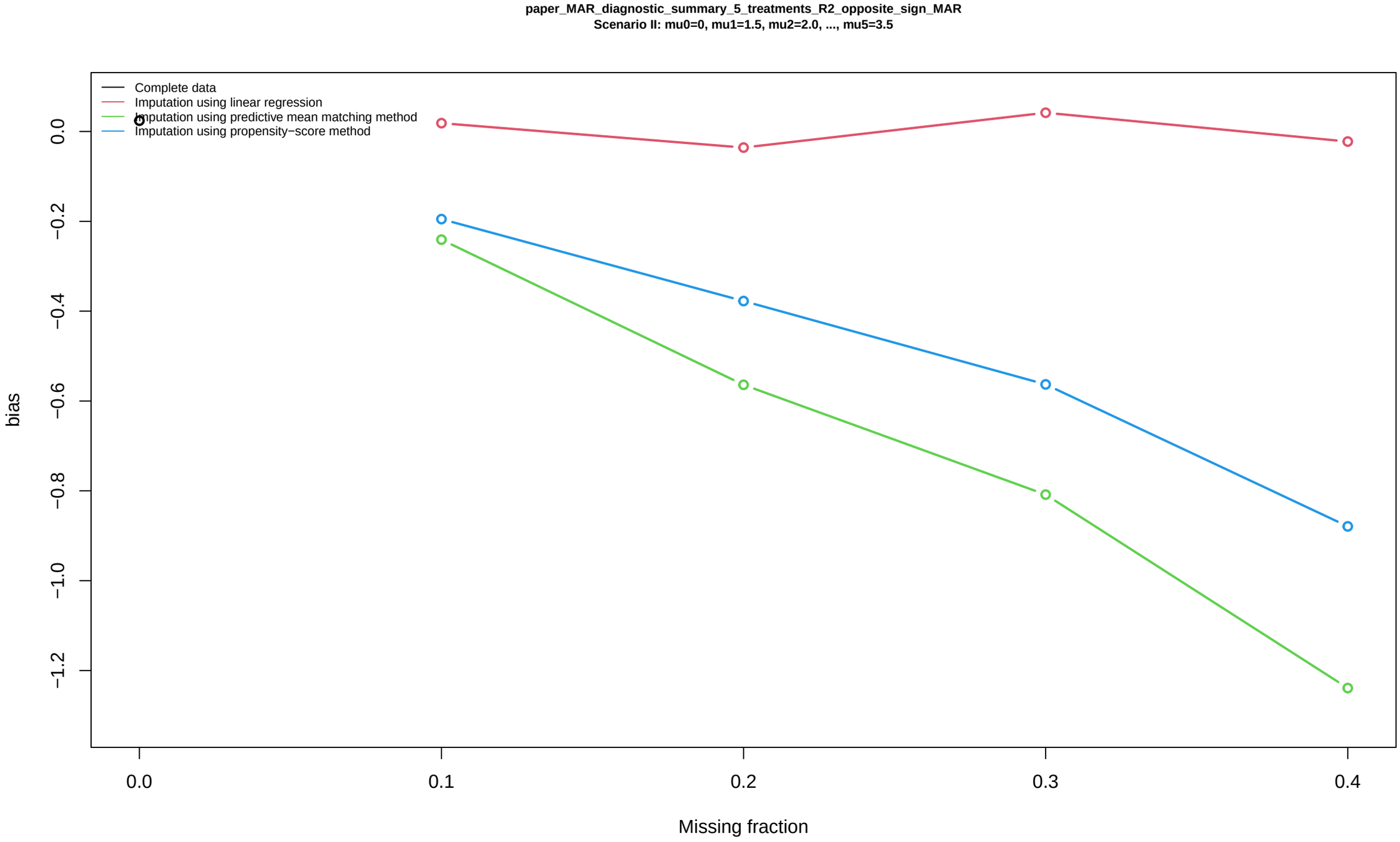

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
0.0
0.1
0.2
0.3
0.4
Missing fraction

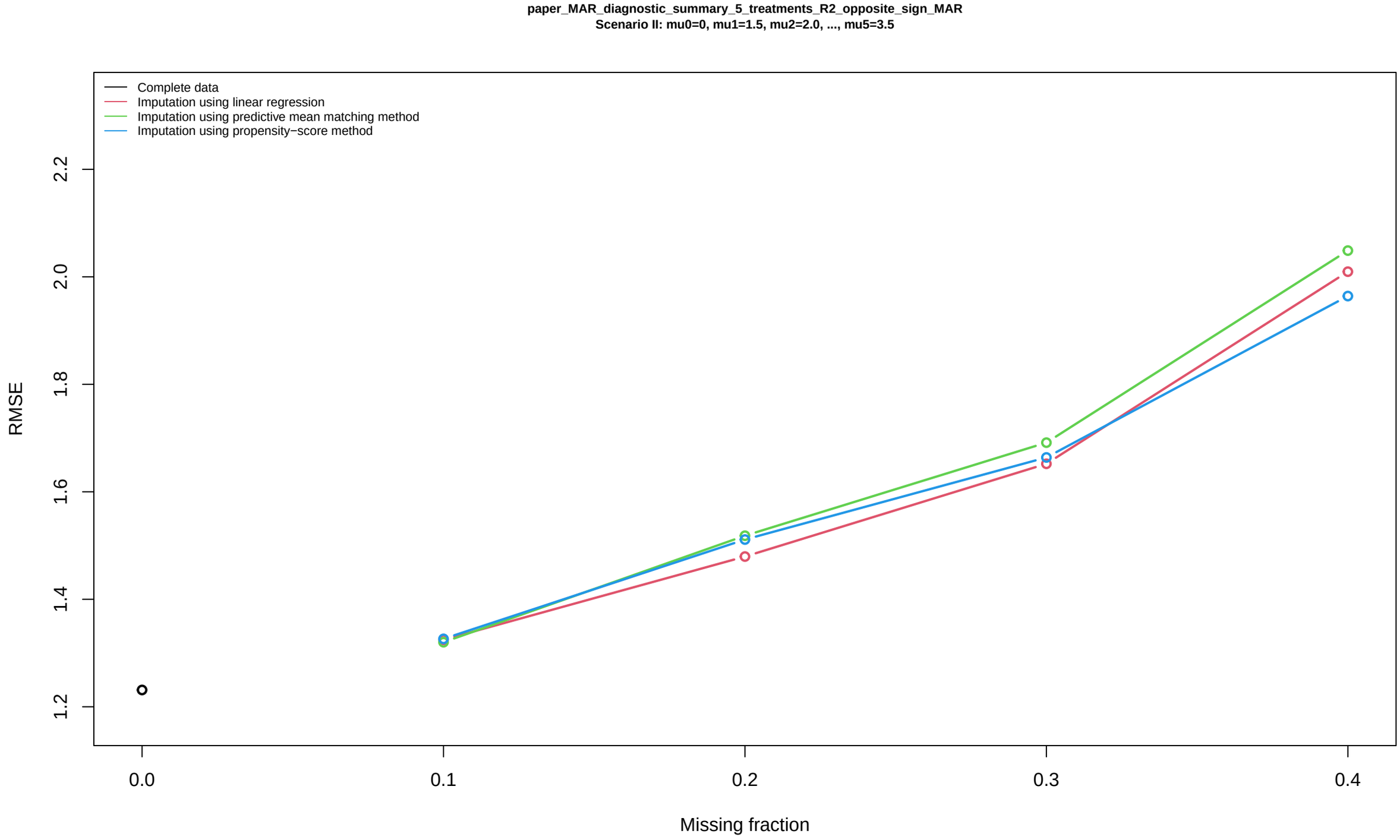

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

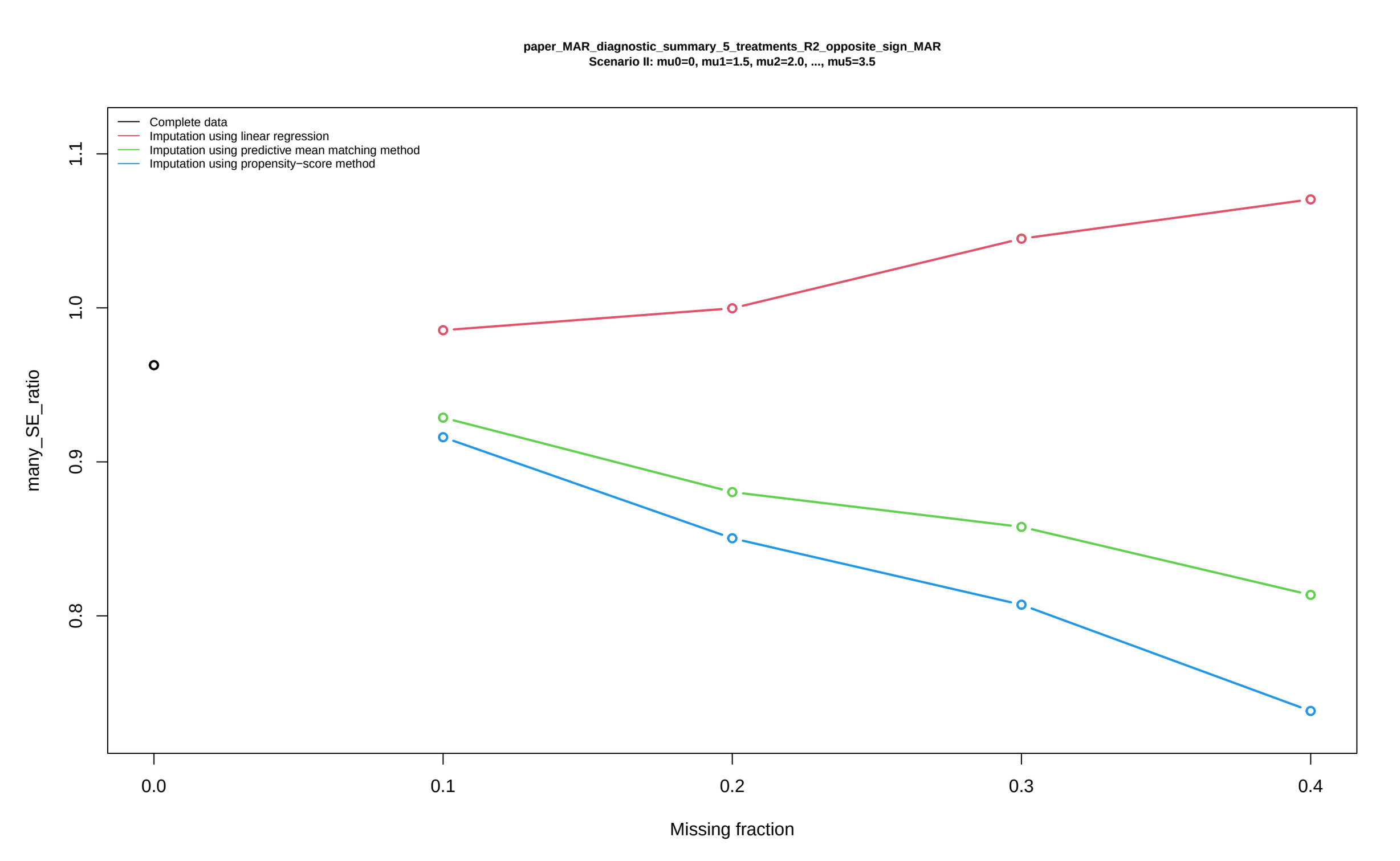

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

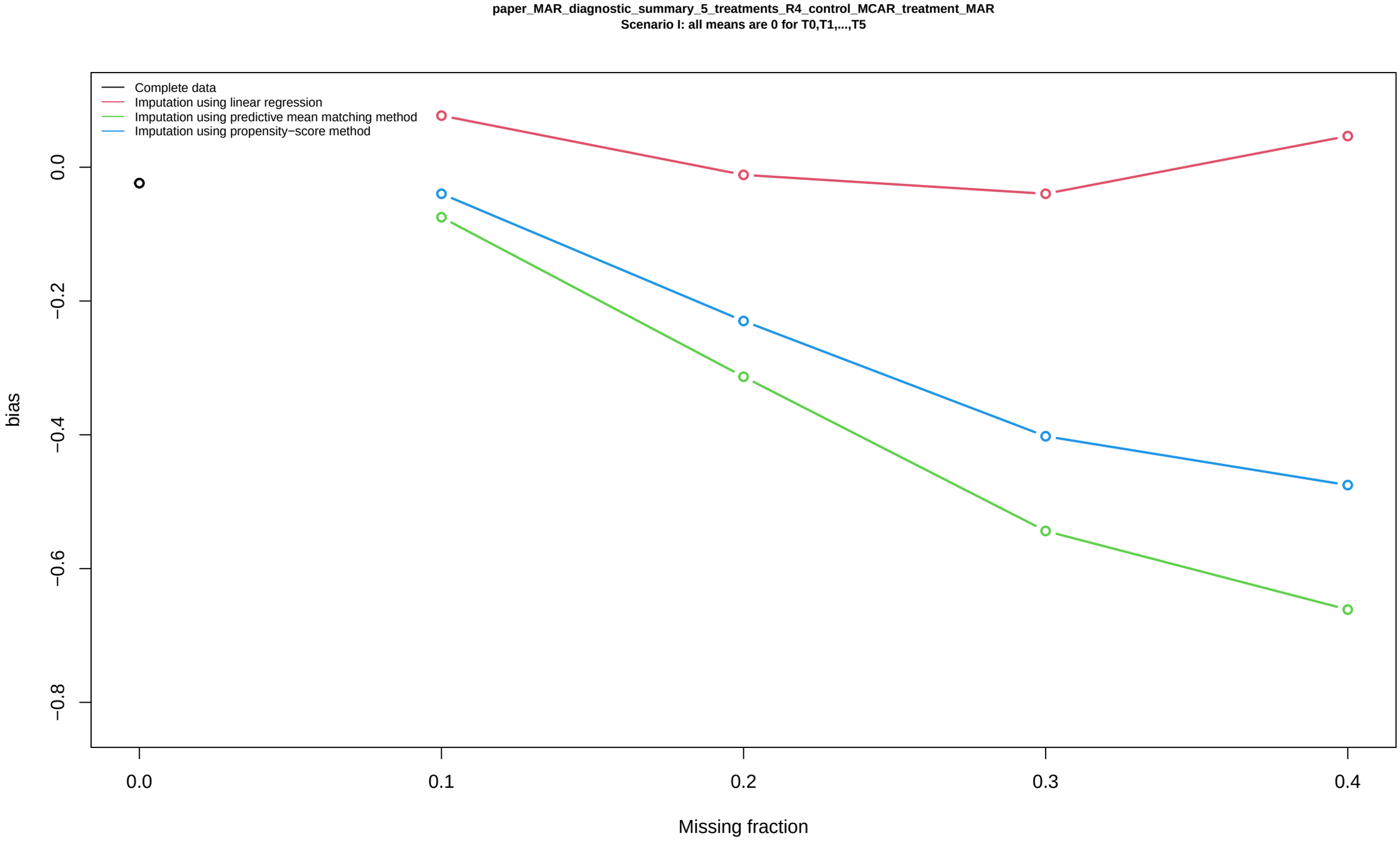

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.4
1.6
1.8
2.0
0.0
0.1
0.2
0.3
0.4
Missing fraction

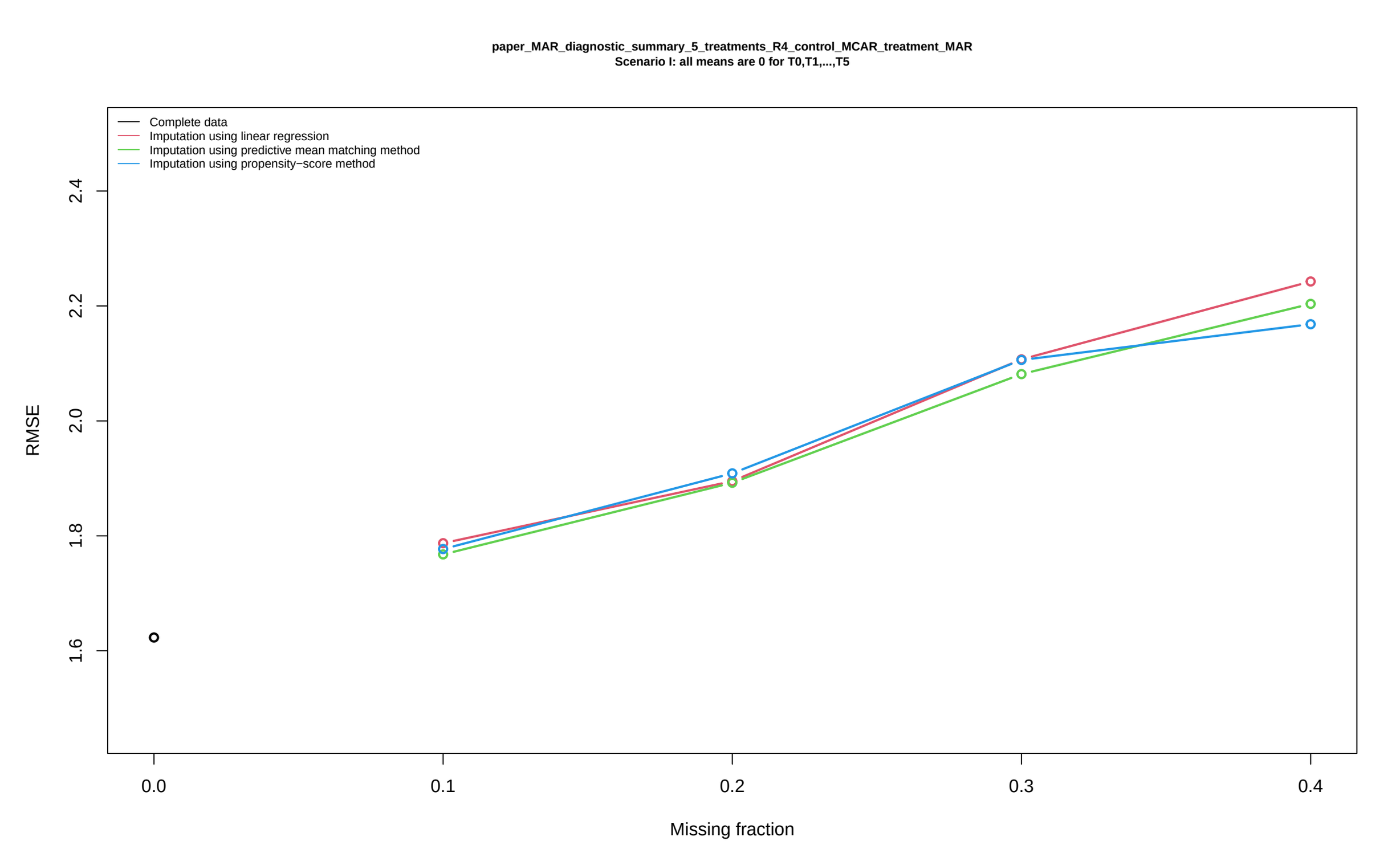

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

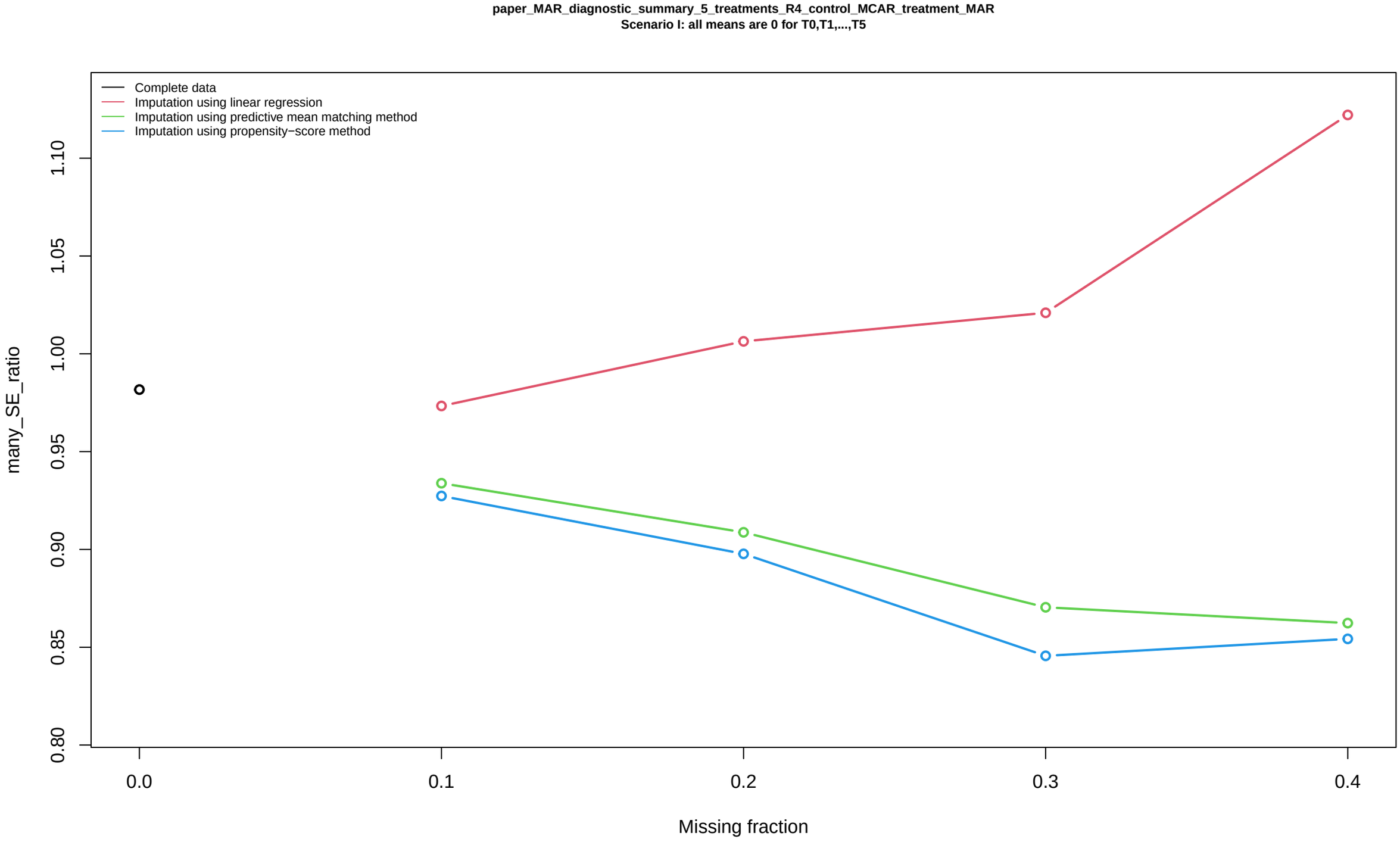

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

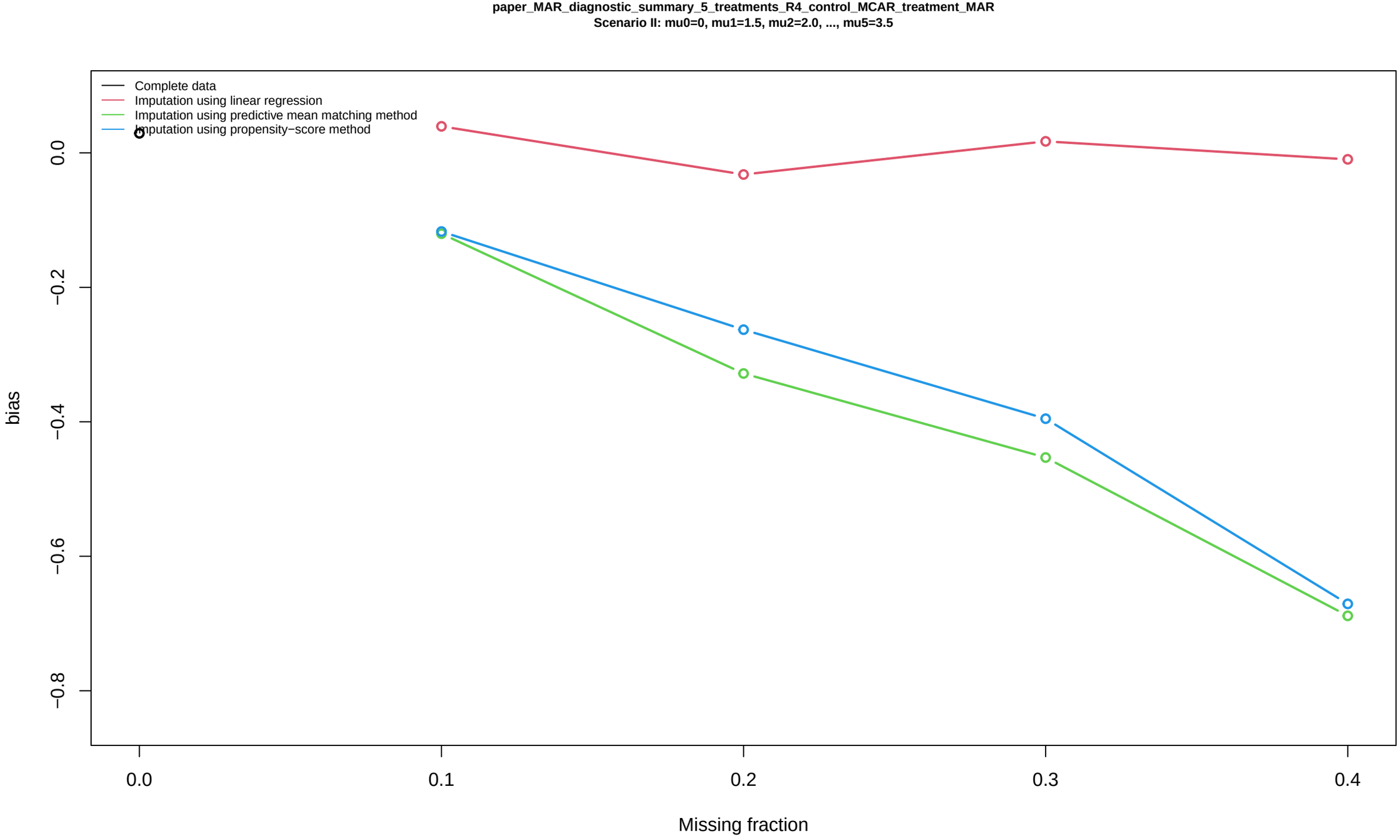

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.4
1.6
1.8
2.0
2.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

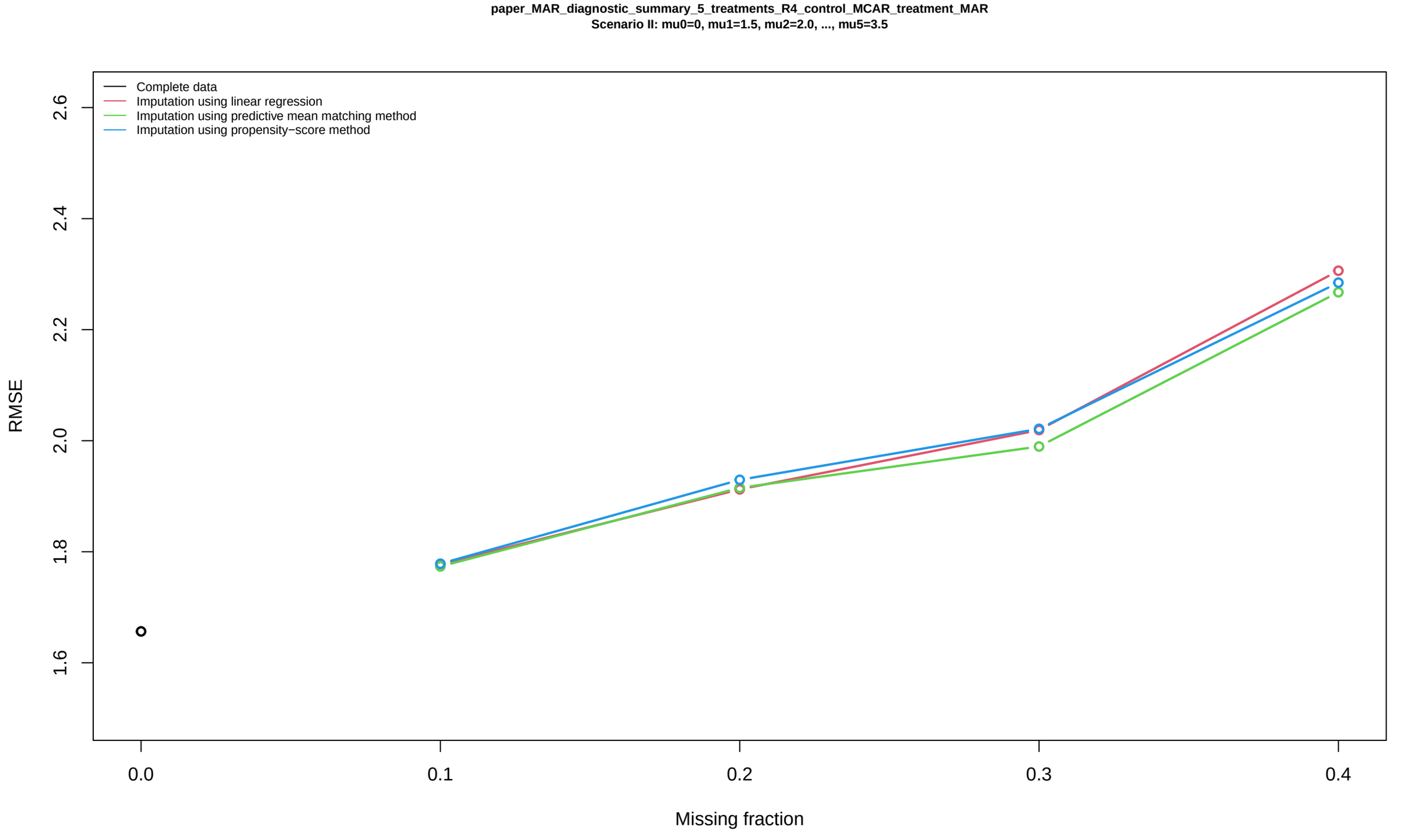

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.8
0.9
1.0
1.1
0.0
0.1
0.2
0.3
0.4
Missing fraction

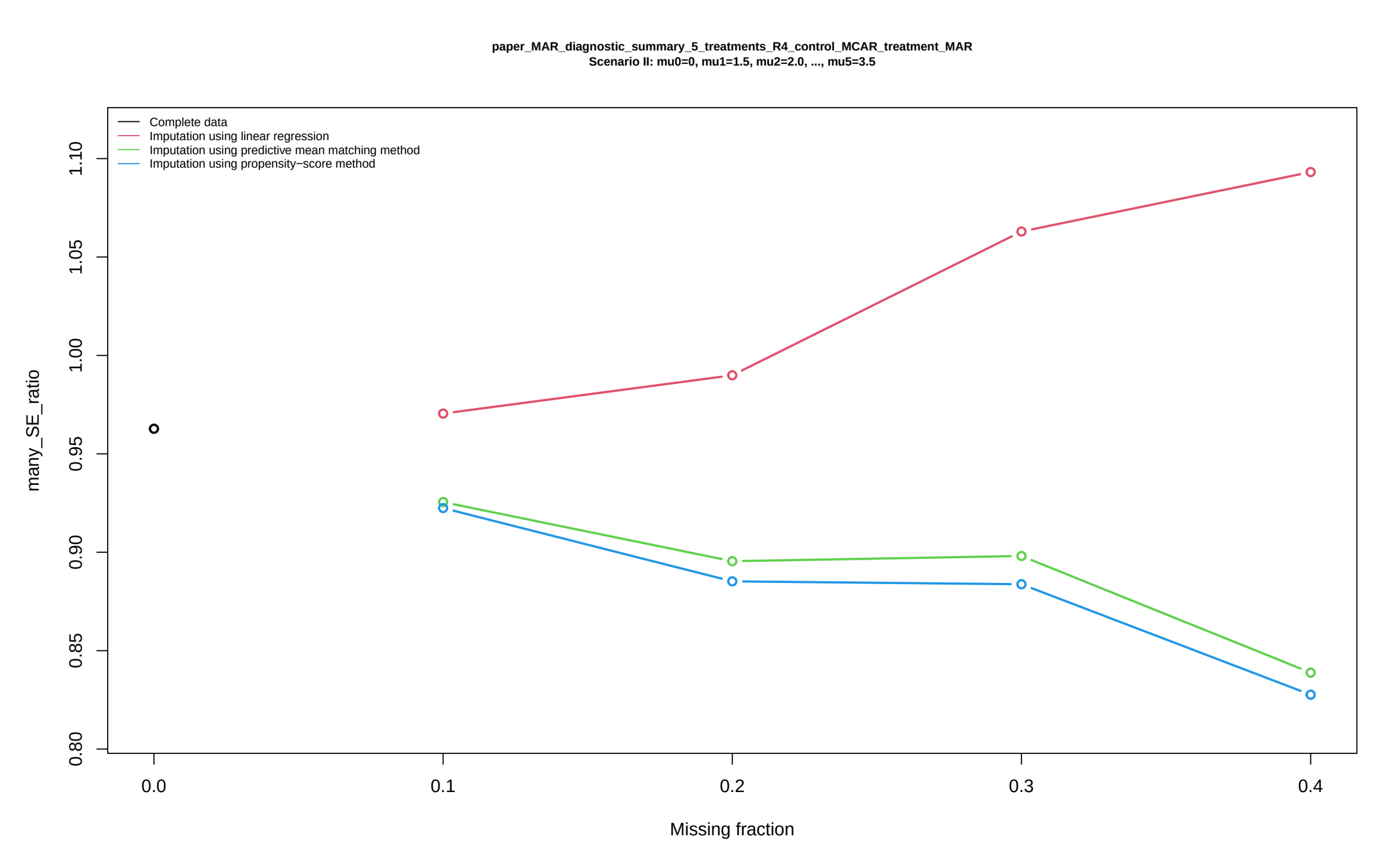

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

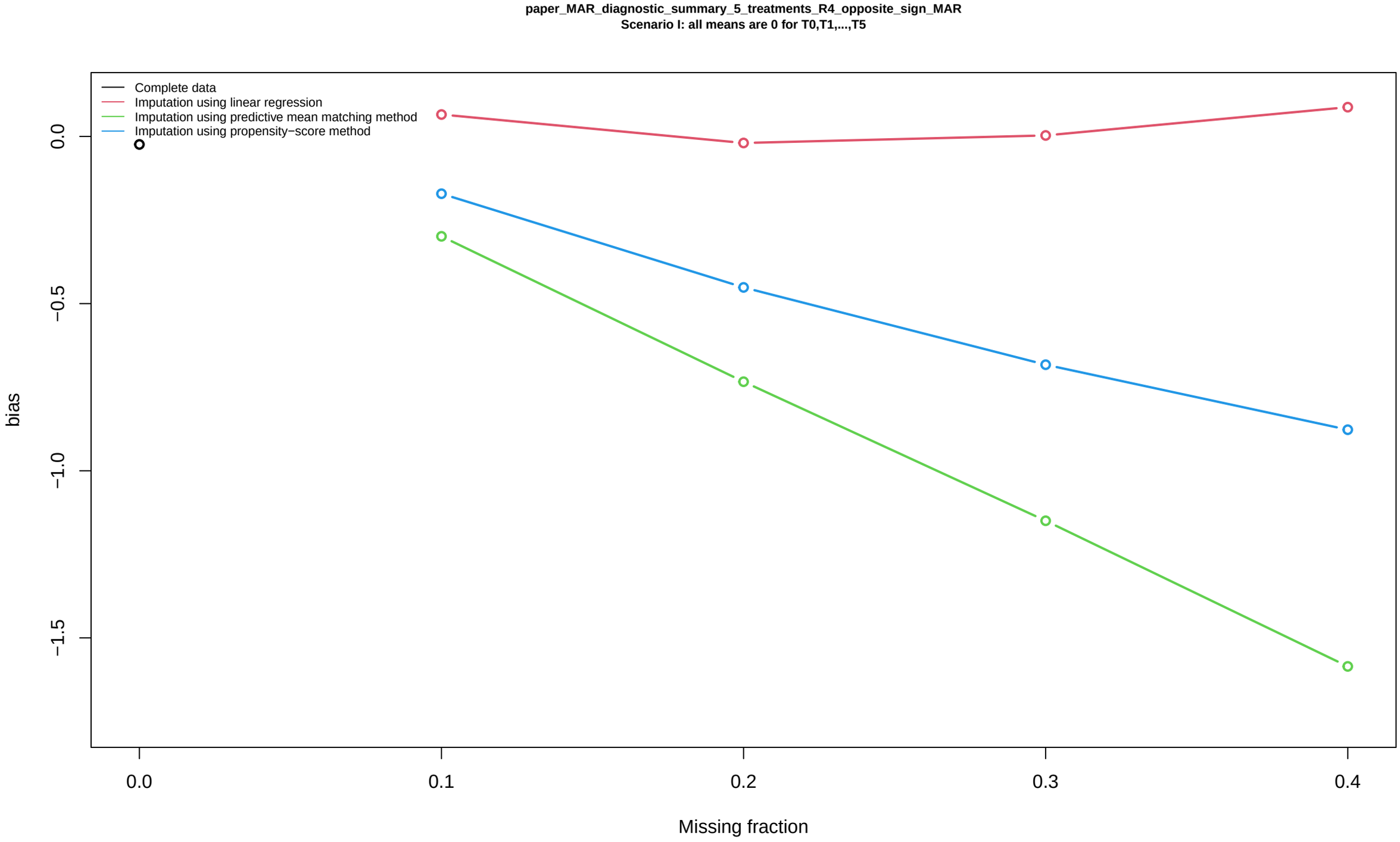

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.6
1.8
2.0
2.2
2.4
0.0
0.1
0.2
0.3
0.4
Missing fraction

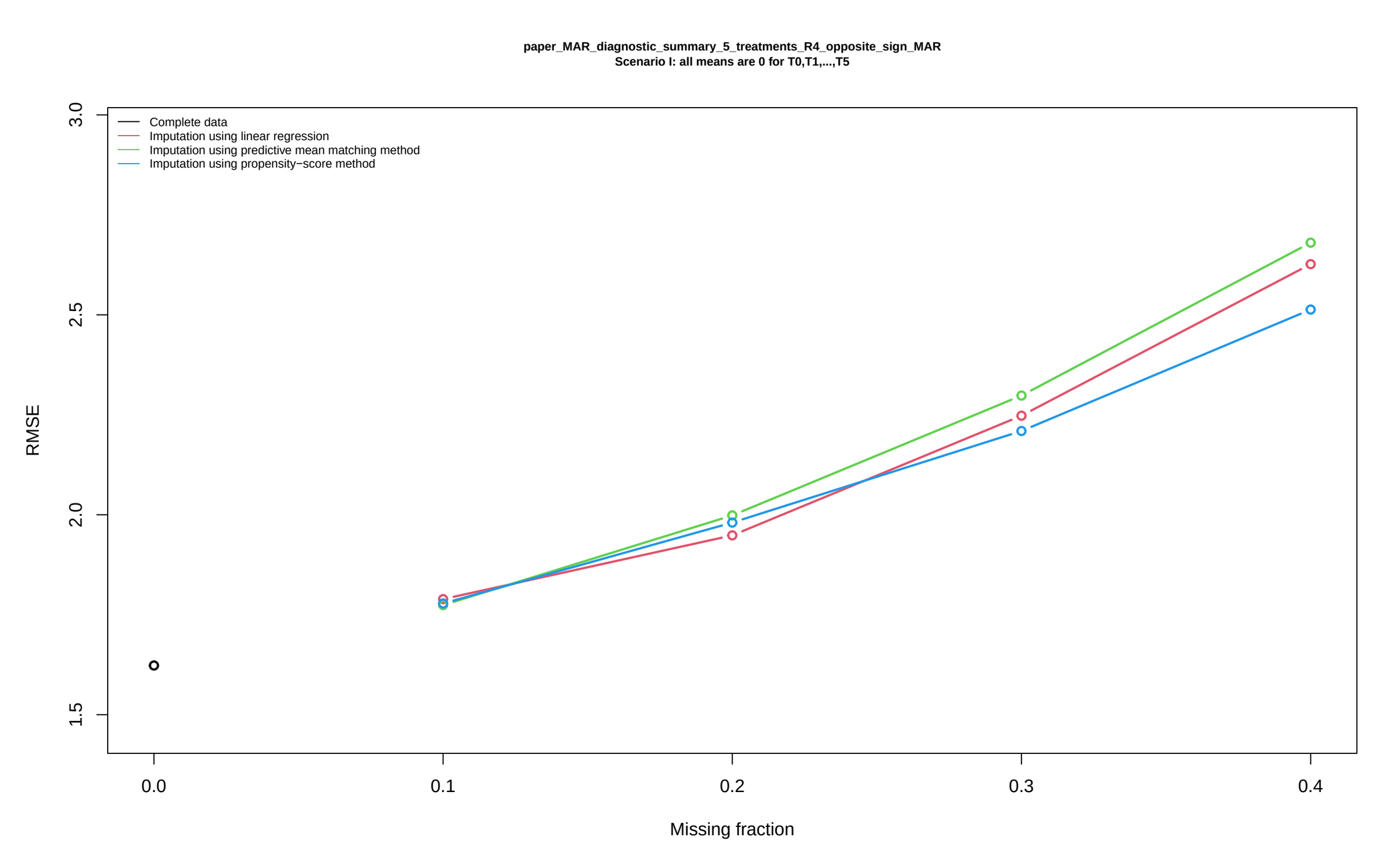

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

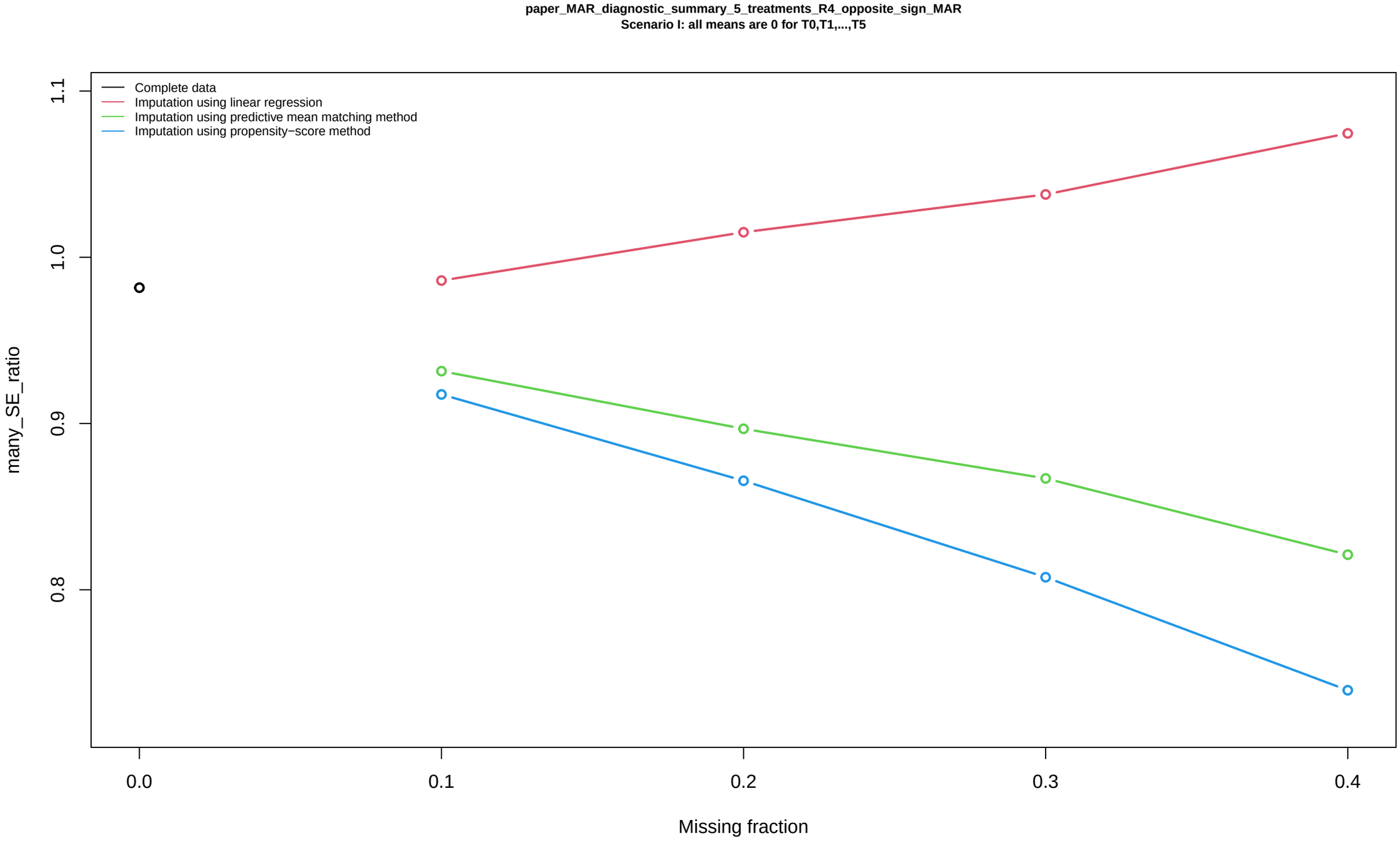

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

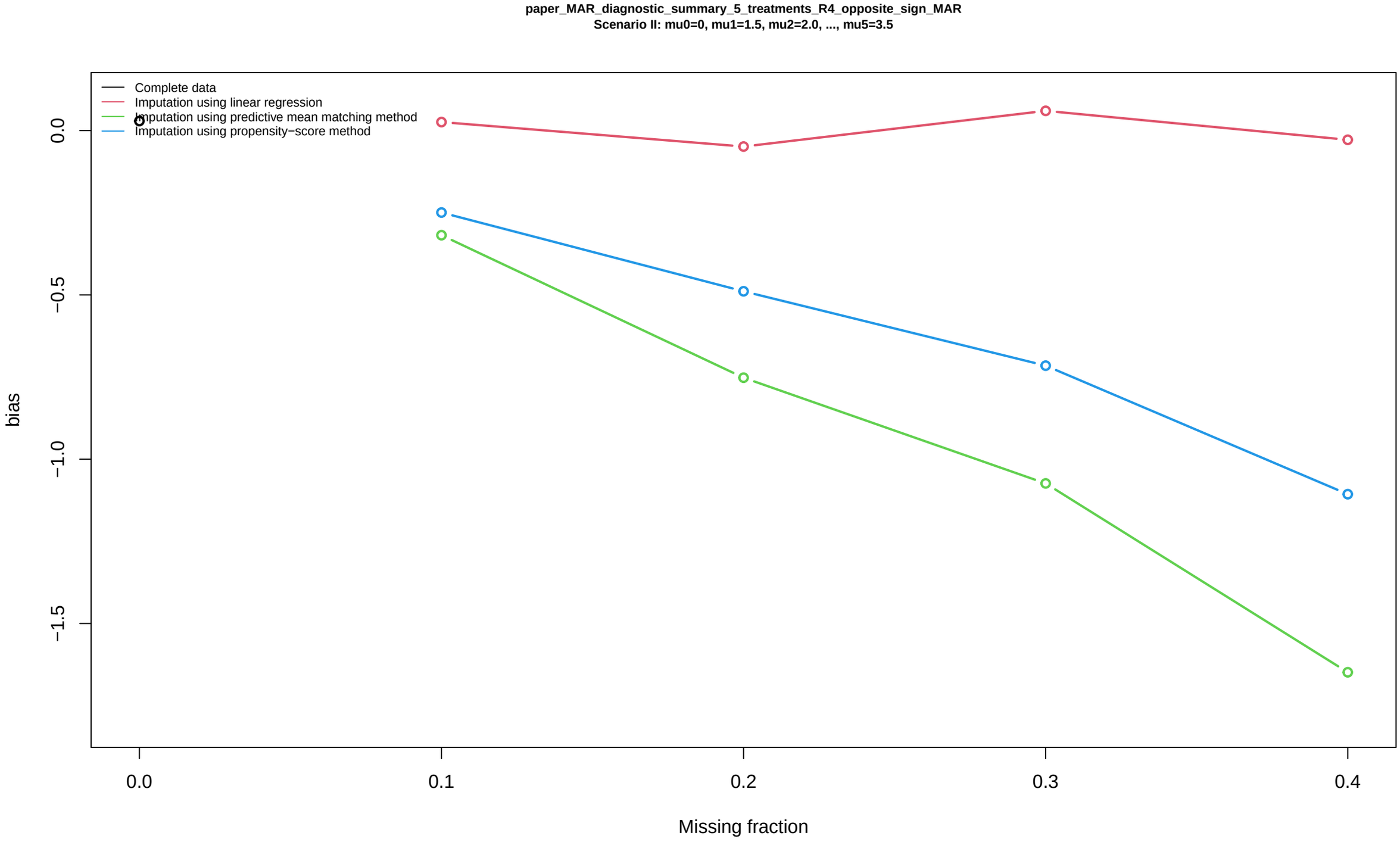

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.6
1.8
2.0
2.2
2.4
2.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

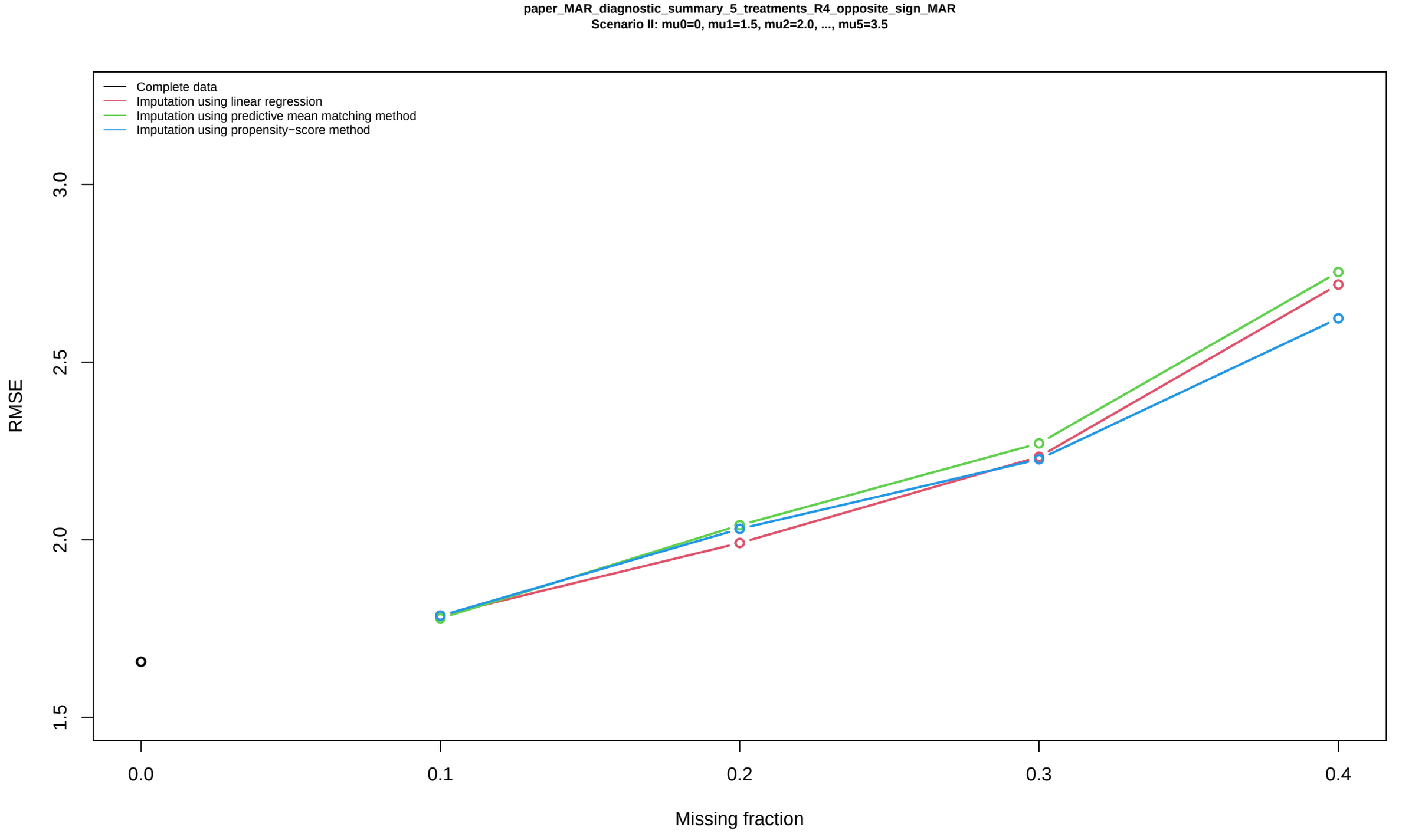

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

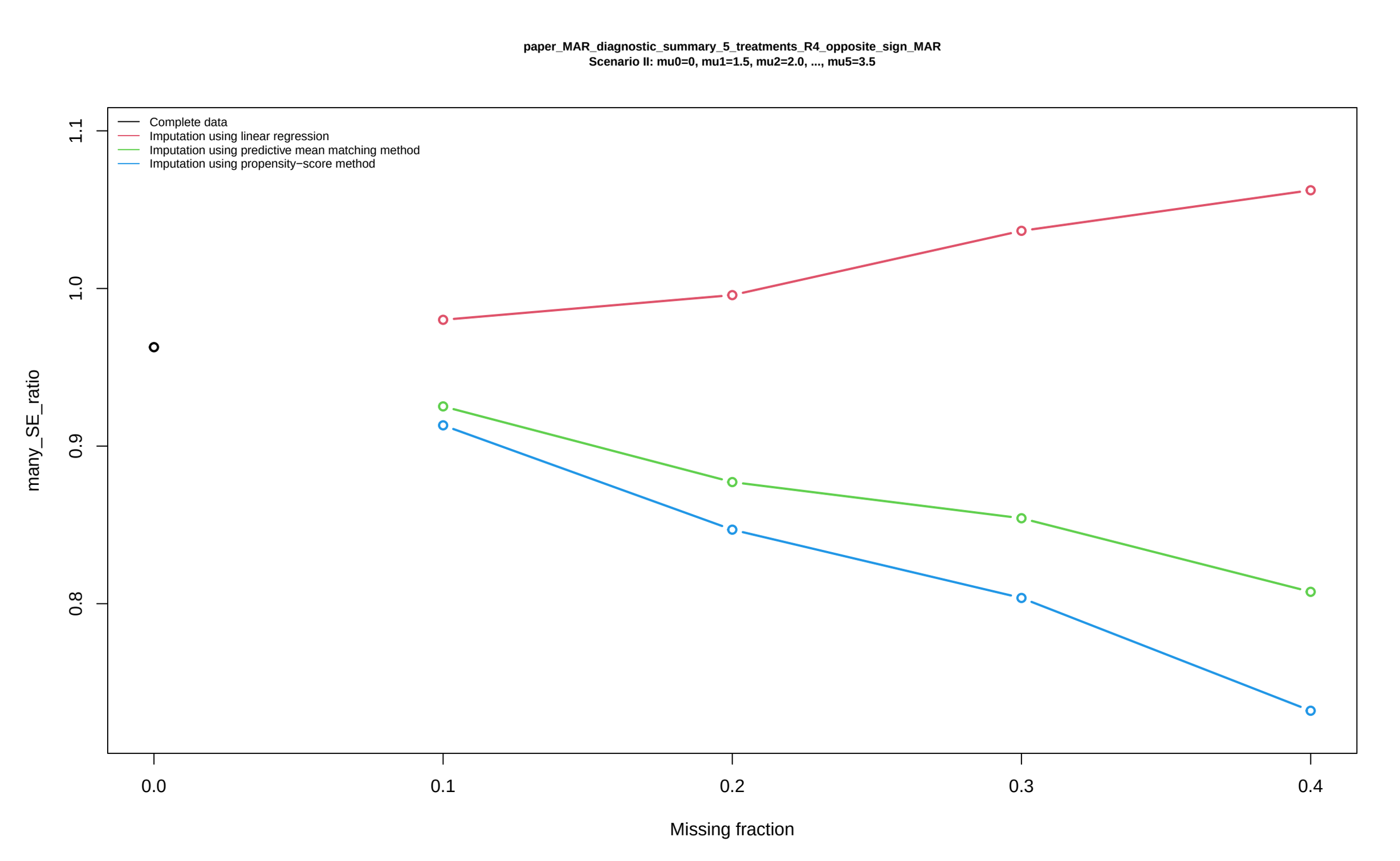

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

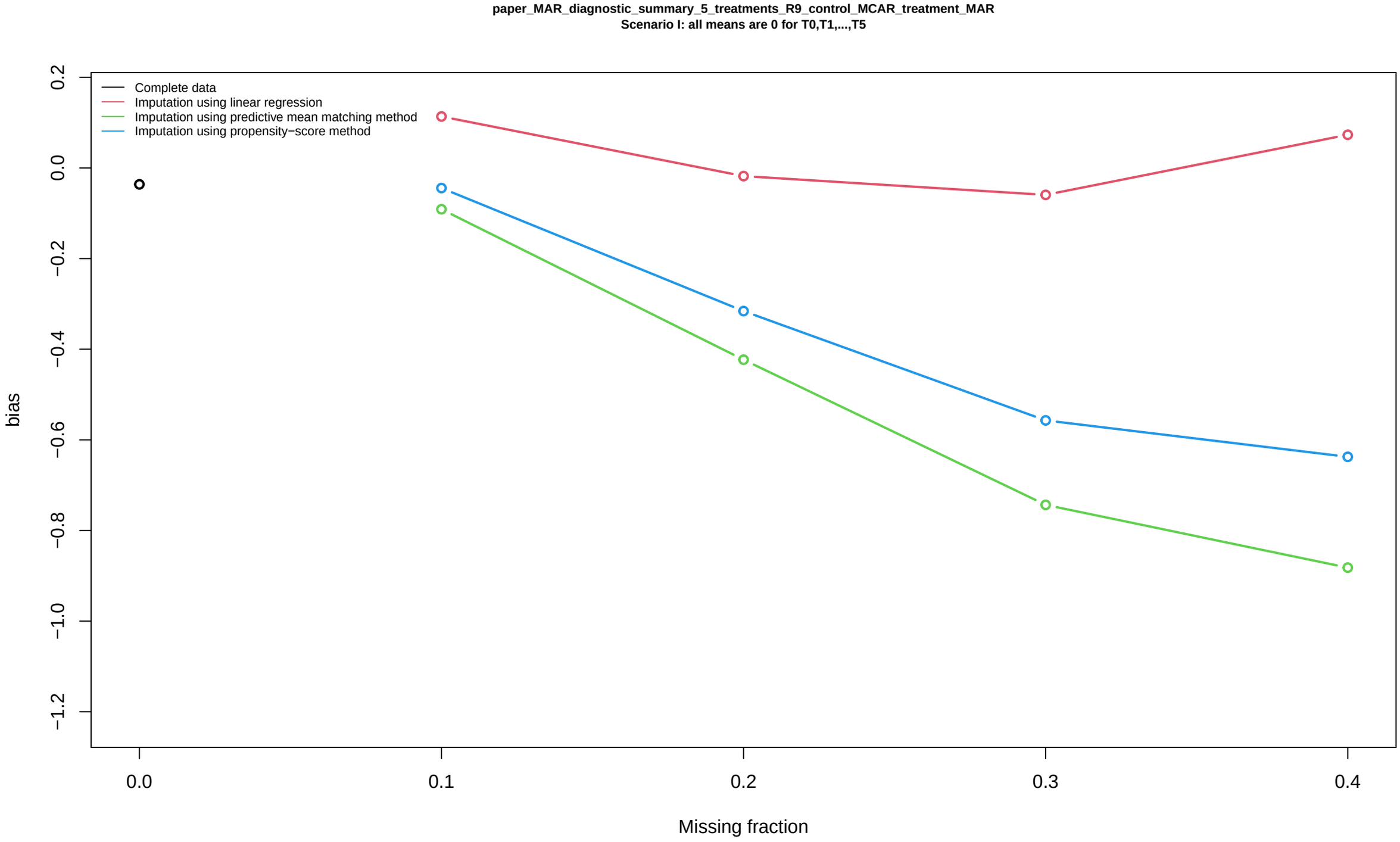

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
3.0
2.5
2.0
1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

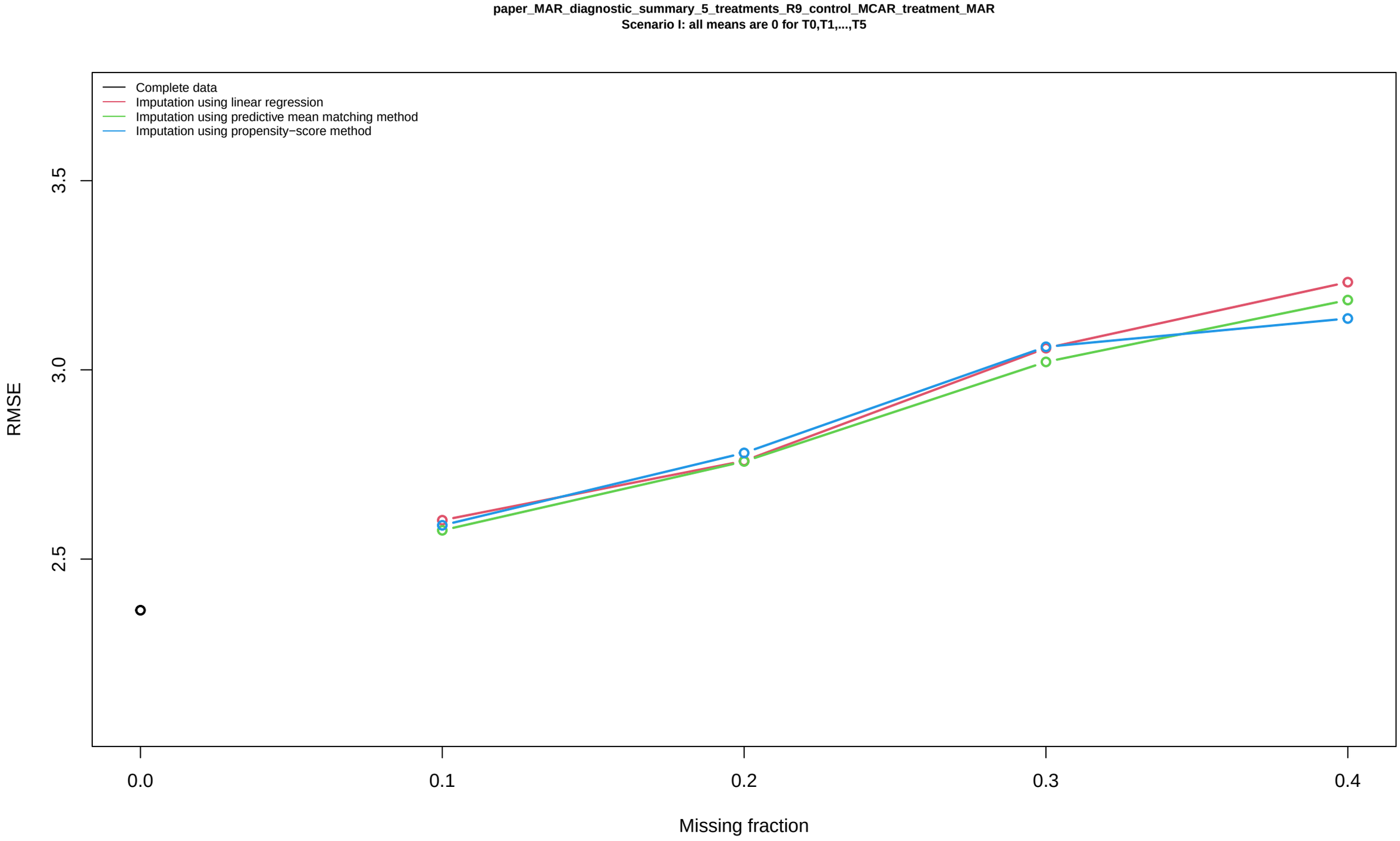

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

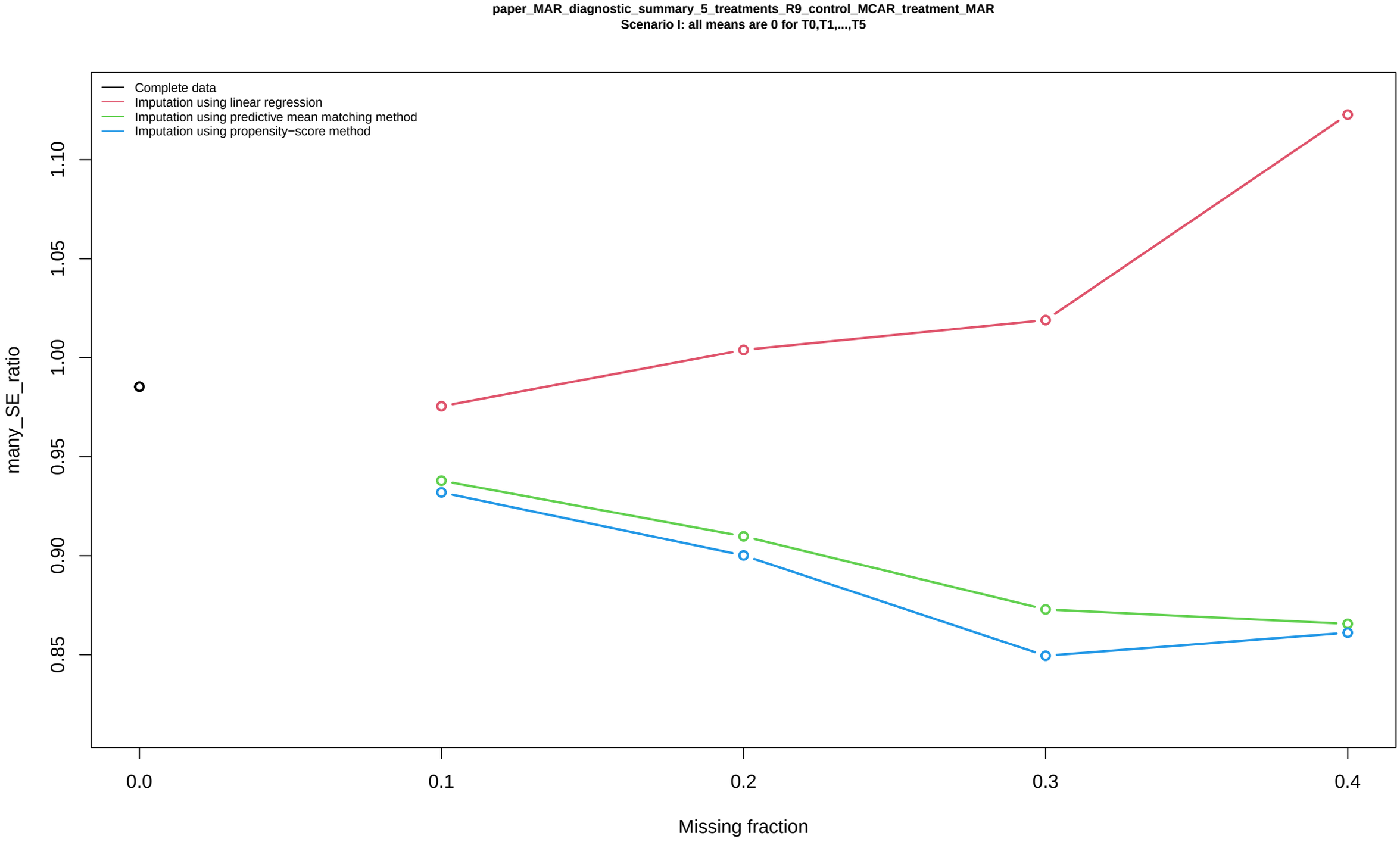

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

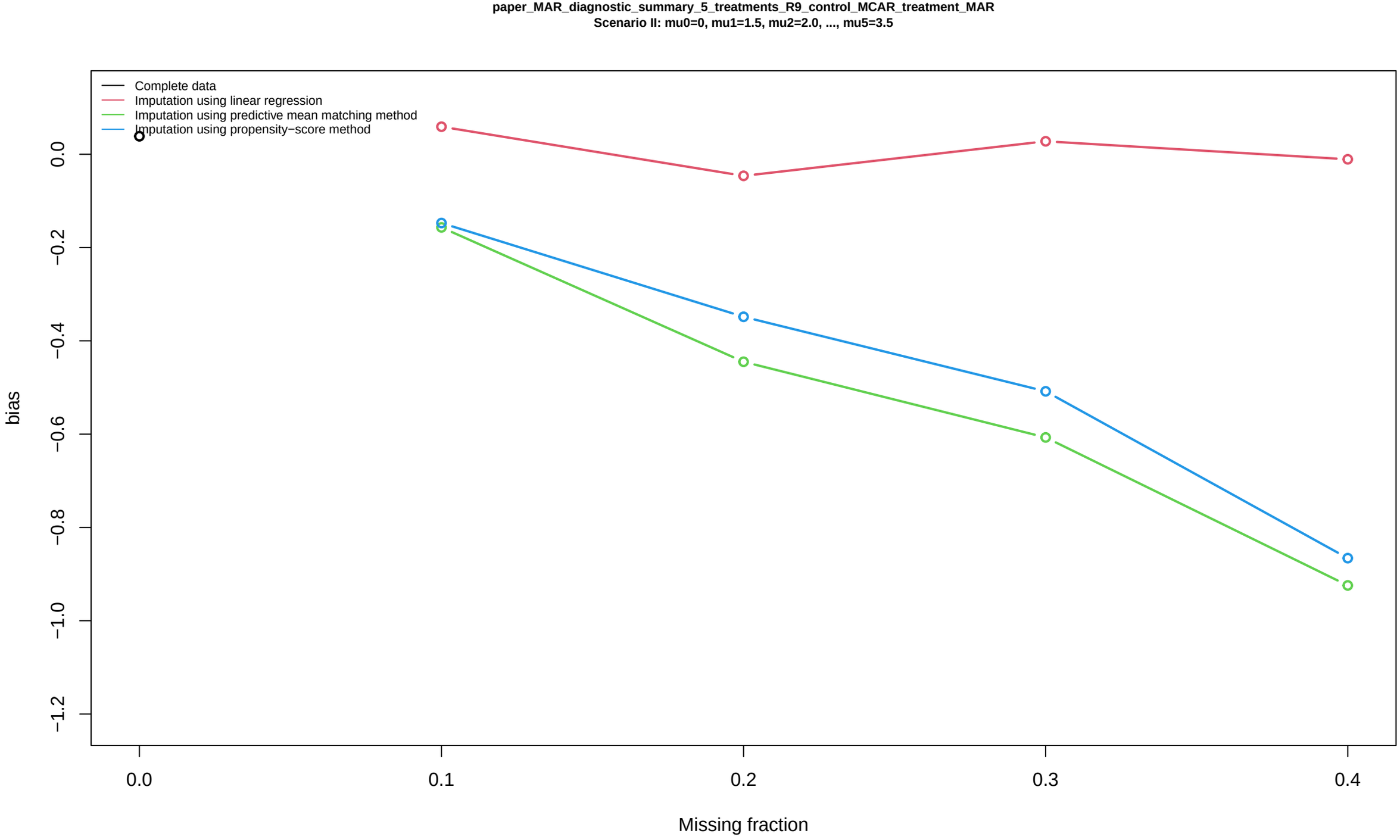

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.5
2.0
2.5
3.0
0.0
0.1
0.2
0.3
0.4
Missing fraction

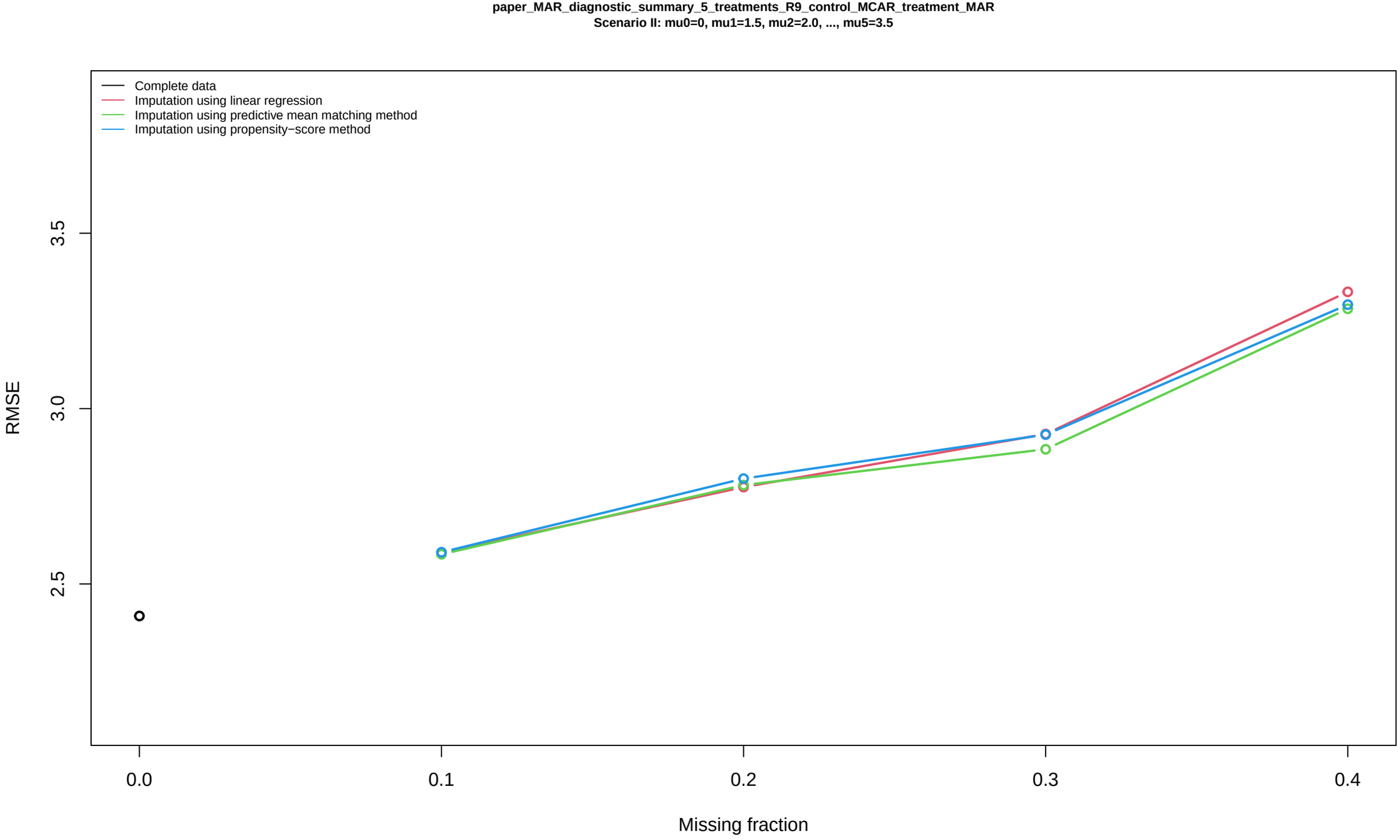

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

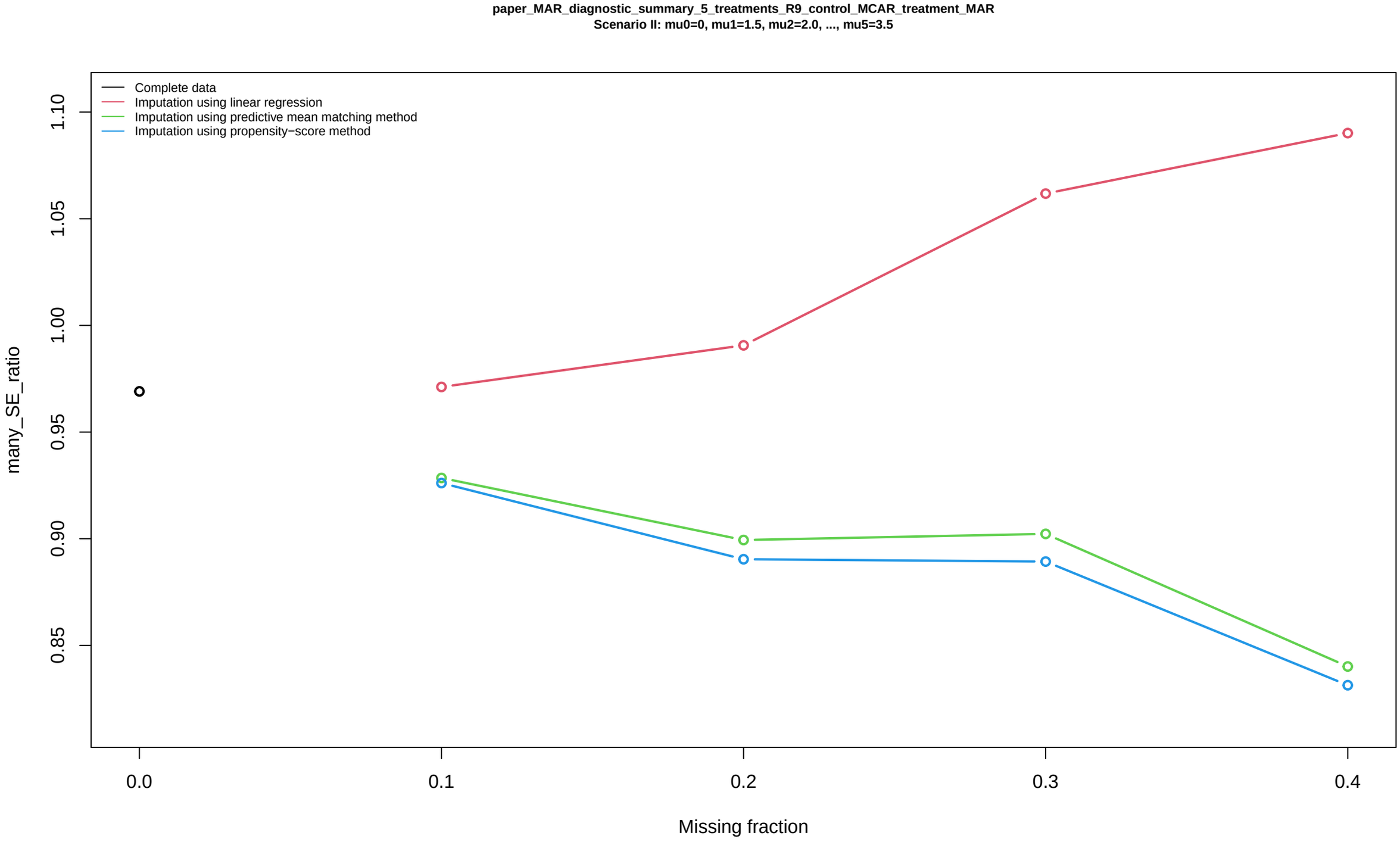

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

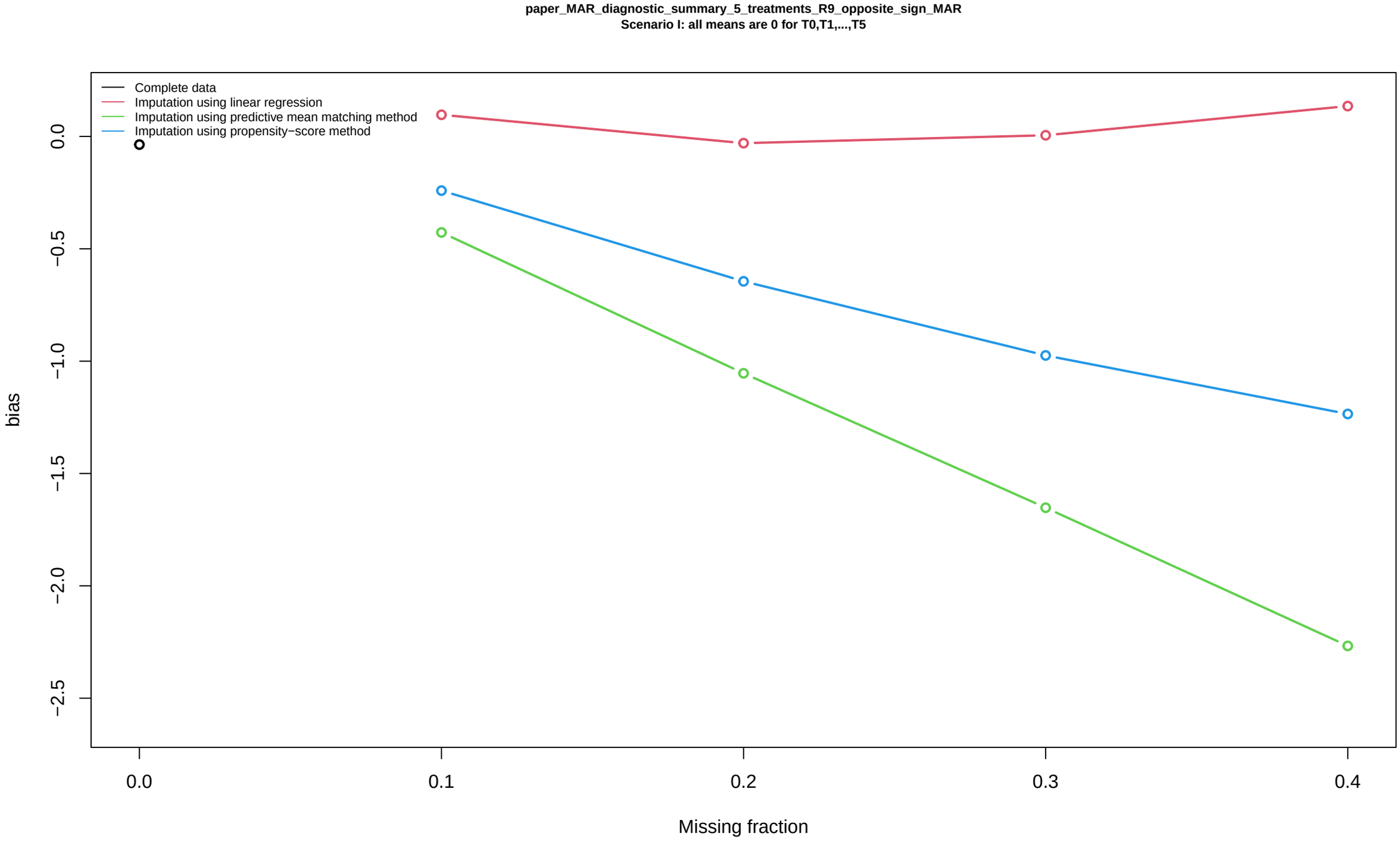

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

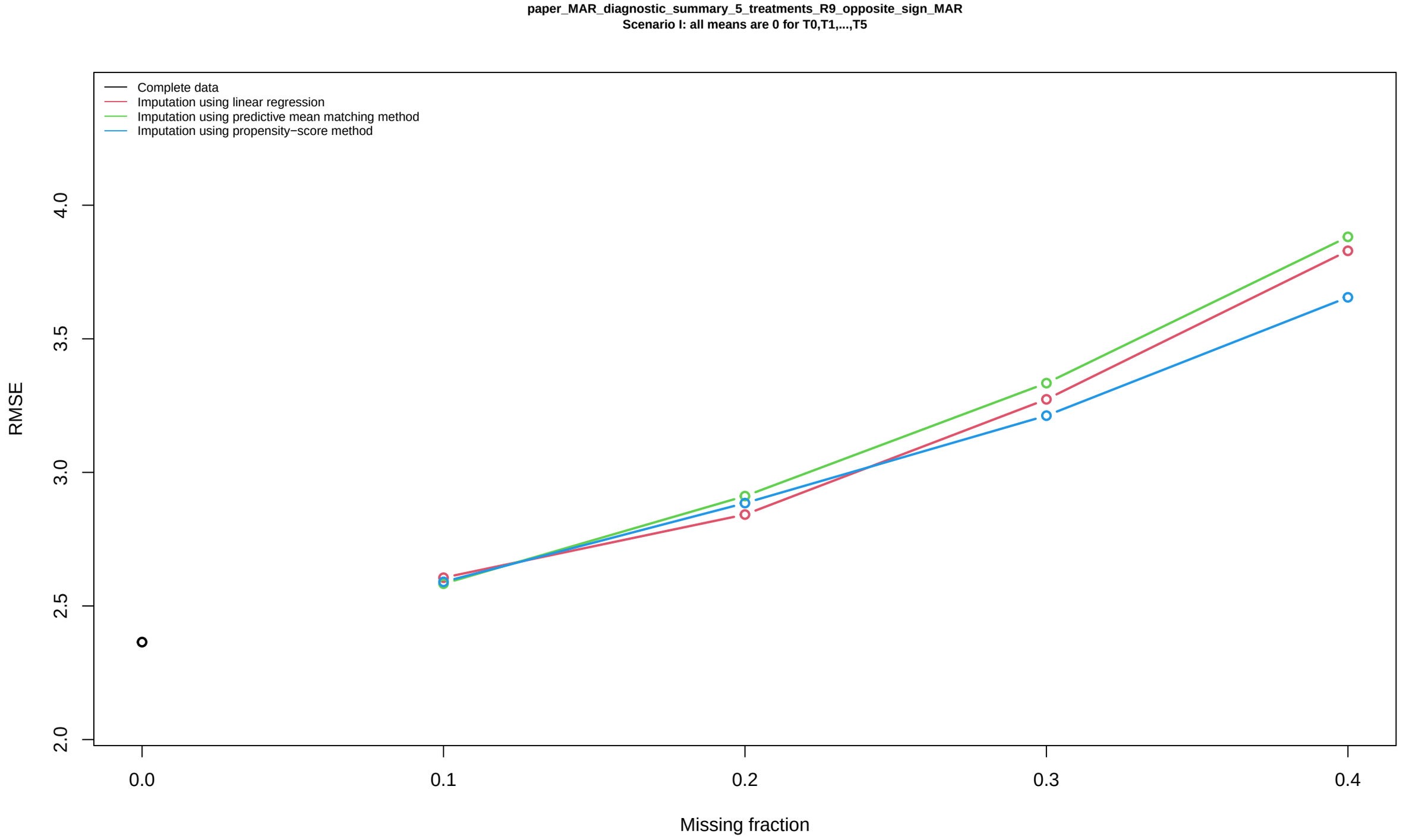

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.10
1.05
1.00
0.95
0.90
0.85
0.0
0.1
0.2
0.3
0.4
Missing fraction

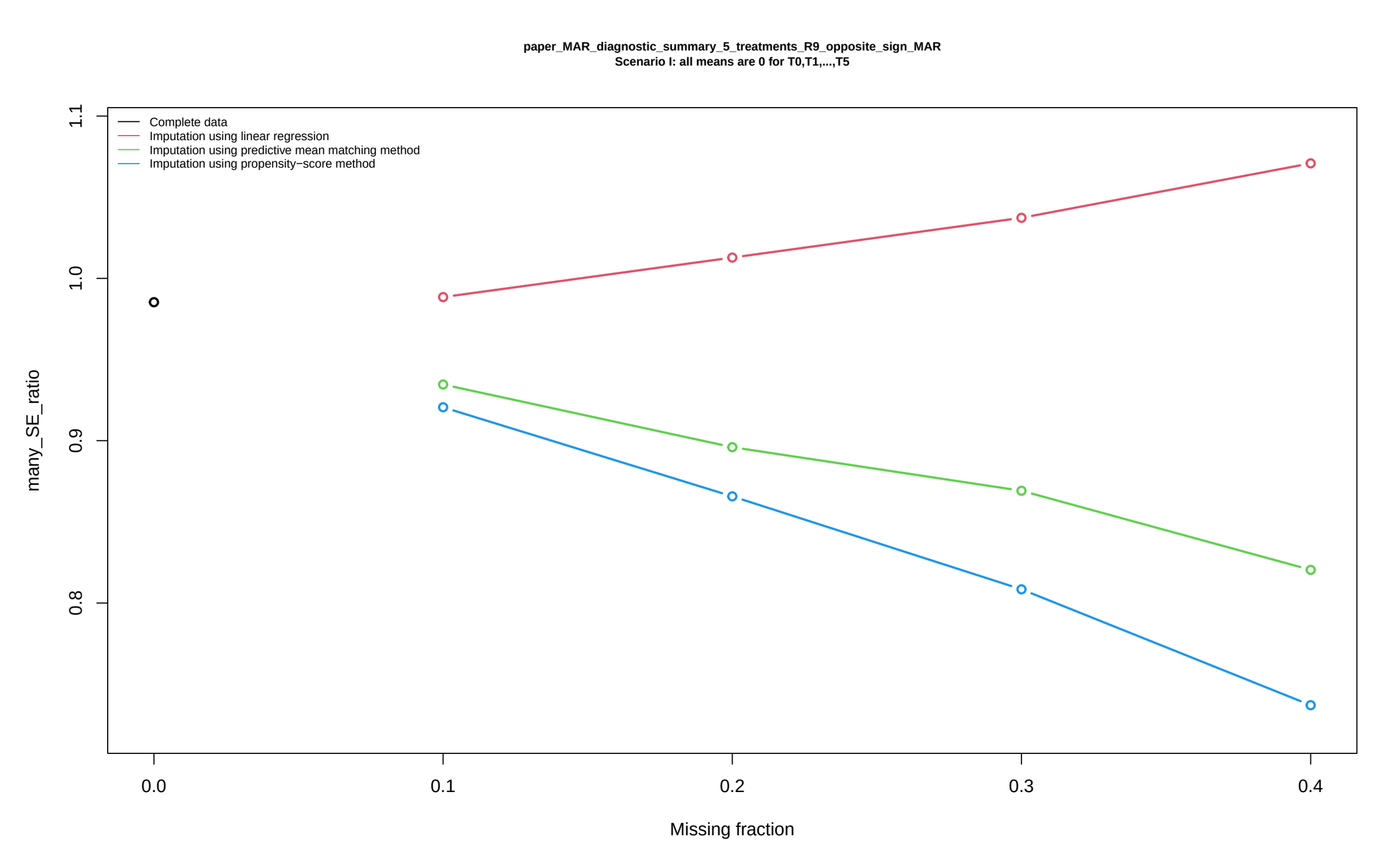

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

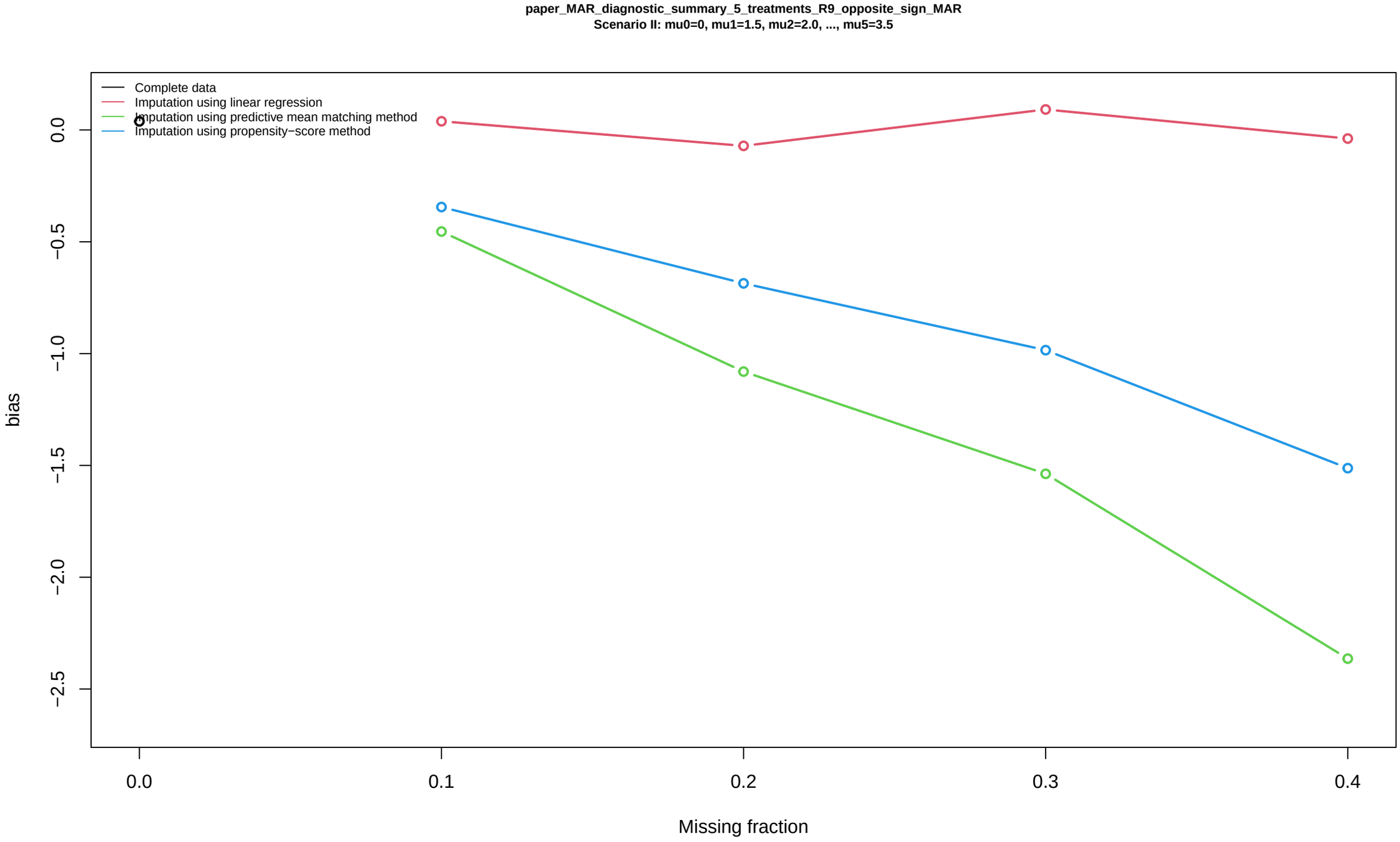

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

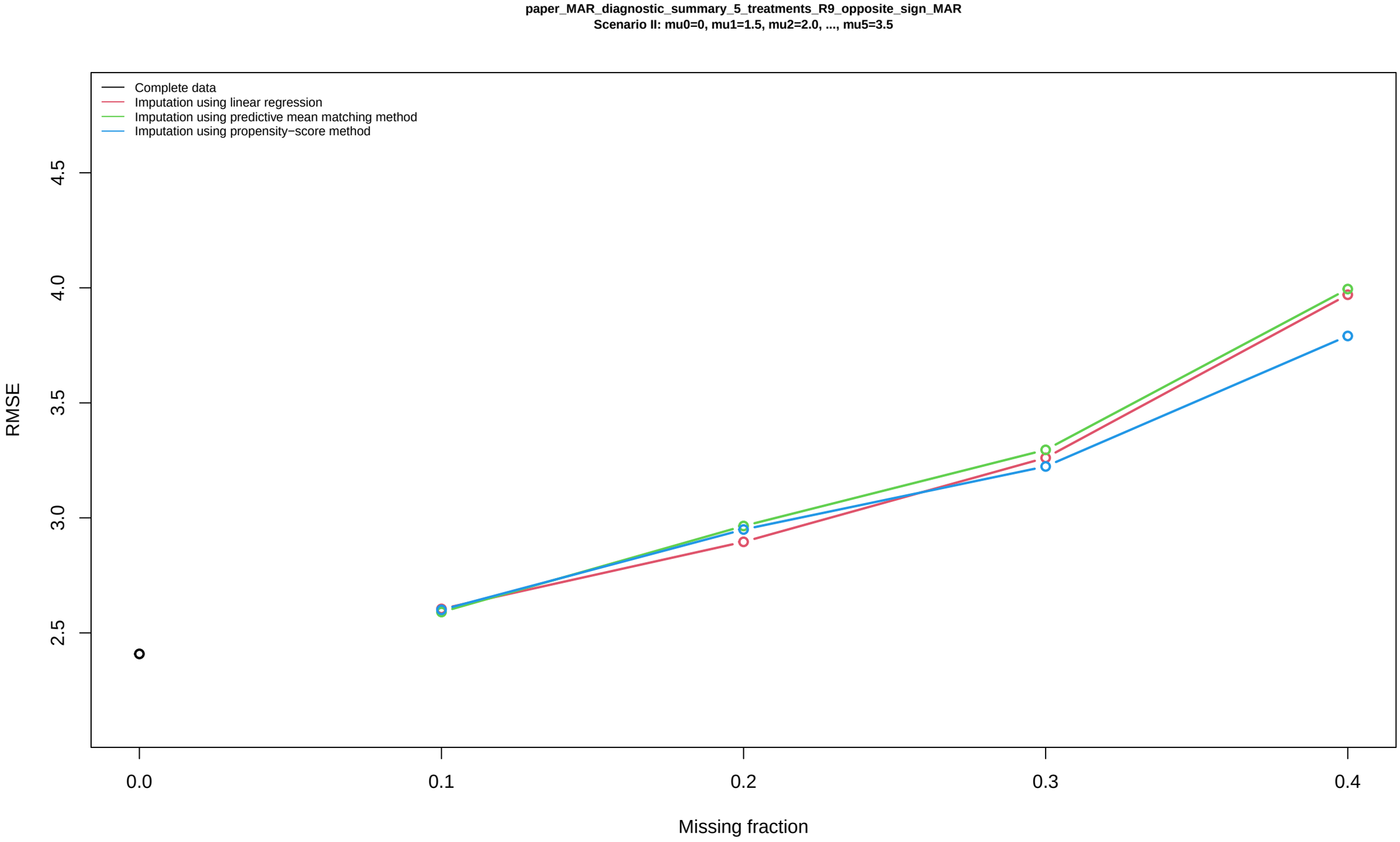

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.10
1.05
1.00
0.95
0.90
0.85
0.0
0.1
0.2
0.3
0.4
Missing fraction

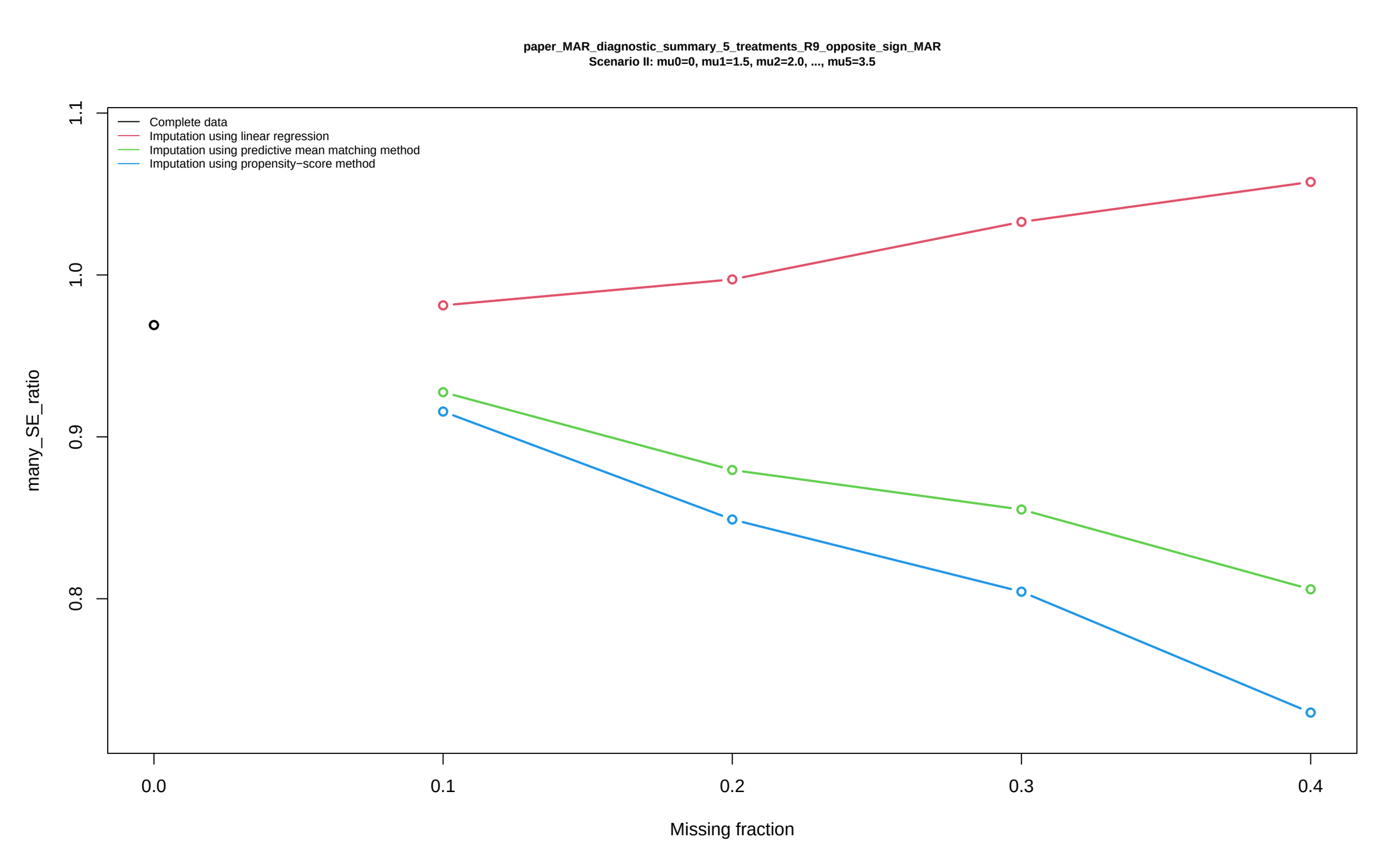

paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.0
2.5
3.0
3.5
4.0
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
4.0
4.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction